The Future of Work: Remote Opportunities and Female Labor Force Participation

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This paper examines the effects of the expansion of work-from-home (WFH) opportunities accelerated by the COVID-19 pandemic on female labor market outcomes. Using a difference-in-differences and event study design, this paper compares individuals in Metropolitan Statistical Areas with higher WFH potential to those in areas with lower WFH potential before and after the pandemic. The results indicate that relative to men, access to WFH opportunities significantly increases the likelihood of female labor force participation, employment, and full-time work. These effects are particularly strong among college-educated mothers and mothers with young children. More broadly, WFH opportunities account for approximately 13 percent of the observed convergence in the male–female labor force participation gap, corresponding to roughly 136,000 additional mothers in the U.S. labor force during the stable post-pandemic period. Furthermore, the mechanism analysis demonstrates that mothers' own WFH opportunity is the primary driver of the effect, while the spillover from a husband's WFH remains a secondary but meaningful force. The effect operates by relaxing a caregiving constraint: it is largest where childcare is most costly and works mainly by drawing nonemployed mothers into the labor force. College mothers benefit most because they have greater access to high-teleworkability occupations.

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1 Introduction

The landscape of the labor force has undergone a significant transformation in recent decades, characterized by an overall upward trend in female labor force participation (LFP) (Fernández, 2013; Goldin, 1990; Goldin and Katz, 2002). This growth pattern in female LFP began in the early 20th century and has since been extensively studied (Albanesi and Olivetti, 2016; Attanasio et al., 2008; Bailey, 2006; Costa, 2000; Goldin and Katz, 2002; Greenwood et al., 2005; Vidart, 2024). More recently, the COVID-19 pandemic has brought about significant changes to societal and economic structures, with the labor market undergoing some of the most profound shifts. One notable transformation has been the rapid adoption of remote and hybrid work arrangements. Data from the U.S. Census Bureau's Household Pulse Survey (2022-2023) and the Survey of Working Arrangements and Attitudes reveal that full days worked from home made up 28 percent of paid workdays in June 2023, a fourfold increase compared to 2019 (Barrero et al., 2023). Importantly, this shift to new work arrangements has proven to be persistent and is likely to become permanent in the post-pandemic era, even as distancing mandates have ended and COVID-19 health risks have diminished (Aksoy et al., 2022; Barrero et al., 2021, 2023; Bick et al., 2023).

This paper examines the effects of the expansion of work-from-home (WFH) opportunities accelerated by the COVID-19 pandemic on female labor market outcomes. It explores whether the rise in remote work can explain the observed resilience and recent growth in female labor force participation, and whether women were more adversely impacted or ultimately benefited from these arrangements. I construct a measure of exposure to WFH opportunities at the Metropolitan Statistical Area (MSA) level, building on the methodology of Gupta et al. (2022) and the occupation-level WFH scores developed by Dingel and Neiman (2020). This measure reflects the share of jobs in each MSA that could be performed remotely, based on the weighted average of occupational teleworkability scores during the pre-pandemic period. Using a difference-in-differences (DiD) and event study design, I then exploit variation across MSAs with higher versus lower pre-existing WFH potential to examine changes in female labor market outcomes before and after the pandemic.

This paper is motivated by the remarkable changes in female labor force participation that coincided with the rapid growth of work-from-home arrangements during the COVID-19 pandemic. Figure 1 illustrates the labor force participation rates for the working-age population (ages 25 to 64) in the United States from 2012 to 2024, based on data from the American Community Survey (ACS). The data reveal three key trends: (1) Male LFP experienced a sharper and larger decline in 2020, recovering quickly initially but taking longer to return to pre-pandemic levels. (2) Female LFP exhibited a smaller decline during the

pandemic, followed by a slower recovery but steady growth in the post-pandemic period, ultimately reaching its highest recorded levels. (3) Across the entire span, male LFP remained relatively stable overall, whereas female LFP demonstrated a sustained upward trend. These findings are consistent with existing literature and various reports,[1] which highlight that prime-age women have driven the labor market recovery and have been a primary source of post-pandemic labor supply growth.

However, this smaller decline and steady rise in female LFP is particularly notable given that early research predicted women would be disproportionately harmed by the COVID-19 pandemic. Termed the "she-cession" (Goldin, 2022), these concerns were grounded in women's heightened caregiving responsibilities due to school closures and eldercare demands (Alon et al., 2020a; Goldin, 2022), and their overrepresentation in disrupted service-sector occupations (Albanesi and Kim, 2021; Alon et al., 2020b). At the same time, prior research has also shown that women are more likely to be employed in occupations with higher remote-work potential and tend to place greater value on occupational flexibility (Bang, 2022; Garro-Marín et al., 2025; Tito, 2024). Figure 2 further validates that women not only matched but exceeded men in actual uptake of remote work arrangements during the pandemic. Importantly, the gender gap in work-from-home rates widened in the post-pandemic years, suggesting that women have continued to leverage these opportunities at higher rates than men. This trend calls for a deeper examination of gender disparities in the context of remote work and underscores the possibility that increased access to flexible work arrangements may have supported and even strengthened female labor force attachment beyond the immediate crisis.

My findings reveal that WFH opportunities have a significant and positive impact on female labor force participation, employment rates, and full-time work. The effects are more pronounced for college-educated mothers and are largest for mothers with young children. At the MSA level, a one–standard deviation increase in WFH opportunities is associated with a 0.74 percentage point increase in labor force participation among mothers with young children, a 0.42 percentage point increase in employment, and a 0.75 percentage point increase in full-time work. The estimates indicate that when local opportunities lead mothers with young children to take up WFH, they are about 32 percent more likely to participate in the labor force, 19 percent more likely to be employed, and 36 percent more likely to

[1] See, for example, U.S. Bureau of Labor Statistics (2023), which reported that the labor force participation rate for women aged 25 to 54 reached 77.6% in May 2023, the highest level since January 2007; NBC News (2023), which highlighted the role of prime-age women in driving the labor market recovery, noting that their participation rates returned to pre-pandemic levels by January 2023; and The Hamilton Project (2024), which emphasized that as of January 2024, 78% of women aged 25 to 54 were participating in the labor force, marking the highest rate on record. The report also noted that prime-age women have played a significant role in driving the overall increase in labor force participation over the past five years, despite variations across demographic subgroups.

work full-time. These effects are economically meaningful. The estimates imply that rising WFH opportunities account for approximately 13 percent of the observed convergence in the male–female labor force participation gap, corresponding to roughly 136,000 additional mothers in the U.S. labor force during the stable post-pandemic period.[2] Furthermore, the mechanism analysis demonstrates that mothers' own WFH opportunity is the primary driver of the effect, while the spillover from a husband's WFH remains a secondary but meaningful force. The effect operates by relaxing a caregiving constraint: it is largest where childcare is most costly and works mainly by drawing nonemployed mothers into the labor force. College mothers benefit most because they have greater access to high-teleworkability occupations.

While the event study results exhibit clear parallel trends prior to the pandemic, a fundamental identification challenge arises because the COVID-19 pandemic affected all metropolitan areas simultaneously, so that the expansion of WFH opportunities coincided with many other pandemic-era changes. The concern is that the observed effects may not solely reflect the causal impact of WFH opportunities, but could instead be driven by differential pandemic-era effects across demographic groups that happen to correlate with WFH availability. To address this, I use time-varying controls following Jack et al. (2023) and Ross et al. (2024), directly control for local COVID-19 severity and school closure intensity, and conduct a falsification exercise around the 2008 financial crisis. In addition, the validity of my findings is supported by a series of robustness checks. I address concerns in the continuous DiD literature (Callaway et al., 2024) by showing that results are robust to alternative treatment constructions, including a median-split binary indicator and quintile bins. I further show that the results are robust to an alternative WFH exposure index using the occupational WFH score of Hansen et al. (2023), which is based on job postings and captures both fully remote and hybrid work arrangements.

This paper contributes to several strands of literature. First, it provides some of the first population-level causal evidence that WFH opportunities can shift aggregate labor supply. A growing body of research examines remote work from various angles, including its effects on productivity and worker outcomes (Bloom et al., 2024, 2015; Emanuel and Harrington, 2024; Gibbs et al., 2023), its gendered impacts in experimental or firm-level settings (Ho et al., 2024; Jalota and Ho, 2024; Mas and Pallais, 2017; Sherman, 2020), intra-household spillovers (Filippo et al., 2026; Pabilonia and Vernon, 2024), and the motherhood penalty and fertility responses (Aksoy et al., 2026; Harrington and Kahn, 2025; Zarate, 2025). While these studies provide important insights, much less is known about the aggregate labor supply effects of remote work. This paper fills this gap by showing that WFH opportunities enable

[2]The results section (Section 4.3) provides a more detailed discussion of the calculation and interpretation of these numbers.

marginalized groups, particularly mothers, to participate more fully in the labor market.

Several closely related papers inform this study but differ in research question or design. Dingel and Neiman (2020) develop the occupation-level teleworkability scores that I use as an input, Gupta et al. (2022) provide the methodology for aggregating these scores to the geographic level, and Mas and Pallais (2017) establish that workers value schedule flexibility, together supplying important inputs and micro-foundations but not estimating causal labor market effects. The closest work, Filippo et al. (2026), focuses on intra-household spillovers by conditioning on husbands' occupations, while Harrington and Kahn (2025) examines the motherhood penalty using degree-level telework feasibility. This paper instead estimates the broader labor supply effect of WFH opportunities leveraging geographic variation in a DiD design. Nevertheless, my findings are consistent with Harrington and Kahn (2025) in showing that college-educated mothers and mothers with young children experienced the largest gains from WFH, and the spousal channel remains meaningful as documented by Filippo et al. (2026).

Second, this paper identifies work-from-home opportunities as a newly important driver of female LFP and a significant contributor to narrowing gender gaps. Historical growth in female LFP has been attributed to structural transformations (Costa, 2000; Goldin, 1990), cultural shifts (Fernández, 2013; Fogli and Veldkamp, 2011), technological advances reducing household burdens (Greenwood et al., 2005; Vidart, 2024), and changes in fertility behavior driven by the contraceptive pill (Bailey, 2006; Goldin and Katz, 2002) and reduced childcare costs (Attanasio et al., 2008). Prior studies remained uncertain whether recent female LFP gains would persist absent pandemic disruptions and whether remote work truly drives these improvements. This paper resolves both questions by providing causal evidence that flexible work arrangements significantly enhance women's labor supply and account for approximately 13% of the narrowing gender gap.

Third, this paper addresses concerns that the observed improvements in gender inequality during the pandemic may reflect temporary responses rather than lasting changes. Prior work documents that women experienced disproportionate job losses and faced increased caregiving burdens during COVID-19, raising the possibility that these factors could offset potential gains from remote work (Albanesi and Kim, 2021; Alon et al., 2020a,b; Goldin, 2022). This paper distinguishes between short-run pandemic effects and more persistent post-pandemic changes, and shows that women's labor force participation proved more resilient than initially predicted. Importantly, the results indicate that the expansion of work-from-home opportunities contributed to sustained improvements beyond the pandemic period.

The rest of the paper is organized as follows: Section 2 describes the data and sample

construction. Section 3 details the empirical strategy and methods for addressing identification concerns. Section 4 presents the main results, heterogeneity analyses, an interpretation of the effect magnitudes, and robustness checks. The final sections discuss mechanisms and conclusions.

2 Data

2.1 Sample Construction

American Community Survey. This study uses the American Community Survey (ACS), a nationally representative, repeated cross-sectional survey of approximately 1% of the U.S. population conducted annually by the U.S. Census Bureau. The ACS offers national representativeness, a large sample size, and consistent annual coverage, making it particularly suitable for providing population-level, nationally aggregate evidence on remote work and labor market outcomes across the United States. The sample spans 2012 to 2024, capturing the labor market landscape before, during, and after the pandemic.[3] The primary outcome variables are labor force participation, employment, and full-time work (defined as working 35 or more usual hours per week). The sample is restricted to working-age adults (ages 25–64) to avoid distortions from schooling or retirement decisions.[4] Military occupations are excluded as WFH scores are not available for those categories.

Metropolitan Statistical Areas. The WFH exposure index is constructed at the metropolitan statistical area (MSA) level using the 2013 Office of Management and Budget (OMB) delineations. This variable provides a consistent geographic definition across all years in the sample without requiring crosswalk construction, making it the cleanest available measure for capturing local variation in WFH opportunity.[5] I restrict the analysis to the

[3]The sample begins in 2012 because the U.S. Census Bureau redraws Public Use Microdata Area (PUMA) boundaries every ten years based on the most recent decennial census, and the 2010-based PUMAs were first implemented in the ACS for 2012 respondents. The sample ends in 2024, as this represents the most recent ACS data available at the time of the study.

[4]As a robustness check, I restrict the sample to ages 25–54 and find similar results.

[5]Alternative geographic units such as commuting zones (CZs) and combined statistical areas (CSAs) are not directly identified in the ACS and require PUMA-to-CZ or PUMA-to-CSA crosswalks. Most importantly, because the Census Bureau redraws PUMA boundaries every ten years, with the 2010 definitions used for 2012–2021 and the 2020 definitions used for 2022–2024, these crosswalks introduce measurement error during the years in which remote work expanded most rapidly, potentially confounding the estimated effects. In addition, MSAs are the standard geographic unit in the WFH literature (Dingel and Neiman, 2020; Gupta et al., 2022; Ramani and Bloom, 2021). They also align with the geographic units at which the auxiliary measures used in this paper, including COVID-19 severity, school closures, childcare prices, and local economic conditions, are available or can be reliably constructed. I verify that the results are robust to using commuting zones as the geographic unit; see Section 4.4.6.

242 MSAs that are consistently identifiable throughout the entire sample period.[6] Figure A1 presents the geographic distribution of pre-pandemic WFH exposure across these MSAs, revealing substantial variation across regions.

2.2 Occupational WFH Score

To evaluate geographic variation in WFH exposure across local labor markets, a key input is occupational teleworkability scores. In the baseline analysis, I leverage the score developed by Dingel and Neiman (2020), which assigns a binary classification to each occupation indicating whether the job can feasibly be performed entirely from home. The classification is based on detailed occupational characteristics from the O*NET database, and the measure remains stable across O*NET database versions, confirming that it reflects structural features of occupations rather than temporal shifts in classification.[7]

The Dingel and Neiman (2020) (DN) measure represents a theoretical upper bound on WFH feasibility and does not fully capture actual employer behavior or distinguish between fully remote and hybrid work arrangements. I enhance the analysis in the robustness checks by employing the measure of Hansen et al. (2023), which is based on job postings and captures both types of arrangements (see Section 4.4.5).

I link the occupational WFH scores to individuals in the ACS based on their reported occupation codes. The ACS uses the 2010 Standard Occupational Classification (SOC) system for 2012–2017 and the 2018 SOC system for 2018 onward. Since the DN scores are based on the 2010 SOC, I use a consistent classification for the 2012–2017 period and apply a crosswalk for subsequent years. Because some ACS occupations are reported at broader categories than the 6-digit SOC codes used by Dingel and Neiman (2020), I aggregate the original scores using employment weights from the Bureau of Labor Statistics (BLS). These weights are averaged across multiple pre-period years to reduce year-to-year fluctuations in occupational composition.

2.3 Summary Statistics

Table 1 presents summary statistics for the key variables, separately by gender. Women have a higher pre-period WFH exposure than men, reflecting their concentration in more teleworkable occupations. They also exhibit lower labor force participation, employment, and full-time work rates, with greater dispersion in each outcome. In terms of education, women

[6] In the sample period, the 2013 MSA definitions encompass 282 MSAs. Due to PUMA boundary changes, some MSAs are not identifiable in all years.

[7] A detailed description of the measure and replication results are provided in the Appendix and Figure A18.

show higher attainment, with a larger college-educated share and a smaller noncollege share than men. They are more likely to be parents of children under 18, while the share with young children under 5 and other demographic characteristics are similar across genders.

Table B1 reports MSA characteristics separately for MSAs with high and low WFH exposure. Above-median WFH MSAs have higher college shares and median household income, are more concentrated in management, business, and computer occupations, and tend to be larger in population. Below-median MSAs have larger shares in service, production, and farming occupations. Table B2 lists the ten MSAs with the highest and lowest WFH exposure values along with their key characteristics. The top ten MSAs are characterized by a large share of workers in management, technology, and professional occupations, while the bottom ten include agricultural and manufacturing-intensive metros with large shares in natural resources, construction, and production occupations.

3 Empirical Strategy

3.1 Measuring Exposure to Work-From-Home Opportunity

$$\text{WFH Measure}_{m,pre} = \frac{1}{T_{pre}} \sum_{t \in T_{pre}} \sum_{o} \left(\frac{\text{Number of jobs}_{m,o,t}}{\sum_o \text{Number of jobs}_{m,o,t}} \times \text{Teleworkability score}_o \right) \tag{1}$$

Equation (1) provides the formula for constructing the WFH measure at the geographic-area level based on the methodology of Gupta et al. (2022). The idea is to calculate the weighted average teleworkability score for each MSA m over the pre-period T_{pre}. This is a two-step process. In the first step, the measure is constructed by calculating the occupation weight for each occupation by dividing the number of jobs in each occupation o by the total number of jobs in the MSA for that year. This weight is then multiplied by the corresponding teleworkability score for that occupation, as developed by Dingel and Neiman (2020).[8] This step outputs a WFH score for each MSA for each year in the pre-period.

The second step is to average these yearly values across all years in the pre-period. I designate the years 2012 to 2019 as the pre-period in the baseline analysis, but results are robust to using a shorter pre-period (2015 to 2017). This produces a stable, time-invariant measure of WFH potential for each MSA during the pre-pandemic period.

Because men and women differ in their occupational distributions, a WFH exposure

[8]To count the number of jobs in each MSA, I apply ACS person weights, which indicate how many persons in the U.S. population are represented by a given person in the sample.

index constructed using one gender's occupational shares may not accurately capture the remote-work opportunities faced by the other gender. To more accurately capture the WFH opportunities available to each group, the WFH index is constructed to match the analysis sample. Specifically, the female WFH index is used for female-only regressions, the male WFH index for male-only regressions, and a pooled index for regressions using the full sample.[9] The baseline measure uses occupational shares among the employed population, which most directly reflects the distribution of jobs that determine remote-work opportunities. As shown in Section 4.4.5, the results are robust to alternative definitions of the WFH index and alternative sampling criteria.

3.2 Empirical Specifications

To explore the causal impact of WFH opportunities on female labor market outcomes, I employ difference-in-differences and event study designs that leverage variation in WFH potential across local labor markets and over time. Specifically, I estimate the following difference-in-differences specification:

$$Y_{imt} = \delta \cdot (\text{WFH}_{m,\text{pre}} \times \text{Post}_t) + \mathbf{X}'_{it}\beta + \alpha_m + \lambda_t + \varepsilon_{imt} \tag{2}$$

Here, Y_{imt} represents the labor market outcome of individual i in MSA m at time t including labor force participation (LFP), employment, and full-time work status. $\text{WFH}_{m,\text{pre}}$ measures the pre-existing WFH potential in MSA m, calculated using Equation (1) and standardized to have mean zero and unit standard deviation. $Post_t$ is an indicator for the post-pandemic period, covering the years 2020 to 2024. MSA fixed effects α_m and year fixed effects λ_t control for time-invariant differences across MSAs and common shocks affecting all individuals in a given year, respectively. The vector of individual-level controls $\mathbf{X}_{it}$ includes variables such as age, race, marital status, educational attainment, and the presence of children under age five. Standard errors are clustered at the MSA level to account for within-region correlation in the error terms.

A key assumption for this analysis is the parallel trends assumption. This assumption posits that, in the absence of the treatment (i.e., the COVID-19 pandemic and the associated increase in WFH opportunities), the labor force participation trends for individuals in MSAs with different levels of pre-pandemic WFH potential would have followed similar trajectories

[9]The female (male) WFH index is constructed following Equation (1) using occupation shares calculated from the female (male) population in each MSA. The pooled index uses occupation shares from the full population.

over time. To assess the validity of this identification strategy and to examine the dynamics of the effect, I estimate the following event study specification:

$$Y_{imt} = \sum_{\substack{s=2012 \\ s\neq 2019}}^{2024} \delta_s \cdot (\text{WFH}_{m,\text{pre}} \times \mathbf{1}\{t = s\}) + \mathbf{X}'_{it}\beta + \alpha_m + \lambda_t + \varepsilon_{imt} \tag{3}$$

The COVID-19 pandemic, which began in 2020, is considered to be the first "treatment" year in this analysis. The year 2019 serves as the reference year, with its coefficient normalized to zero. All other variables are defined as in Equation (2). Additionally, I separate the effects into a pandemic period (2020–2021) and a post-pandemic period (2022–2024) to distinguish transitory pandemic-era responses from more persistent effects.

3.3 Threats to Identification

3.3.1 Validity of the WFH Measure and COVID-19 Activation

Beyond the parallel trends assumption, a key concern is whether the WFH measure actually predicts remote work behavior and whether COVID-19 was the primary driver of remote work expansion. To assess this, I regress actual WFH behavior on the constructed WFH measure following the same DiD and event study design. Actual WFH behavior is measured as an indicator for whether the individual's primary mode of commuting is working from home, based on the ACS transportation-to-work variable.

Table 2 Panel A presents the pseudo first-stage results across demographic groups, where males and fathers use a male-specific WFH index and females and mothers use a female-specific one. The coefficients are positive and statistically significant across all subgroups, with high F-statistics confirming strong predictive power.[10] Men exhibit slightly higher WFH take-up rates than women, and college-educated mothers substantially higher than non-college mothers.

Figure 3 presents the event study results, which further verify that COVID-19 activated latent WFH capacity. There were no differential pre-trends between high and low WFH potential areas before the pandemic, and remote work adoption increased sharply beginning in 2020 and persisted through 2024. This finding is consistent with Barrero et al. (2021), who document that WFH adoption was induced by COVID-19 and has largely persisted.[11]

[10] Appendix Table B5 presents a placebo test in which the cross-gender index is applied to each sample (e.g., the female index predicting male WFH takeup). The estimates are nearly identical across index definitions, with pairwise correlations of 0.984, 0.955, and 0.895, confirming that men and women within the same MSA face broadly similar occupational structures.

[11] Harrington and Kahn (2025) argue that access to WFH opportunities increased prior to the pandemic.

Notably, the pseudo first-stage analysis is best understood as an illustrative tool rather than a structural estimate. It serves two purposes: validating the predictive power of the WFH measure and providing a rough scaling tool for contextualizing the magnitudes of the reduced-form estimates. This paper uses a DiD design rather than an instrumental variables framework, and thus does not require the stronger assumptions associated with a formal IV approach.[12]

3.3.2 Pandemic-Era Confounders

While the event study analysis demonstrates no evidence of differential pre-trends, parallel pre-trends alone may be insufficient when analyzing a shock that affects all units simultaneously (Ross et al., 2024). The pandemic may have changed how individual demographic and MSA characteristics predict female labor market outcomes, which could bias the estimates if those changes are correlated with WFH exposure. At the individual level, for example, the returns to education or marital status may have shifted during the pandemic, and if high-WFH MSAs have a larger share of college-educated women, the estimated WFH effect could partly reflect these changing relationships. At the MSA level, areas with lower remote work potential may have experienced more severe COVID-19 outbreaks or greater school and childcare closures, suppressing female labor force participation through channels other than WFH itself.

To address this, I adopt several strategies. First, following Altonji et al. (2005), Jack et al. (2023), and Ross et al. (2024), I allow all individual-level demographic controls to vary by year, permitting the relationship between observable characteristics and labor market outcomes to shift freely across the pandemic period. Second, I augment the specification with MSA-level characteristics interacted with year dummies to absorb differential trends across areas with different socioeconomic profiles. Third, I directly control for local COVID-19 severity and school closure intensity to isolate the WFH effect from pandemic-related disruptions. Fourth, as a falsification exercise, I replicate the design around the 2008 financial crisis to verify that the WFH index does not predict differential labor market effects during a non-WFH recession. Detailed results from these approaches are presented in the robustness analysis.

To ensure that such pre-existing trends do not drive the results, I include a robustness check controlling for differential pre-existing linear trends in Section 4.4.7.

[12] A formal IV approach would use the predicted WFH rate as the second-stage regressor, as in Mondragon and Wieland (2022), but would require an exclusion restriction that the WFH index affects female labor supply only through actual WFH takeup. In addition, the pseudo first stage is estimated on employed individuals only, since the ACS commuting variable is undefined for non-employed workers. This sample mismatch should be kept in mind when scaling the reduced-form effects for labor force participation; the employment and full-time results are less affected. Finally, the pseudo first-stage coefficients should also be interpreted with the caveat that the Dingel and Neiman (2020) WFH index captures the theoretical upper bound on WFH feasibility, while the ACS *TRANWORK* variable captures only fully remote work.

3.3.3 Additional Identification Concerns

Continuous treatment and linearity. The baseline specification uses a continuous WFH exposure index, which implicitly assumes that the treatment effect scales proportionally with treatment intensity. Figure A9 provides direct evidence supporting this assumption: the change in actual WFH rates between the pre-pandemic and post-pandemic periods increases approximately linearly with the WFH exposure measure across the full distribution. I also show that results are robust to alternative treatment constructions, including a median-split binary indicator and quintile bins, in Section 4.4.3. In addition, I estimate a triple-difference specification comparing women to men within the same MSAs, treating men as effectively never treated under the assumption that men's labor supply is not affected by the WFH opportunity measure. This provides a valid within-MSA control group that directly addresses concerns raised in the continuous DiD literature (Callaway et al., 2024).

Occupational composition. The WFH exposure index is built from pre-pandemic occupation shares. This is not because occupational composition is assumed stable across the pandemic. Rather, the post-pandemic structure is jointly determined with the labor market outcomes studied here, and the counterfactual structure absent the pandemic is unobservable, so the pre-pandemic structure is the only admissible basis for constructing exposure. The design requires only that pre-period exposure remain a strong predictor of post-period WFH opportunity across MSAs. Section 4.4.4 shows that this condition holds, with coefficients close to one when the post-period index is regressed on the pre-period index (Appendix Table B4). Results are also robust to an industry-based exposure index. Any measurement error from relying on the pre-period structure rather than the unobservable counterfactual could bias the estimates downwards, making the main results conservative.

Fully remote vs. hybrid work. The baseline WFH exposure index and the ACS measure of actual remote work both primarily capture fully remote work rather than hybrid arrangements. If hybrid work also affects mothers' labor supply, the baseline estimates may understate the full effect of remote work expansion. Conversely, if fully remote jobs are not tied to a specific location, workers may relocate away from their pre-pandemic MSA, weakening the link between local occupational composition and realized remote work. I assess sensitivity to this distinction using an alternative WFH index based on job postings that captures both fully remote and hybrid arrangements, and find qualitatively similar results in Section 4.4.5.

Other concerns. Several additional concerns are addressed in the robustness analysis. Individuals may relocate to exploit WFH opportunities, introducing selection bias; I address this using WFH measures constructed from prior-year residence. The choice of MSAs as the geographic unit may not fully capture local labor markets; I verify results using commuting

zones under both 1990 and 2020 definitions. I also examine sensitivity to weighted regressions, exclusion of older cohorts, alternative baseline periods, and alternative sampling criteria for the WFH index. These results are presented in Section 4.4.

4 Results

4.1 Main Results

Figure 4 presents the event study estimates of the effects of WFH opportunities on labor force participation for fathers (panel a), mothers (panel b), and college-educated mothers (panel c). Across all three panels, the pre-pandemic coefficients from 2012 to 2018 are small in magnitude and statistically indistinguishable from zero, supporting the parallel trends assumption underlying the DiD design. A clear divergence emerges after 2019. For fathers, the estimates remain close to zero throughout the post-pandemic period, showing no systematic relationship between WFH exposure and labor force participation. For mothers, the coefficients shift upward beginning in 2020 and become statistically significant by 2022, with the positive effects persisting through 2024. College-educated mothers exhibit the largest and most sustained shift. The absence of any response among fathers serves as a placebo check, alleviating concerns that the observed effects are driven by declining male participation in high-WFH areas rather than true increases in maternal labor force participation.

Table 3 reports the DiD estimates across six demographic groups, distinguishing between transition effects (2020–2021) and stable effects (2022–2024). Panel A presents results for labor force participation. For all males, the estimates are precisely estimated zeros in both periods. For all females, the stable effect is positive and marginally significant (0.08 percentage points), indicating a modest aggregate response. The effects concentrate sharply among mothers: a one-standard-deviation increase in MSA-level WFH exposure is associated with a 0.21 percentage point increase in labor force participation during the transition period and a 0.30 percentage point increase in the stable period, both significant at the 1 percent level. College-educated mothers show the largest response, with estimates of 0.31 and 0.36 percentage points for the transition and stable periods, respectively. Noncollege mothers exhibit a significant negative transition effect that attenuates and becomes insignificant in the stable period, consistent with pandemic-related disruptions disproportionately affecting occupations less amenable to remote work. The strengthening of effects from the transition to the stable period across maternal subgroups suggests that the results reflect sustained adoption of WFH arrangements rather than temporary pandemic responses.

Panels B and C of Table 3 extend the analysis to employment and full-time work.

Employment effects are not statistically significant for all females or all mothers, but college-educated mothers show positive and significant effects in both periods. For all males, fathers, and noncollege mothers, the stable effects on employment are negative and statistically significant.[13] The most pronounced results emerge for full-time work in Panel C. Mothers show significant effects of 0.44 percentage points during the transition period and 0.64 percentage points in the stable period, and college-educated mothers show effects of 0.65 and 0.83 percentage points, respectively. These full-time effects are approximately twice the magnitude of the corresponding labor force participation effects, indicating that WFH opportunities enable mothers not only to enter the labor force but also to increase work intensity. Event study estimates for employment and full-time work, presented in Appendix Figures A3 and A4, confirm these patterns.

Table 4 and Figure 5 examine parents with young children, defined as those with at least one own child under age five residing with the individual. The effects for mothers with young children are approximately two to three times larger than the corresponding estimates for all mothers. During the post-pandemic period, a one-standard-deviation increase in WFH exposure is associated with a 0.74 percentage point increase in labor force participation, a 0.42 percentage point increase in employment, and a 0.75 percentage point increase in full-time work, all significant at the 1 percent level. Fathers with young children show a modest positive effect on labor force participation in the stable period, but no positive effects on employment or full-time work. The event study estimates in Figure 5 display flat pre-trends and clear post-pandemic increases across all three outcomes for mothers with young children. The magnification of effects among mothers facing the most intensive caregiving demands is consistent with WFH reducing the work-family constraints that are most binding for this group.

4.2 Heterogeneous Effects

The effects of WFH opportunities may vary with household structure, life stage, and socioeconomic background. To examine these distributional effects, I estimate the baseline specification separately by marital status, age, and race. Figure 6 summarizes the results for labor force participation. The effects are generally larger for married women, younger age groups, and White and Asian women, and these patterns sharpen as the sample narrows from all females to mothers to college mothers. The corresponding results for employment and

[13]These negative effects likely reflect distinct forces. For noncollege mothers, limited access to remote-compatible occupations constrains their ability to benefit from WFH expansion. For males and fathers, pre-existing employment rates near the ceiling leave limited room for extensive margin responses; WFH changes how and where they work rather than whether they work. See Mechanism Section 5 for a detailed discussion.

full-time work, reported in Appendix Figures A6 and A7, show broadly consistent patterns.

Further, the larger effects among younger age groups reflect parental status more than age itself. Table B6 confirms that the first-stage relationship between WFH exposure and actual remote work takeup decreases monotonically with age among females, but Figure A8 shows that this age gradient in the reduced-form effects is driven specifically by parental status. Once the sample is split by mothers and non-mothers, only mothers in the two youngest age groups show positive and significant effects, while non-mothers show little response regardless of age.[14] This is consistent with younger mothers being more likely to have young children requiring active caregiving, making WFH particularly valuable for their labor force participation. The effect fades for older mothers, whose children are largely independent.

4.3 Interpreting the Effects

Table 5 provides a framework for gauging the magnitudes of the main estimates. Panel B reports the pseudo first stage for each group, and Panel C divides the reduced-form estimates (2022–2024) by the corresponding first-stage estimates to obtain a Wald estimator.[15] For college-educated mothers, a one-standard-deviation increase in MSA-level WFH exposure is associated with a 0.36 percentage point increase in labor force participation, a 0.19 percentage point increase in employment, and a 0.83 percentage point increase in full-time work in the post-pandemic period. These effects are larger and more robust for mothers with young children: the Wald estimates suggest that when local WFH opportunities lead mothers with young children to take up remote work, they are about 32 percent more likely to participate in the labor force, 19 percent more likely to be employed, and 36 percent more likely to work full-time.

To place these effects in a broader context, I compare them to the recent convergence in male-female labor force participation rates shown in Figure 1. Between 2019 and 2024, the gender gap in labor force participation narrowed by approximately 1.8 percentage points.[16] A two-standard-deviation increase in MSA-level WFH exposure, roughly the difference between a high and low WFH metropolitan area, raises mothers' labor force participation by 0.60

[14]The exception is college-educated non-mothers aged 25–34, who preserve a modest positive effect.

[15]As discussed in Section 3.3.1, the Wald estimator is best understood as an illustrative tool for contextualizing the reduced-form estimates rather than a structural causal estimate. A formal IV approach would use the predicted WFH takeup rate as the regressor in the second stage, which differs from the design employed here. In addition, the pseudo first stage is estimated on employed individuals only, since the ACS commuting variable is undefined for non-employed workers. This sample mismatch should be kept in mind when scaling the reduced-form effects for labor force participation; the employment and full-time results are less affected.

[16]In 2019, male and female labor force participation rates were 81.39 and 72.96 percent, respectively, yielding a gap of 8.43 percentage points. By 2024, the corresponding rates were 81.23 and 74.59 percent, narrowing the gap to 6.64 percentage points.

percentage points.[17] Since mothers comprise approximately 38 percent of the female sample, this translates to roughly a 0.23 percentage point increase in aggregate female labor force participation, accounting for about 13 percent of the observed convergence.[18] This two-standard-deviation increase in WFH also corresponds to approximately 136,000 additional mothers in the U.S. labor force during the stable post-pandemic period.[19] This calculation is necessarily approximate, but it suggests that the expansion of remote work opportunities represents a nontrivial contributor to both the narrowing of the gender participation gap and aggregate U.S. labor supply in the post-pandemic period.

4.4 Robustness Analysis

4.4.1 Robustness Analysis Using Time-Varying Controls

As discussed in Section 3.3.2, a key concern is that the pandemic may have changed how individual demographic and MSA characteristics predict female labor market outcomes, which could bias the estimates if those changes are correlated with WFH exposure. To address this concern, I augment the baseline specification by using time-varying controls following Altonji et al. (2005); Jack et al. (2023); Ross et al. (2024). This approach allows the effects of all control variables to vary flexibly across years, thereby accounting for potential shifts in how observable characteristics relate to labor market outcomes, particularly during the pandemic period. Specifically, I compare estimates from models that include standard controls, no controls, and year-varying controls.

Table B7 reports results across alternative sets of control variables. Labor force participation estimates are stable across all specifications. The employment effect for mothers becomes significant once MSA-level controls are included.[20] Full-time employment is somewhat sensitive to the inclusion of MSA median income interacted with year dummies, but is fully restored once median household income is excluded from the time-varying MSA controls (Panel F).[21] Overall, allowing controls to vary flexibly over time leads to only modest

[17] The stable effect for mothers' LFP is 0.30 percentage points per one-standard-deviation increase. A two-standard-deviation increase yields $2 \times 0.30 = 0.60$ percentage points. The two-standard-deviation spread captures the approximate distance between a low-WFH MSA (one SD below the mean) and a high-WFH MSA (one SD above the mean).

[18] The calculation is $0.60 \times 0.38 \approx 0.23$ percentage points attributed to WFH expansion. Dividing by the 1.8 percentage point narrowing of the gender gap yields $0.23/1.8 \approx 0.13$, or 13 percent.

[19] The weighted population of mothers aged 25–64 in the United States is approximately 22.7 million in 2024 based on ACS person weights. Multiplying by the estimated 0.60 percentage point increase in mothers' labor force participation yields $22{,}700{,}000 \times 0.006 \approx 136{,}000$ additional mothers in the labor force.

[20] This likely reflects that non-college mothers' employment is more sensitive to local labor demand conditions than that of college-educated mothers, whose employment decisions are driven more by individual-level factors.

[21] Full-time status is a margin of adjustment that is more sensitive to cyclical fluctuations in local economic

attenuation in effect sizes, indicating that a small portion of the estimated effects may reflect pandemic-related shifts, but the core finding that WFH opportunities improve mothers' labor market outcomes remains robust and economically meaningful.

4.4.2 Robustness to COVID-19 Severity and Childcare Disruption

A further concern is that pandemic-era disruptions correlated with WFH exposure may independently affect female labor market outcomes. MSAs with lower remote work potential may have experienced more severe COVID-19 outbreaks and, consequently, greater school and childcare closures during the pandemic, suppressing female labor force participation through channels unrelated to WFH access. To assess whether such confounders drive my estimates, I augment the baseline specification with two MSA-level pandemic controls. The first is a local COVID-19 severity measure, constructed as cumulative deaths per 100,000 residents through the end of 2021 from county-level data provided by The New York Times (2021), aggregated to the MSA level following Hansen et al. (2024). The second is a school closure intensity measure, constructed as the average share of schools experiencing at least a 50 percent year-over-year decline in in-person visits during 2020–2021, from the U.S. School Closure and Distance Learning Database of Parolin and Lee (2021).[22] I estimate three specifications: controlling for COVID severity alone, school closure alone, and both simultaneously.[23]

Table B8 Panels B–D report results across these three specifications. The coefficients on the WFH exposure index remain positive and statistically significant throughout, with magnitudes closely tracking the baseline estimates. These results indicate that neither COVID-19 severity nor school and childcare closure intensity across MSAs accounts for the estimated effects of WFH opportunities on mothers' labor market outcomes.

conditions than the extensive margins of labor force participation and employment. High-income MSAs tend to have tighter labor markets, stronger professional norms, and a higher concentration of industries where full-time work is the default. Moreover, high-income MSAs had distinctly different business cycle dynamics post-pandemic, recovering faster and experiencing stronger wage growth, which causes the MSA median income interacted with year dummies to absorb part of the identifying variation.

[22]School closure serves as the primary observable proxy for childcare disruption because the public K–12 system functions as the principal source of subsidized daytime childcare for school-age children. Parolin and Lee (2021) report a correlation of 0.9 between school closure rates and childcare center closure rates across states, confirming that the two measures capture the same underlying disruption. This approach follows Hansen et al. (2024), who characterize K–12 school availability as a direct shock to mothers' childcare costs.

[23]The school closure database covers January 2020 through December 2021, and the NYT COVID data covers January 2020 through March 2023. To maintain consistency across the two measures, both are constructed as cross-sectional MSA-level averages over 2020–2021 and interacted with a year-2021 indicator, the only post-period year during which these disruptions were meaningfully present and both data sources are available.

4.4.3 Robustness to Alternative Treatment Constructions

The baseline specification uses a continuous WFH exposure index, which implicitly assumes that the effect of WFH potential on labor market outcomes scales proportionally with treatment intensity. Figure A9 provides supporting evidence for this assumption by plotting the change in actual WFH rates between the pre-pandemic and post-pandemic periods against the WFH exposure measure. The relationship is approximately linear across the full range of the WFH distribution for both the pooled sample and gender-specific subsamples.

To further assess robustness, I replace the continuous WFH index with alternative treatment constructions. Table B10 compares estimates from the continuous specification against a median-split binary indicator, with the binary coefficients scaled to be comparable in magnitude to the continuous estimates.[24] The continuous and binary estimates are closely aligned across subgroups and post-period years. Figure A10 plots the estimated LFP effects across quintiles of WFH exposure, showing effects that increase broadly monotonically for both mothers and college-educated mothers. Table 2 Panels B and C confirm that the WFH measure retains strong predictive power under both the median-split and quintile constructions, with high F-statistics across all demographic groups. These results indicate that the baseline findings are not sensitive to the functional form assumption imposed by the continuous treatment specification.

4.4.4 Robustness to Post-Pandemic Occupational Change

As discussed in Section 3.3.3, the WFH exposure index is built from pre-pandemic occupation shares because the post-pandemic occupation structure is jointly determined with the outcomes studied and the counterfactual structure absent the pandemic is unobservable. The design requires only that pre-period exposure remain a strong predictor of post-period WFH opportunity across MSAs. To assess this condition directly, I construct an analogous WFH index from observed post-period occupation shares using the same procedure and regress it on the pre-period index across the 242 MSAs. Table B4 reports the results for two post-period windows and for indices based on all employed workers, employed men, and employed women. The coefficients are close to one across all specifications, and the pre-period index explains the large majority of the cross-MSA variation in post-period exposure. Moreover, the compositional shifts are modest in magnitude, systematically signed toward teleworkable occupations (Figure A13), and symmetric across high-WFH and low-WFH MSAs, with no

[24] Binary model coefficients are scaled by the difference in mean WFH exposure between above- and below-median MSAs (1.526). Standard errors are scaled accordingly.

evidence that WFH-suitable occupation groups grew differentially in high-exposure areas (Figures A14 and A15).

Beyond preserving the design, these shifts are themselves informative. Figures A11 and A12 trace the mean teleworkability of jobs held by employed workers over time. The measure rises modestly after 2020 and persists through 2024, with broadly similar shifts for women and men and the largest increases among mothers with young children. While descriptive, this is the pattern one would expect if women were responding to expanded remote opportunities in their labor supply decisions; its absence would have worked against a causal interpretation of the main results.[25]

As an additional robustness check, I construct an alternative WFH exposure index using 2-digit NAICS industry shares rather than occupation shares, since large employers and industries that dominate an MSA are unlikely to change substantially over the short period studied.[26] Table B8 Panel E reports the results. The estimated effects on labor force participation and full-time work remain positive, statistically significant, and close in magnitude to the baseline estimates for both mothers and college-educated mothers. The employment effect is slightly attenuated, likely reflecting that the industry-level index is a coarser measure of individual WFH potential.

4.4.5 Robustness Analysis Using Alternative WFH Measures

The baseline analysis primarily captures fully remote work rather than hybrid work arrangements.[27] However, hybrid work arrangements may also be important in this context for two reasons. First, the labor supply implications may differ between fully remote and hybrid work. Hybrid arrangements can still provide flexibility and may make labor market participation easier for mothers with young children, but likely less so than fully remote work that eliminates the need to commute entirely. Second, if fully remote jobs are not tied to a specific location, the occupational structure of an MSA measured before the pandemic

[25]The subgroup gradient is more muted in these figures than in the labor force participation estimates. This is expected: the figures condition on employment, so women who enter the labor force in response to remote opportunities appear only partially in the composition of held jobs, while the participation estimates count both the employed and those searching for work.

[26]For each industry, I use the employment-weighted industry-level teleworkability score from Dingel and Neiman (2020), which aggregates their occupation-level classification using BLS employment data. The MSA-level industry index is then the employment-weighted average of these industry scores, computed from pre-pandemic ACS data and held fixed thereafter.

[27]Specifically, the WFH exposure index is constructed using the teleworkability measure of Dingel and Neiman (2020), which identifies whether an occupation can plausibly be performed entirely from home based on job characteristics. In addition, the empirical proxy for actual remote work is based on the ACS variable *TRANWORK*, which classifies an individual as working remotely only if their primary means of transportation to work is reported as "Worked from home". This variable is widely regarded as capturing fully remote work (e.g., Bick et al. (2025); Buckman et al. (2025); Kmetz et al. (2023); Mondragon and Wieland (2022)).

may become a weaker predictor of actual remote work opportunities after the pandemic. In such cases, workers may live in locations different from where their employers are located, weakening the link between local occupational composition and realized remote work in that MSA.

Although Kmetz et al. (2023) show that major WFH measures are highly correlated and yield similar causal effects,[28] I examine whether the results are sensitive to measures that incorporate hybrid work. Specifically, I construct an alternative WFH exposure index using the occupational WFH score developed by Hansen et al. (2023), which is based on job postings and captures the prevalence of both fully remote and hybrid work arrangements. As shown in Table B9, the results are qualitatively similar, with slightly smaller magnitudes under the Hansen et al. (2023) measure. This is expected: the Dingel and Neiman (2020) baseline measure represents a theoretical upper bound on WFH feasibility, while the Hansen et al. (2023) job-posting-based measure filters out the gap between theoretical feasibility and actual employer behavior.[29]

In addition, as discussed in Section 3.1, the baseline index uses gender-specific occupation shares among the employed. Results are robust to alternative indices based on different population definitions (Table B3 and Figure A17).

4.4.6 Robustness to Alternative Geographic Units

As discussed in the Data Section 2.1, the choice of MSAs as the geographic unit is largely because they provide the cleanest and most consistent geographic definition across the full sample period in the ACS data. However, commuting zones (CZs) may better capture the scope of local labor markets. Implementing a CZ-based analysis with the ACS introduces practical challenges, most notably that the Census Bureau redraws PUMA boundaries every ten years, with the 2010 definitions used for 2012–2021 and the 2020 definitions used for 2022–2024. This boundary change coincides with the post-treatment period and inevitably introduces geographic inconsistencies and measurement error. Nonetheless, I take several steps to mitigate these measurement concerns and compare estimates under all three definitions: MSA, CZ1990, and CZ2020.[30] The results are qualitatively consistent across all three

[28] Kmetz et al. (2023) compare several major WFH measures in the cross-section of the United States, including the Survey of Working Arrangements and Attitudes (Barrero et al., 2021), the Real-Time Population Survey (Bick et al., 2023), the Dingel and Neiman (2020) teleworkability measure implemented in the ACS, and the ACS itself.

[29] Note that this refers to the WFH exposure index used as the independent variable, which is constructed from pre-pandemic occupational composition. This is distinct from the ACS *TRANWORK* variable used as the empirical proxy for actual remote work, which captures only fully remote workers and represents a lower bound on realized WFH behavior.

[30] I construct PUMA-to-CZ crosswalks for both vintages. For the 1990 CZ definition, I follow the methodology of Dorn (2009) and Autor et al. (2019). For the 2020 CZ definition, I use the updated delineation

specifications, with MSA estimates slightly larger in magnitude and CZ estimates somewhat more precise due to larger sample sizes. These results are reported in Table B11.

4.4.7 Robustness to Pre-existing Linear Trends

Harrington and Kahn (2025) argue that pre-existing trends could reflect increased access to WFH before the pandemic. If high-WFH MSAs were already on a rising female LFP trajectory as a result, post-pandemic estimates could capture this pre-existing trajectory rather than a causal effect of the pandemic-driven WFH expansion. The time-varying individual and MSA controls in Section 4.4.1 partially address this concern. To further probe robustness, I augment the baseline specification by controlling for differential pre-pandemic linear trends by MSA.[31] This allows high- and low-suitability MSAs to follow differential linear trajectories before the pandemic without absorbing any post-pandemic variation. Table B8 Panel A confirms that the results are consistent with the baseline findings, with coefficients attenuating slightly for college-educated mothers as the pre-trend control absorbs the modest upward drift visible in the event study figures.

4.4.8 Additional Robustness Checks

2008 financial crisis as falsification. To assess whether the estimated effects are specific to the COVID-19 pandemic and its expansion of WFH opportunities rather than general responses to macroeconomic shocks, I replicate the event study design using the 2008 financial crisis as a placebo event, with ACS data from 2005–2011. Figure A16 shows the results for college-educated mothers' labor force participation. The coefficients are small, statistically insignificant, and show no systematic pattern around the 2008 recession. This indicates that the estimated effects emerge only when a shock expands remote work opportunities, not in response to a general economic downturn.

Prior-year residence. Individual mobility could introduce bias if workers relocate to areas with greater WFH opportunities, or if fully remote workers are not geographically tied to their employer's location. To address this, I construct WFH measures based on individuals' prior-year locations rather than current residence, so that the WFH exposure reflects labor market conditions before any relocation decisions. Table B8 Panel F reports the results, which are consistent with the baseline estimates across all outcomes and subgroups.

Excluding older cohorts. Workers aged 55–64 may respond to WFH opportunities through retirement timing rather than labor force attachment. I restrict the sample to ages

released by the Economic Research Service (ERS). Details are provided in Appendix E.

[31] Specifically, I add the interaction $\text{WFH}_m \times t_{\text{pre}}$, where $t_{\text{pre}} = (t - 2019) \cdot \mathbf{1}[t \leq 2019]$ and equals zero in the post-pandemic period.

25–54 to address this concern. Table B8 Panel G reports the results, which remain stable and, if anything, slightly larger in magnitude.

Shorter baseline period. The baseline WFH index averages occupation shares over 2012–2019, which may encompass longer-run occupational shifts less relevant to the immediate pre-pandemic period. I construct an alternative measure using only 2017–2019. Table B8 Panel H confirms that the results are virtually unchanged.

Weighted regressions. The baseline specification is not weighted by ACS person weights because the unweighted estimates yield more precise results in this setting, consistent with Solon et al. (2015).[32] Table B12 reports the weighted estimates, which have slightly smaller magnitudes and larger standard errors. However, the general pattern is unchanged, confirming that the main findings are not sensitive to the choice of weighting scheme.

5 Mechanisms

5.1 Who Responds? The Role of Children and Caregiving

WFH opportunities ease labor market constraints by reducing the need for physical presence, so the response should be largest among those for whom presence is hardest to maintain. Mothers are the natural candidate: they bear a disproportionate share of childcare responsibilities relative to fathers (Baker et al., 2008; Bertrand et al., 2010; Goldin, 2014; Kleven et al., 2019), which makes caregiving the most plausible binding constraint on their labor supply. The main results support this interpretation. Table 3 shows that the effect is near zero for all women and for fathers, but large and significant for mothers. The response strengthens when the constraint is tightest: Table 4 shows that mothers of children under age five exhibit an effect roughly double that for all mothers, while fathers of young children show no corresponding response. Table 6 tells the same story from within the household, where the wife coefficient is near zero for childless couples, positive and significant for mothers, and largest for mothers of young children.

If WFH eases the labor supply constraint by substituting for paid childcare, its effect on mothers should be largest where childcare is most costly relative to what mothers earn. To test this, I construct an MSA-level measure of childcare cost burden as the ratio of the annual price of young-child center-based care to female median earnings, and split MSAs at the median.[33] Columns 1 and 2 of Table 7 report the result. The stable-period effect is

[32]The loss of precision under weighting likely reflects heteroskedasticity induced by the weighting scheme and the use of experimental weights in the 2020 ACS to address pandemic-era disruptions in data collection.

[33]Childcare prices are from the 2022 National Database of Childcare Prices. I use the median weekly price of full-time center-based care, annualized by multiplying by 52. I average across the three pre-school-

larger in high-burden MSAs than in low-burden MSAs, for mothers and for college-educated mothers alike. Together, these results indicate that WFH operates by relaxing a caregiving constraint, and that the response is largest where the work-family tradeoff is most acute.

5.2 Whose Opportunity Matters? Own and Spousal Channels

The previous results show that WFH raises mothers' participation by easing the caregiving constraint. That constraint can be relaxed through two channels within the household: a mother's own WFH access, which gives her more flexibility to balance work and caregiving, or her husband's WFH access, which can shift caregiving time toward him and loosen the constraint the mother faces. Filippo et al. (2026) find that increases in men's WFH exposure improve wives' labor market outcomes through intra-household reallocation of childcare time, while Zarate (2025) finds that mothers whose partners can work from home are significantly more likely to remain employed after childbirth across Latin American countries. To separate the own-opportunity channel from the spousal spillover channel, I augment the baseline design to explicitly account for the husband's WFH treatment. I link each married woman to her co-resident husband, restrict to opposite-sex couples, and re-estimate equation (2) on this spouse-linked sample. Following Filippo et al. (2026) in spirit, I then add a husband channel by interacting a binary indicator for whether the husband's occupation is above the median teleworkability score with the post-pandemic dummy, and absorb time-invariant differences across husband occupations through husband occupation fixed effects.[34]

Table 6 confirms that the WFH effects reflect both own WFH opportunity and intra-household spillovers, with the former serving as the primary channel. The wife coefficient remains positive and significant overall after adding the husband channel, attenuating by roughly 25 to 30 percent, indicating that own WFH opportunity is the primary driver of the baseline effect. In addition, both the wife and husband coefficients grow with caregiving

age categories, infants (0–23 months), toddlers (24–35 months), and preschoolers (36–54 months), which correspond to children under five. Prices are aggregated from counties to MSAs and weighted by the number of households with children under six, the youngest age cut available in the database. Arkansas, Indiana, Missouri, New Mexico, and Pennsylvania lack 2022 price data and are excluded, leaving 217 of the 242 sample MSAs and about 95.5 percent of mother-year observations.

[34]The husband treatment indicator is constructed by aggregating Dingel and Neiman (2020) teleworkability scores to the 3-digit SOC level using employment weights, and classifying occupations as high-WFH if their score is at or above the median. Formally, the decomposed specification is:

$$\begin{aligned} Y_{imt} = {} & \delta \cdot (\text{WFH}_{m,\text{pre}} \times \text{Post}_t) \\ & + \theta \cdot (\mathbf{1}\{o(H) \in \text{High-WFH Occ.}\} \times \text{Post}_t) \\ & + \mathbf{X}'_{it}\beta + \alpha_m + \lambda_t + \phi_{o(H)} + \varepsilon_{imt} \end{aligned}$$

where $\phi_{o(H)}$ are husband occupation fixed effects at the 3-digit SOC level, which absorb time-invariant differences across husband occupation types.

demands: near zero and insignificant for couples with no children, large and significant once children are present, and largest for those with young children, in both panels. This gradient is consistent with the caregiving mechanism, since husbands' WFH opportunities ease time constraints most when childcare demands are highest.

The results also help explain why fathers, despite having the highest WFH takeup in the pseudo first stage, show no corresponding labor force participation effect. With pre-existing participation rates near the ceiling, fathers face limited room for any extensive-margin response. Instead, their WFH does not change whether they work, but changes how and where they work, freeing up caregiving time for wives. These findings complement the broader literature on intra-household spillovers by showing that when women's own WFH access is the source of variation, the own-opportunity channel dominates while the spousal spillover remains a secondary but meaningful force.

Establishing that women's own WFH access is the primary channel raises a further question: through which margin of labor supply does it operate? A rise in participation can reflect either increased hiring of previously non-employed mothers or reduced separations among those already attached. Columns 3 and 4 of Table 7 separate the two by restricting the sample on prior-year work history. The entry margin, estimated among mothers with weak or no employment in the prior year, shows a positive and significant stable-period effect for both mothers and college-educated mothers. The retention margin, estimated among mothers who worked continuously in the prior year, shows no corresponding effect; with labor force attachment in this group already near universal, there is little room for WFH to raise it further. The participation response therefore operates mainly through entry: WFH expands labor force participation by drawing previously unattached mothers into work.

5.3 Who Has Access? The Role of Occupational Concentration

The preceding analysis shows that WFH raises participation mainly through women's own opportunity and operates by drawing previously unattached mothers into the labor force. Yet access to that opportunity is uneven: college mothers show the largest gains, noncollege mothers show near-zero or negative effects, and fathers show no labor supply response despite high WFH takeup. One explanation is differential access to teleworkable occupations. If the post-pandemic WFH expansion primarily activated latent remote work capacity in occupations that were already teleworkable, then groups concentrated in those occupations before the pandemic were better positioned to benefit. I examine this precondition by comparing the occupation distributions across twelve occupation groups sorted by their

average teleworkability score.[35]

Figure 7, Panel (a) compares the occupation distributions of mothers and fathers. In the pre-period, mothers are disproportionately concentrated in high-teleworkability occupations, with large shares in Education and Arts, Management and Business, and Office and Administrative Support, alongside lower-teleworkability groups such as Service. Fathers, by contrast, are spread more broadly across the teleworkability spectrum: they hold substantial shares in high-teleworkability groups such as Management and Business and Computer and Science, but also in occupations with near-zero teleworkability such as Construction, Transportation, Production, and Installation and Repair.

The post-period distributions show that this pattern persists with no substantial change in the overall composition. Mothers' concentration in more teleworkable occupations meant that the pandemic activated opportunities that were previously latent, turning theoretical teleworkability into actual flexible work arrangements. For fathers, the lack of an obvious labor supply response reflects three reinforcing forces: the larger share of employment in non-teleworkable occupations limits the aggregate reach of WFH opportunities; pre-pandemic labor force participation rates already near the ceiling leave minimal room for an extensive-margin response; and as shown in the intra-household analysis, fathers' WFH translates into spousal spillovers that benefit wives' labor supply rather than their own.

The comparison between college-educated and noncollege mothers in Panel (b) uncovers the source of this within-mother variation. The previous panel shows that mothers as a group are spread across both high- and low-teleworkability occupations; Panel (b) reveals that this spread maps sharply onto education. In the pre-period, college mothers are overwhelmingly concentrated in the highest-teleworkability groups, with the bulk of their employment in professional, managerial, and technical occupations such as Education and Arts, Management and Business, Computer and Science, and Office and Administrative Support, consistent with the broader literature on gendered occupational sorting (Flabbi and Moro, 2012; Harrington and Kahn, 2023; Hu et al., 2024). Noncollege mothers, by contrast, are predominantly employed in Service occupations, which have a teleworkability score near zero and require physical presence that makes remote work infeasible. The post-period distributions confirm that this separation persisted. This occupational divide creates a structural barrier: when the pandemic activated remote work in teleworkable occupations, college mothers were positioned

[35]Occupation groups follow the 2018 SOC intermediate aggregation (Table 5, BLS SOC 2018 User Guide). The twelve groups are: Management, Business, and Financial; Computer, Engineering, and Science; Education, Legal, Community Service, Arts, and Media; Healthcare Practitioners and Technical; Service; Sales and Related; Office and Administrative Support; Farming, Fishing, and Forestry; Construction and Extraction; Installation, Maintenance, and Repair; Production; and Transportation and Material Moving. Military occupations are excluded from the sample.

to benefit; noncollege mothers, concentrated in occupations where remote work is infeasible, could not take advantage of the expanded opportunities.

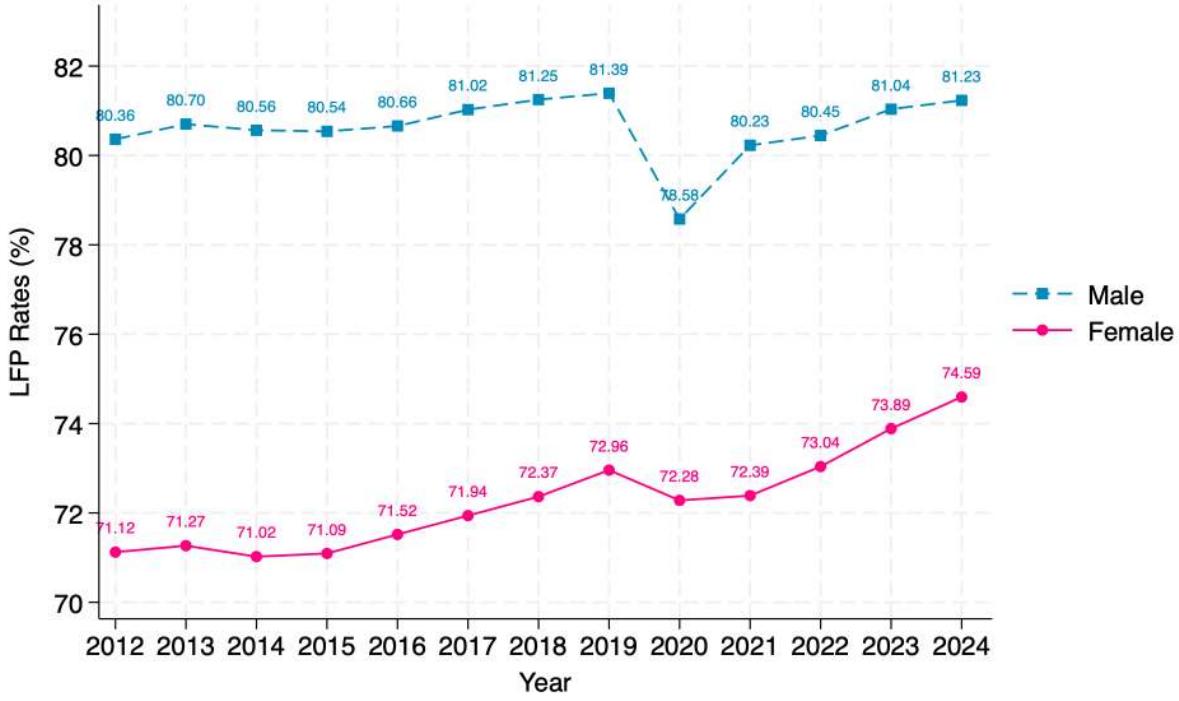
6 Conclusion

This paper analyzes how the expansion of remote work accelerated by COVID-19 affected women's labor market outcomes, exploiting variation in Metropolitan Statistical Areas' baseline occupational suitability for remote work. The results show that, relative to men, women experienced more favorable shifts in labor force participation, employment, and full-time work. The effects are concentrated among mothers, particularly those with college degrees and young children. More broadly, WFH opportunities account for approximately 13 percent of the observed convergence in the male–female labor force participation gap, corresponding to roughly 136,000 additional mothers in the U.S. labor force during the stable post-pandemic period. The mechanisms analysis further demonstrates that mothers' own WFH opportunity is the primary channel driving these effects, while spousal spillovers play a secondary but meaningful role. College mothers gained the most because their concentration in high-teleworkability occupations gave them greater access to the expanded opportunities.

More generally, this paper provides new empirical evidence on the transformative potential of flexible work arrangements in promoting gender equity in the labor market. The heterogeneous effects suggest that while WFH arrangements can serve as a powerful tool for reducing the labor supply costs of motherhood among advantaged women, they may inadvertently widen existing inequalities if access remains concentrated in higher-skilled occupations. In addition, the research design and findings may be useful for examining similar questions in other contexts. For example, the same approach could study the effects of other flexible work policies, such as flexible scheduling, compressed workweeks, or job-sharing arrangements, across demographic groups. It could also examine how WFH opportunities affect outcomes such as fertility decisions, childcare arrangements, wage growth, or career advancement. The framework may further extend to other technology shocks with uneven occupational exposure, most notably the adoption of artificial intelligence. Because AI exposure is concentrated in many of the same teleworkable occupations that expanded under remote work, studying its effects will require accounting for how the two shocks interact. Looking ahead, remote work is becoming a permanent feature of the post-pandemic labor market. The persistent effects observed in this paper suggest that its role in shaping gender dynamics will therefore continue to evolve, warranting ongoing investigation of both its benefits and its potential unintended consequences across different segments of the workforce.

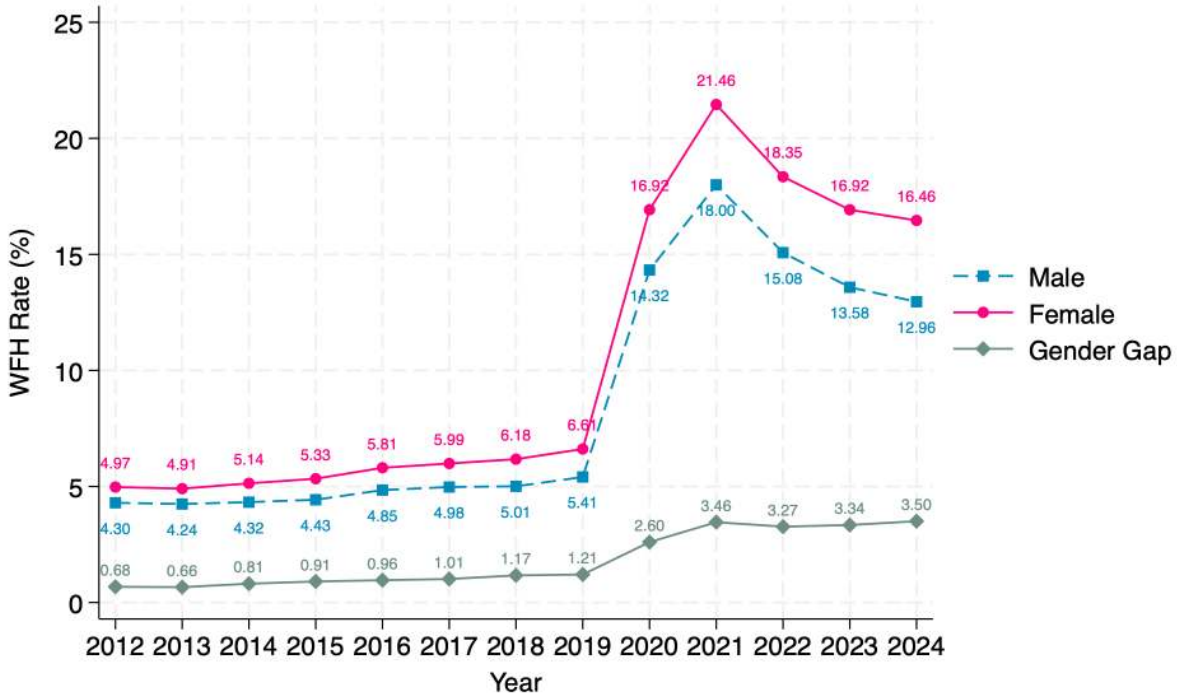
7 Figures and Tables

Figure 1: Labor Force Participation Rates by Sex



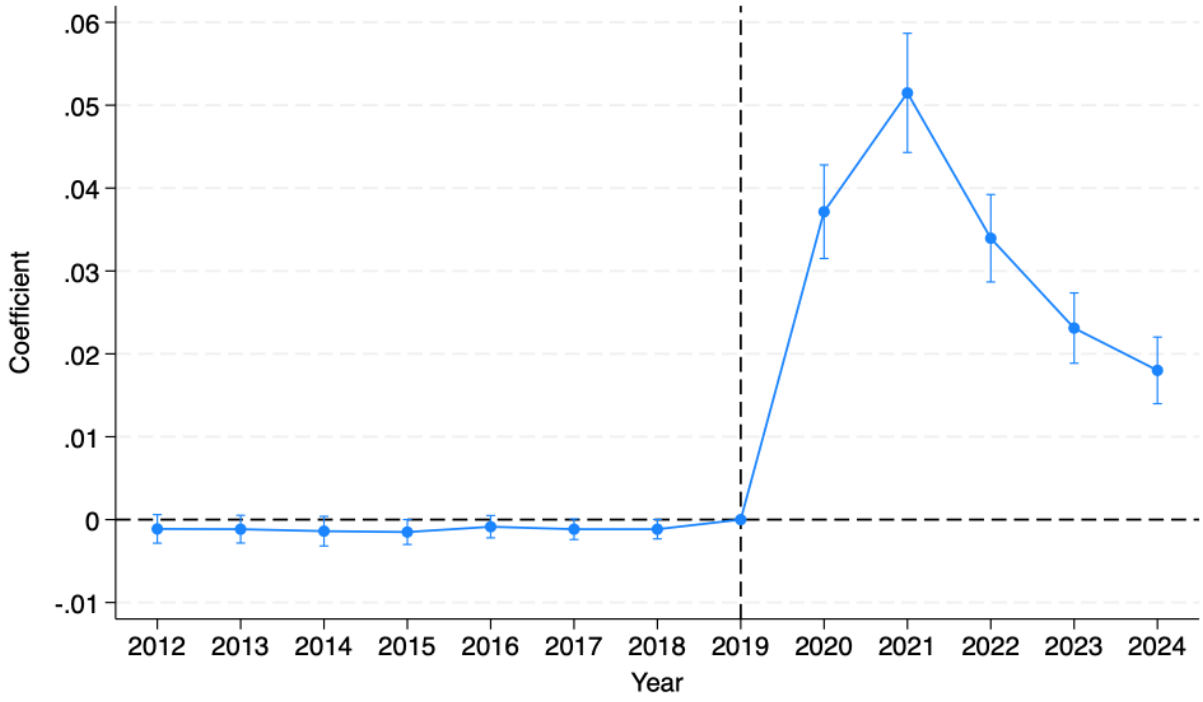
Notes: This figure presents the labor force participation rates from 2012 to 2024, separately for males and females. The data source is the American Community Survey (ACS) and reflects the working-age population (ages 25 to 64) in the United States. The values are expressed as percentages and are not seasonally adjusted.

Figure 2: Actual Work-From-Home Rates by Sex



Notes: This figure reports the share of individuals working fully remotely from 2012 to 2024, separately for males and females. Actual WFH behavior is measured using the ACS variable *TRANWORK*, which records the respondent's primary means of transportation to work. An individual is classified as working from home if the reported mode is "Worked from home." This variable is widely understood to capture fully remote work rather than hybrid arrangements. The sample includes individuals ages 25–64 in the labor force in the United States.

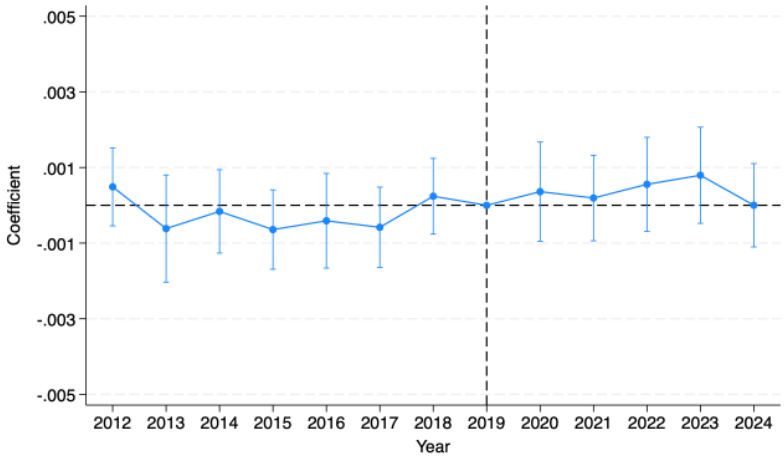
Figure 3: Validity of the WFH Measure and COVID-19 Activation



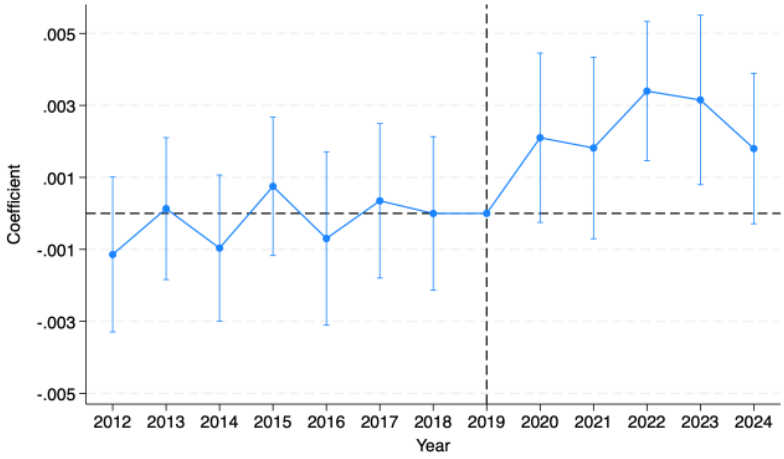
Notes: This figure presents an event study analysis examining the relationship between WFH measures and actual work-from-home behavior for females before, during, and after the COVID-19 pandemic. The specification follows the same event study design as in Equation (3), with actual WFH behavior indicator as the dependent variable. The actual WFH behavior leverages the transportation-to-work variable of the ACS, which surveys each employed individual about their primary mode of commuting, including an option for working from home.

Figure 4: Effects of WFH Opportunities on LFP

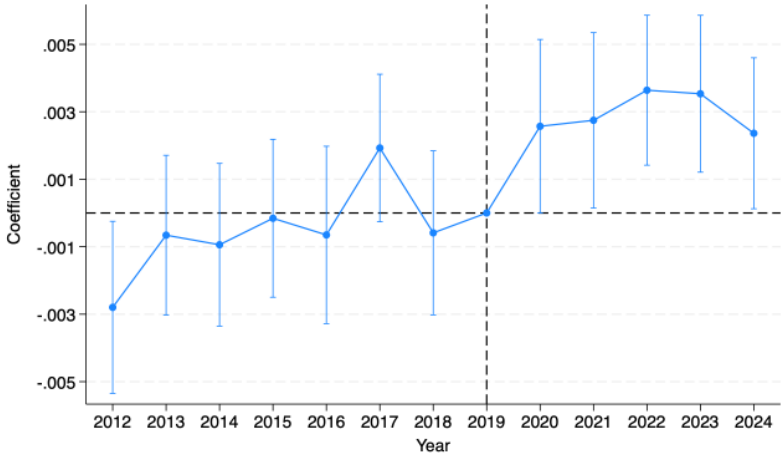
(a) Fathers



(b) Mothers



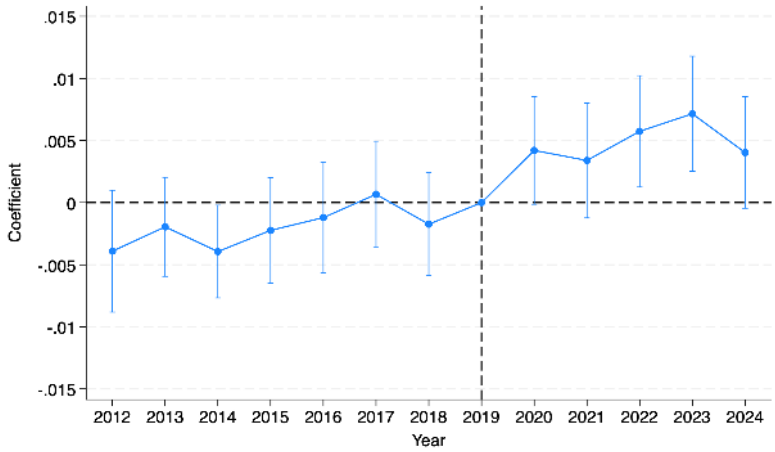
(c) College Mothers



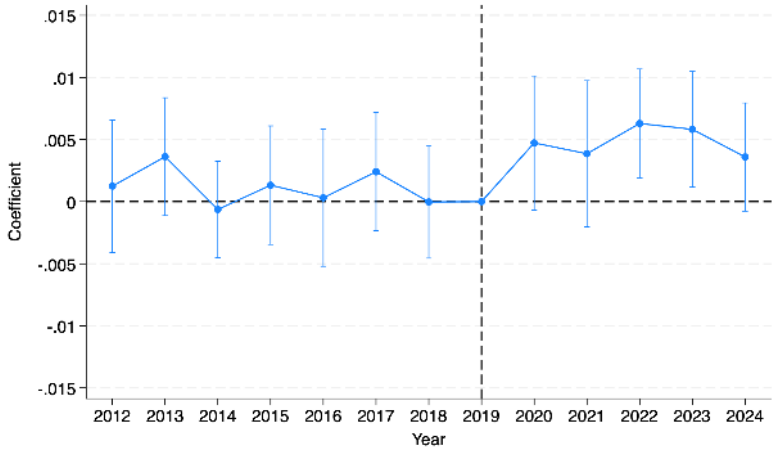
Notes: This figure displays the effects of WFH opportunities on labor force participation for (a) fathers, (b) mothers, and (c) college-educated mothers. Mothers (fathers) are defined as women (men) with at least one own child under age 18 residing with the individual. Each panel plots the coefficients and corresponding 95% confidence intervals from Equation (3). The year 2019 is used as the reference point and normalized to zero. The WFH measure is standardized to have a mean of 0 and a standard deviation of 1. Standard errors are clustered at the MSA level.

Figure 5: Effects of WFH on Mothers with Young Children

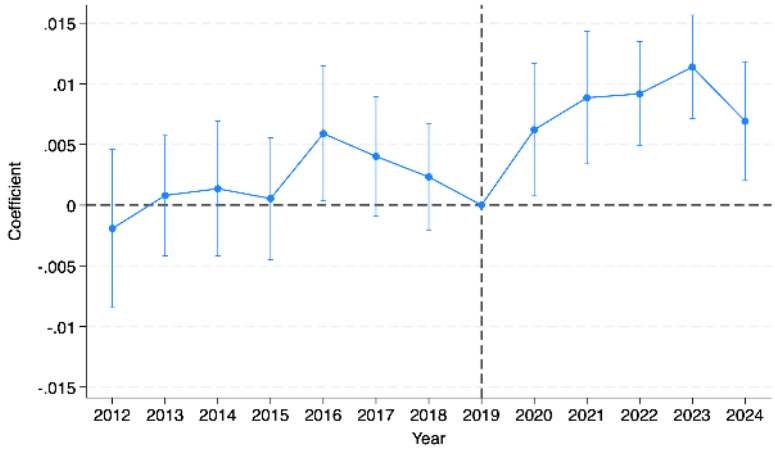
(a) Labor Force Participation



(b) Employment

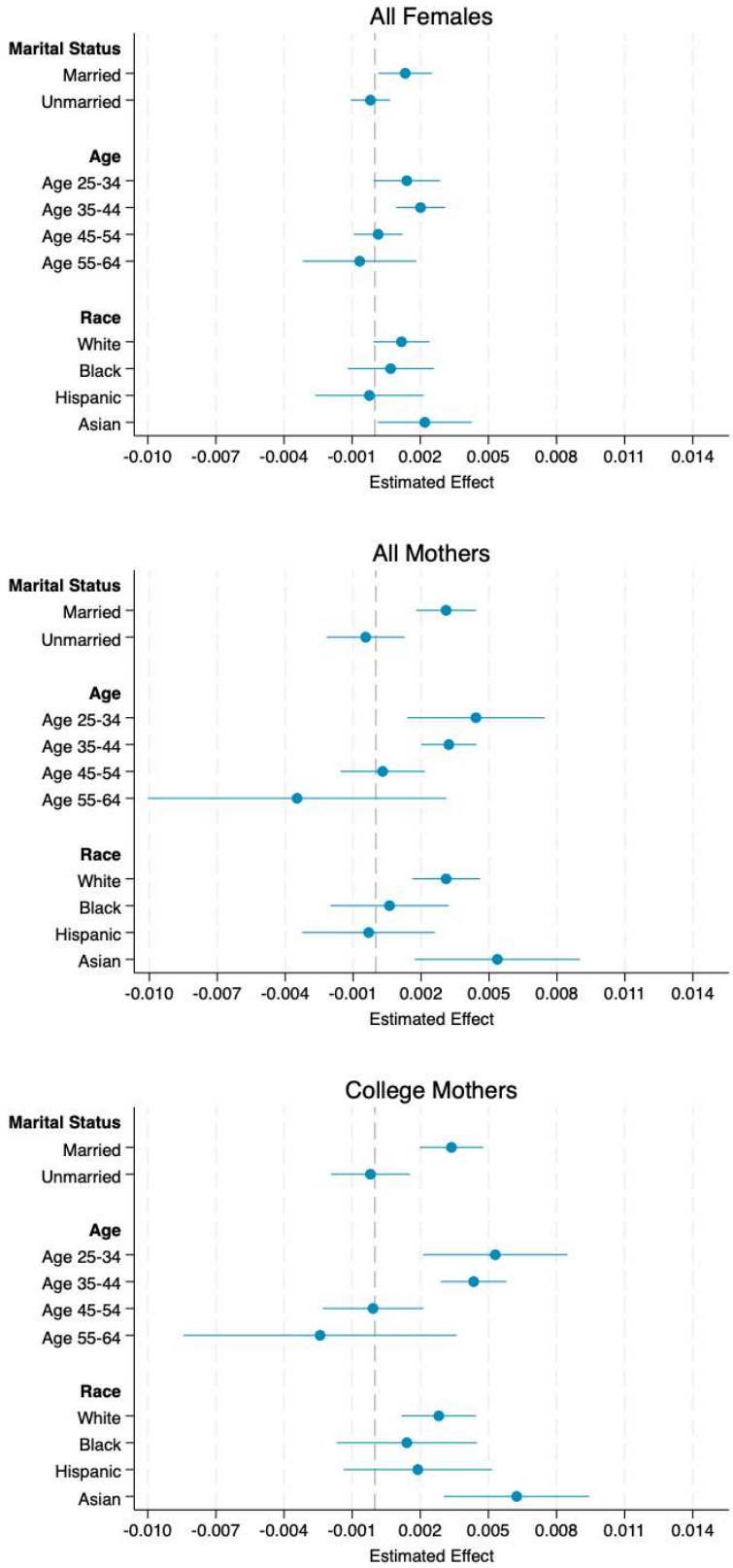


(c) Full-Time Work



Notes: This figure displays the effects of WFH opportunities on (a) labor force participation, (b) employment, and (c) full-time work for mothers with young children. Mothers with young children are defined as women with at least one own child aged 4 and under residing with the individual. Each panel plots the coefficients and corresponding 95% confidence intervals from Equation (3). The year 2019 is used as the reference point and normalized to zero. The WFH measure is standardized to have a mean of 0 and a standard deviation of 1. Standard errors are clustered at the MSA level.

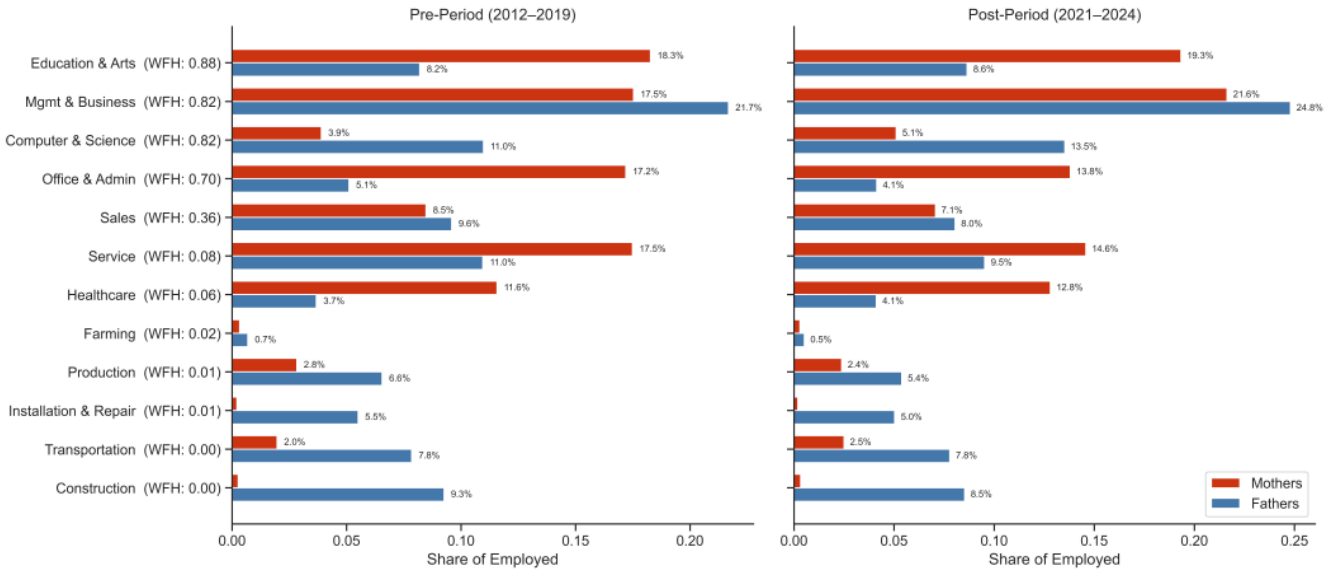
Figure 6: Heterogeneous Effects of WFH Opportunities on LFP



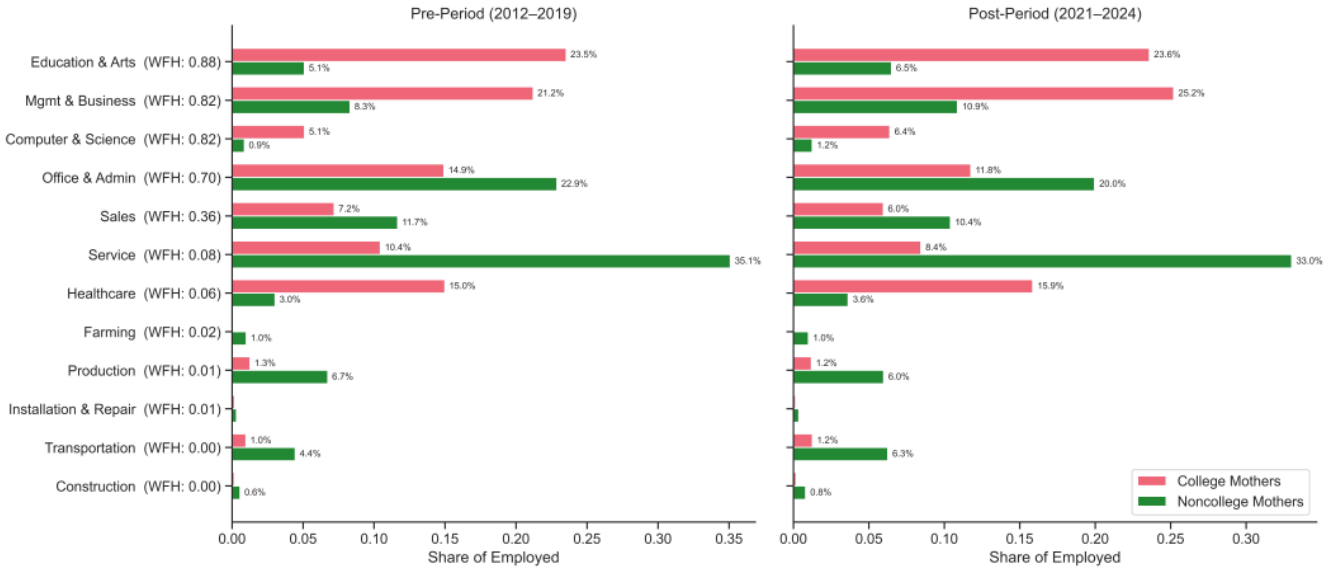
Notes: This figure plots the heterogeneous effects of WFH opportunities on labor force participation by demographic groups. Each panel reports estimated coefficients and 95% confidence intervals from equation (2). Standard errors are clustered at the MSA level.

Figure 7: Occupation Distribution by Demographic Group

(a) Mothers vs. Fathers



(b) College Mothers vs. Noncollege Mothers



Notes: This figure plots the share of employed workers in each of twelve intermediate occupation groups, following the 2018 SOC intermediate aggregation (Table 5, BLS SOC 2018 User Guide). Occupation groups are sorted by their average Dingel and Neiman (2020) teleworkability score, shown in parentheses on the vertical axis. Shares are computed from ACS microdata over the pre-period (2012–2019) and post-period (2021–2024), excluding the transition year of 2020. The twelve groups are: Management, Business, and Financial; Computer, Engineering, and Science; Education, Legal, Community Service, Arts, and Media; Healthcare Practitioners and Technical; Service; Sales and Related; Office and Administrative Support; Farming, Fishing, and Forestry; Construction and Extraction; Installation, Maintenance, and Repair; Production; and Transportation and Material Moving. Military occupations are excluded from the sample.

Table 1: Summary Statistics

	Female		Male	
Variable	Mean	SD	Mean	SD
WFH Measure	0.50	0.04	0.36	0.06
Labor Force Participation	0.87	0.33	0.93	0.26
Employment	0.84	0.37	0.89	0.31
Full-Time Work	0.77	0.42	0.90	0.31
White	0.61	0.49	0.63	0.48
Black	0.11	0.31	0.08	0.28
Hispanic	0.16	0.37	0.17	0.38
Asian	0.08	0.28	0.08	0.27
Married	0.62	0.49	0.64	0.48
Parent	0.38	0.48	0.36	0.48
Parent with Young Children	0.13	0.33	0.13	0.34
College Education	0.69	0.46	0.62	0.49
Noncollege Education	0.31	0.46	0.38	0.49
Age 25-34	0.25	0.43	0.25	0.43
Age 35-44	0.24	0.43	0.25	0.43
Age 45-54	0.26	0.44	0.26	0.44
Age 55-64	0.25	0.43	0.25	0.43
Observations	6,579,056		6,667,785	

Notes: This table presents summary statistics for key variables used in the analysis at the individual level from 2012 to 2024. The sample is restricted to working-age individuals aged 25–64, and individuals with military occupations are excluded. Parent is defined as having at least one own child under age 18 residing with the individual. Parent with Young Children is defined as having at least one own child aged 4 and under residing with the individual. Full-time work is conditional on employment. College education refers to college-level attainment and above; noncollege education includes high school or below. In the baseline analysis, the female WFH index is used for female-only regressions, the male WFH index for male-only regressions, and a pooled index for regressions using the full sample. The WFH measures are reported here in raw units; in the regression analysis, they are standardized to have a mean of zero and a standard deviation of one.

Table 2: Predictive Power of WFH Measure

	(1) All Males	(2) All Females	(3) Fathers	(4) Mothers	(5) College Mothers	(6) Noncollege Mothers
Panel A: Continuous WFH Measure						
Coefficient	0.0356***	0.0289***	0.0404***	0.0291***	0.0310***	0.0094***
	(0.0018)	(0.0022)	(0.0018)	(0.0022)	(0.0025)	(0.0015)
F-statistic	235.47	181.79	136.82	164.41	197.44	135.41
Panel B: Median Split of WFH Measure						
Above Median	0.0636***	0.0524***	0.0740***	0.0516***	0.0554***	0.0186***
	(0.0052)	(0.0056)	(0.0057)	(0.0062)	(0.0069)	(0.0040)
F-statistic	235.23	183.35	142.28	182.60	207.80	134.98
Panel C: Quintiles of WFH Measure						
Quintile 2	0.0200***	0.0066	0.0209***	0.0112*	0.0101	0.0051
	(0.0039)	(0.0065)	(0.0039)	(0.0061)	(0.0075)	(0.0048)
Quintile 3	0.0247***	0.0294***	0.0289***	0.0360***	0.0375***	0.0129**
	(0.0039)	(0.0075)	(0.0049)	(0.0075)	(0.0089)	(0.0056)
Quintile 4	0.0585***	0.0483***	0.0669***	0.0511***	0.0557***	0.0187***
	(0.0046)	(0.0072)	(0.0051)	(0.0068)	(0.0078)	(0.0070)
Quintile 5	0.0961***	0.0780***	0.1093***	0.0810***	0.0863***	0.0269***
	(0.0075)	(0.0101)	(0.0087)	(0.0100)	(0.0111)	(0.0065)
F-statistic	180.50	177.68	113.33	138.25	162.38	105.37
N	5934998	5522785	2238592	2068180	1505739	562441

Notes: This table presents pseudo first-stage results comparing the predictive power of the WFH measure across continuous and binned constructions. The continuous WFH measure in Panel A is constructed at the MSA level using Equation (1). Panel B replaces the continuous measure with an indicator for MSAs above the median WFH exposure, while Panel C replaces it with indicators for each quintile of WFH exposure, with the lowest quintile omitted as the reference group. Regressions for males and fathers use a male-specific index constructed from the employed male population; regressions for females and mothers use a female-specific index constructed from the employed female population. Actual WFH behavior is measured using the ACS commuting variable, which captures each employed individual's primary mode of travel to work, including an option for working from home. All specifications employ a difference-in-differences design with MSA and year fixed effects; standard errors are clustered at the MSA level. F-statistics indicate strong predictive power of the WFH measure across all constructions. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3: Effects of WFH Opportunities on Labor Market Outcomes

	(1)	(2)	(3)	(4)	(5)	(6)
	All Males	All Females	Fathers	Mothers	College Mothers	Noncollege Mothers
Panel A: Labor Force Participation						
Transition Effects (2020–2021)	-0.0000	0.0003	0.0005	0.0021***	0.0031***	-0.0037**
	(0.0004)	(0.0005)	(0.0004)	(0.0008)	(0.0007)	(0.0018)
Stable Effects (2022–2024)	-0.0001	0.0008*	0.0007*	0.0030***	0.0036***	-0.0018
	(0.0003)	(0.0005)	(0.0004)	(0.0007)	(0.0007)	(0.0013)
Control Mean	0.9286	0.8722	0.9666	0.8690	0.8813	0.8409
N	6667785	6579056	2385866	2477570	1754457	723113
Panel B: Employment						
Transition Effects (2020–2021)	-0.0018*	-0.0009	-0.0007	0.0007	0.0026**	-0.0076***
	(0.0010)	(0.0012)	(0.0008)	(0.0013)	(0.0011)	(0.0029)
Stable Effects (2022–2024)	-0.0025***	-0.0010	-0.0016***	0.0004	0.0019***	-0.0048***
	(0.0005)	(0.0008)	(0.0005)	(0.0009)	(0.0007)	(0.0018)
Control Mean	0.8881	0.8352	0.9373	0.8295	0.8520	0.7782
N	6667785	6579056	2385866	2477570	1754457	723113
Panel C: Full-Time Employment						
Transition Effects (2020–2021)	-0.0018***	0.0005	-0.0020***	0.0044***	0.0065***	-0.0043
	(0.0007)	(0.0009)	(0.0007)	(0.0011)	(0.0011)	(0.0027)
Stable Effects (2022–2024)	-0.0003	0.0015	-0.0005	0.0064***	0.0083***	-0.0030
	(0.0006)	(0.0010)	(0.0006)	(0.0013)	(0.0014)	(0.0024)
Control Mean	0.8947	0.7573	0.9315	0.7184	0.7361	0.6759
N	6249415	5907945	2313788	2222120	1599723	622397
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table reports the effects of WFH opportunities on labor market outcomes, separately estimated for the transition period (2020–2021) and the stable post-pandemic period (2022–2024). The dependent variables are labor force participation (Panel A), employment (Panel B), and full-time employment (Panel C). Mothers (fathers) are defined as women (men) with at least one own child under age 18 residing with the individual. Regressions for male samples use a male-specific WFH index constructed from the employed male population; regressions for female samples use a female-specific WFH index constructed from the employed female population. All specifications include MSA and year fixed effects. Standard errors are clustered at the MSA level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 4: Effects of WFH on Parents with Young Children

	(1) Fathers children < 5	(2) Mothers children < 5
Panel A: Labor Force Participation		
Transition Effects (2020–2021)	0.0011	0.0055***
	(0.0007)	(0.0014)
Stable Effects (2022–2024)	0.0017***	0.0074***
	(0.0005)	(0.0015)
Control Mean	0.9694	0.7996
N	864951	846022
Panel B: Employment		
Transition Effects (2020–2021)	-0.0008	0.0032*
	(0.0012)	(0.0018)
Stable Effects (2022–2024)	-0.0017**	0.0042***
	(0.0007)	(0.0014)
Control Mean	0.9409	0.7583
N	864951	846022
Panel C: Full-Time Employment		
Transition Effects (2020–2021)	-0.0020*	0.0060***
	(0.0010)	(0.0016)
Stable Effects (2022–2024)	-0.0010	0.0075***
	(0.0009)	(0.0016)
Control Mean	0.9314	0.7084
N	841829	708852
Controls	Yes	Yes
Fixed Effects	Yes	Yes

Notes: This table reports the effects of WFH opportunities on labor market outcomes for parents with young children, separately estimated for the transition period (2020–2021) and the stable post-pandemic period (2022–2024). The dependent variables are labor force participation (Panel A), employment (Panel B), and full-time employment (Panel C). The sample is restricted to parents with at least one own child under age five residing with the individual. Regressions for fathers use a male-specific WFH index; regressions for mothers use a female-specific WFH index. All specifications include MSA and year fixed effects. Standard errors are clustered at the MSA level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 5: Interpreting the Effects

	(1) Mothers	(2) College Mothers	(3) Mothers children < 5
Panel A: Stable Effects (2022–2024)			
Labor Force Participation	0.0030***	0.0036***	0.0074***
	(0.0007)	(0.0007)	(0.0015)
Employment	0.0004	0.0019***	0.0042***
	(0.0009)	(0.0007)	(0.0014)
Full-Time	0.0064***	0.0083***	0.0075***
	(0.0013)	(0.0014)	(0.0016)
Panel B: Pseudo First Stage			
2022-2024	0.0263***	0.0281***	0.0293***
	(0.0023)	(0.0026)	(0.0029)
F-statistic	154.84	183.70	203.65
Panel C: Wald Estimator			
Labor Force Participation	0.1141	0.1281	0.2526
Relative effect (%)	13.13	14.54	31.59
Employment	0.0152	0.0676	0.1434
Relative effect (%)	1.83	7.93	18.91
Full-Time	0.2433	0.2954	0.2560
Relative effect (%)	33.87	40.13	36.14
Control Mean (LFP)	0.8690	0.8813	0.7996
Control Mean (Emp)	0.8295	0.8520	0.7583
Control Mean (FT)	0.7184	0.7361	0.7084

Notes: Panel A reproduces the stable effects (2022–2024) from the main results. Panel B reports the pseudo first stage for each corresponding group. Panel C presents the Wald estimator, computed as the ratio of the reduced-form estimate to the first-stage estimate, along with the relative effect expressed as a percentage of the pre-pandemic control mean. Mothers are defined as women with at least one own child under age 18 residing with the individual. Mothers with young children are defined as women with at least one own child under age five residing with the individual. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 6: Household WFH Effects on Labor Force Participation: Separating Own and Spousal Channels

Panel A: All Education Levels

	All		No Children		Children		Young Children	
	Baseline	Decomposed	Baseline	Decomposed	Baseline	Decomposed	Baseline	Decomposed
Wife	0.0017***	0.0013**	0.0003	0.0004	0.0029***	0.0020**	0.0058***	0.0042**
	(0.0006)	(0.0006)	(0.0008)	(0.0008)	(0.0008)	(0.0008)	(0.0017)	(0.0017)
Husband		0.0079***		0.0007		0.0147***		0.0238***
		(0.0010)		(0.0012)		(0.0015)		(0.0025)
Control Mean	0.8624	0.8624	0.8708	0.8708	0.8545	0.8545	0.7825	0.7825
N	3261043	3261043	1578289	1578289	1682754	1682754	616803	616803

Panel B: College-Educated

	All		No Children		Children		Young Children	
	Baseline	Decomposed	Baseline	Decomposed	Baseline	Decomposed	Baseline	Decomposed
Wife	0.0020***	0.0017**	0.0009	0.0009	0.0028***	0.0021**	0.0056***	0.0043**
	(0.0007)	(0.0007)	(0.0010)	(0.0010)	(0.0009)	(0.0009)	(0.0018)	(0.0017)
Husband		0.0066***		0.0006		0.0116***		0.0185***
		(0.0011)		(0.0013)		(0.0016)		(0.0025)
Control Mean	0.8750	0.8750	0.8847	0.8847	0.8669	0.8669	0.8035	0.8035
N	2373517	2373517	1078232	1078232	1295285	1295285	504058	504058
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
MSA F.E.	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Husb. Occ. F.E	No	Yes	No	Yes	No	Yes	No	Yes

Notes: This table presents within-household WFH effects on labor force participation (2022–2024) for married women across education levels and family compositions. The sample is restricted to married women with co-resident husbands in opposite-sex couples. Baseline models include only the wife's WFH exposure index; decomposed models augment the baseline by adding the husband's WFH treatment and husband occupation fixed effects to isolate the spousal channel from own-opportunity effects. Children refers to women with at least one own child under age 18 residing in the household; young children refers to women with at least one own child under age five. Standard errors clustered at the MSA level are reported in parentheses. All models include year and MSA fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 7: Mechanisms of WFH Effects on LFP

	Childcare Cost Burden		Entry vs Retention Margin	
	(1)	(2)	(3)	(4)
	High	Low	Entry	Retention
Panel A: Mothers				
LFP (2022–2024)	0.0033***	0.0017*	0.0073***	0.0002
	(0.0009)	(0.0010)	(0.0021)	(0.0002)
Control Mean	0.8733	0.8625	0.4921	0.9891
N	1438820	926481	580017	1897553
Panel B: College Mothers				
LFP (2022–2024)	0.0039***	0.0027**	0.0073***	0.0003*
	(0.0009)	(0.0011)	(0.0026)	(0.0002)
Control Mean	0.8844	0.8759	0.4876	0.9906
N	1025408	651309	365874	1388583
Controls	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes

Notes: This table reports stable-period (2022–2024) effects of WFH opportunities on labor force participation for mothers (Panel A) and college-educated mothers (Panel B). The Childcare Cost Burden samples split MSAs at the median ratio of annual young-child center-based care prices to female median earnings, constructed from the 2022 National Database of Childcare Prices; MSAs in states without 2022 price data are excluded. The Entry sample is restricted to women who worked fewer than 40 weeks in the prior year, capturing those with weak or no employment; the Retention sample is restricted to those who worked 40 or more weeks, capturing those who were continuously employed. All models include year and MSA fixed effects and the full set of controls. Standard errors clustered at the MSA level are in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

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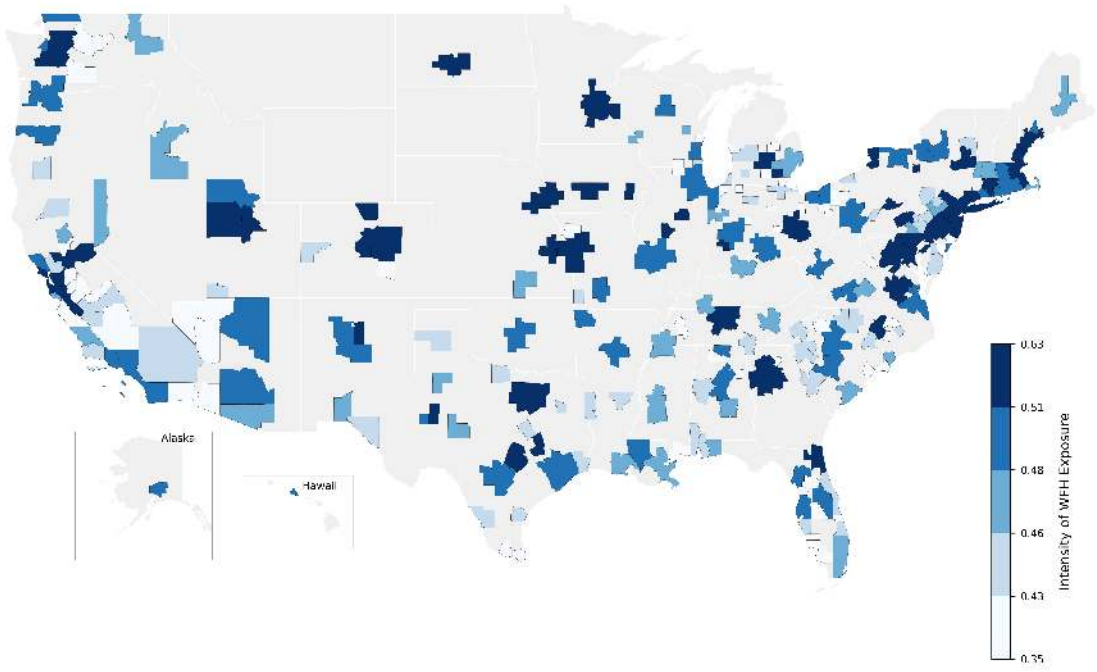
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Supplemental Appendix

"The Future of Work: Remote Opportunities and Female Labor Force Participation"

Appendix A Supplemental Figures

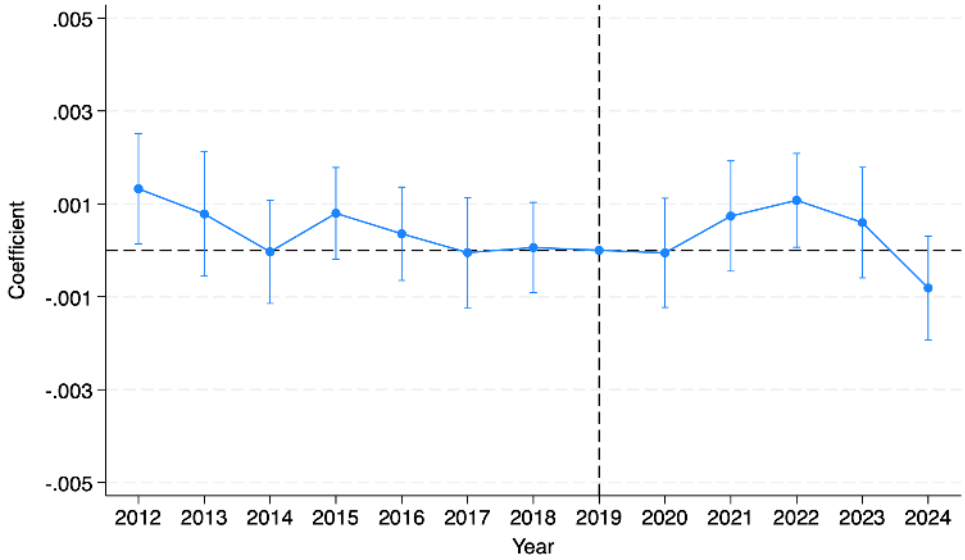
Figure A1: Geographic Variation in Pre-Pandemic Exposure to Work-from-Home Opportunity



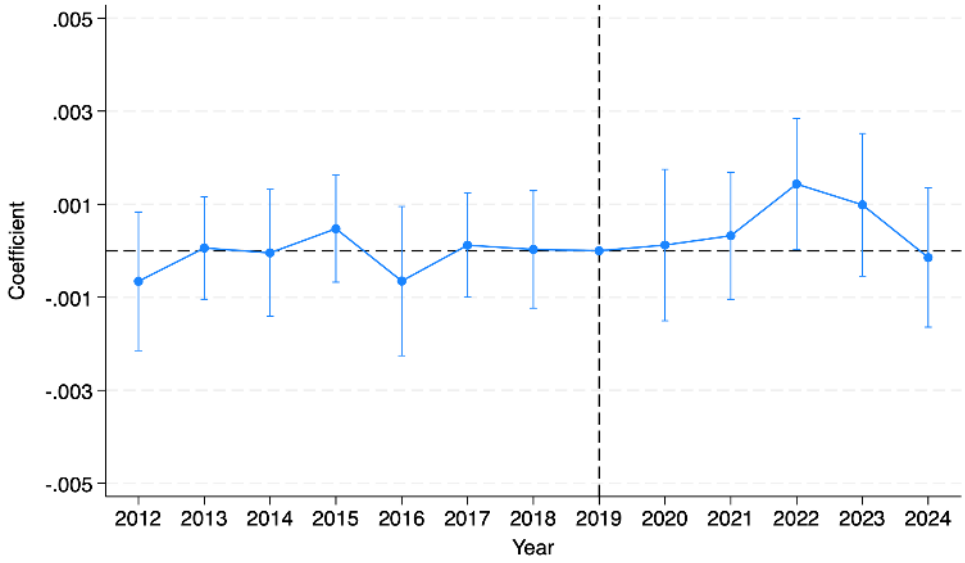
Notes: This figure shows pre-pandemic work-from-home exposure across 242 MSAs from 2012-2019, based on Dingel and Neiman (2020) teleworkability scores weighted by local occupational distributions. Darker shades indicate higher WFH exposure intensity.

Figure A2: Effects of WFH Opportunities on LFP

(a) Male



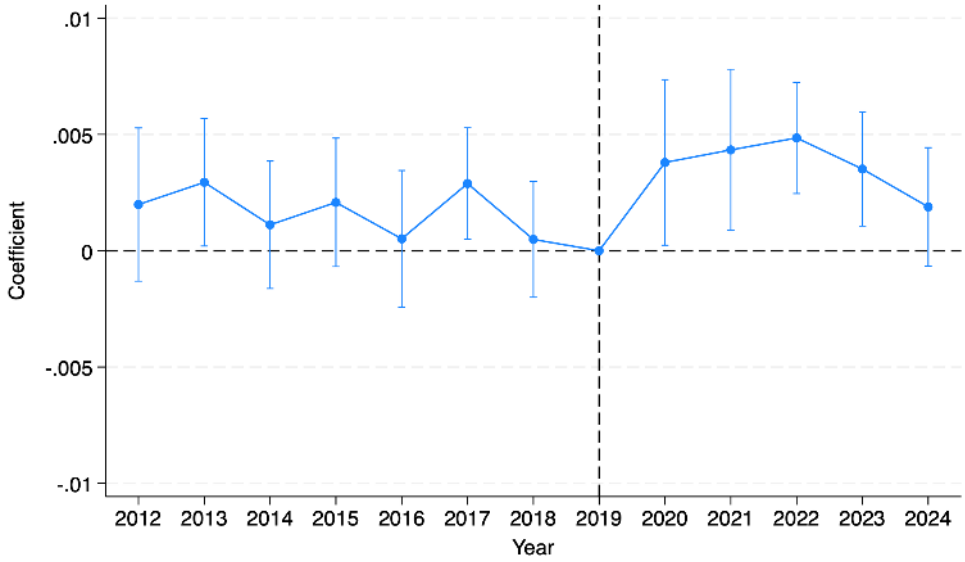
(b) Female



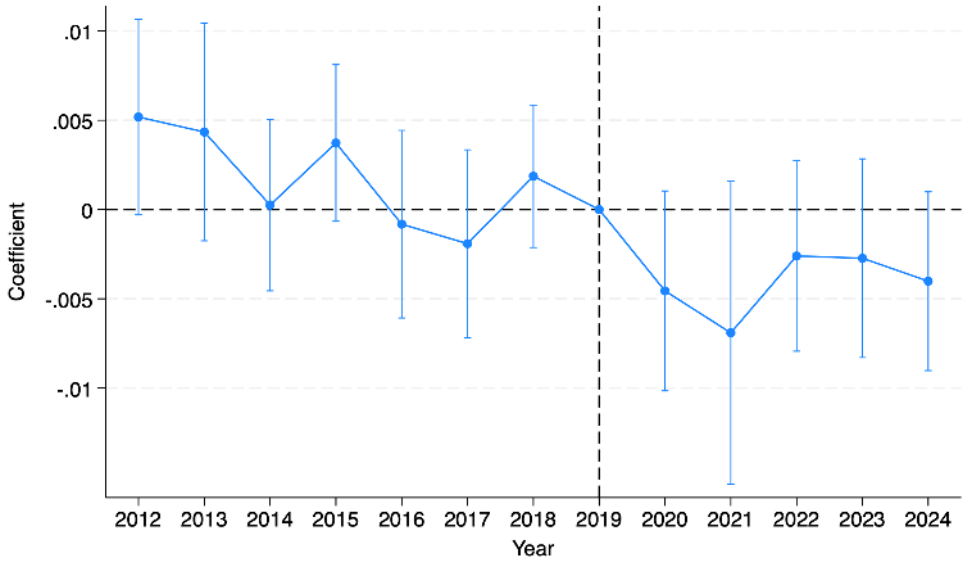
Notes: This figure compares the effects of individual-level exposure to WFH opportunities on labor force participation for (a) male and (b) female. It displays the coefficients and corresponding 95% confidence intervals for the Equation (3). The year 2019, one year prior to the onset of the COVID-19 pandemic, is used as the reference point and normalized to zero. Standard errors are clustered at the MSA level. The WFH measure is standardized to have a mean of 0 and a standard deviation of 1.

Figure A3: Effects of WFH Opportunities on Employment

(a) College Mothers



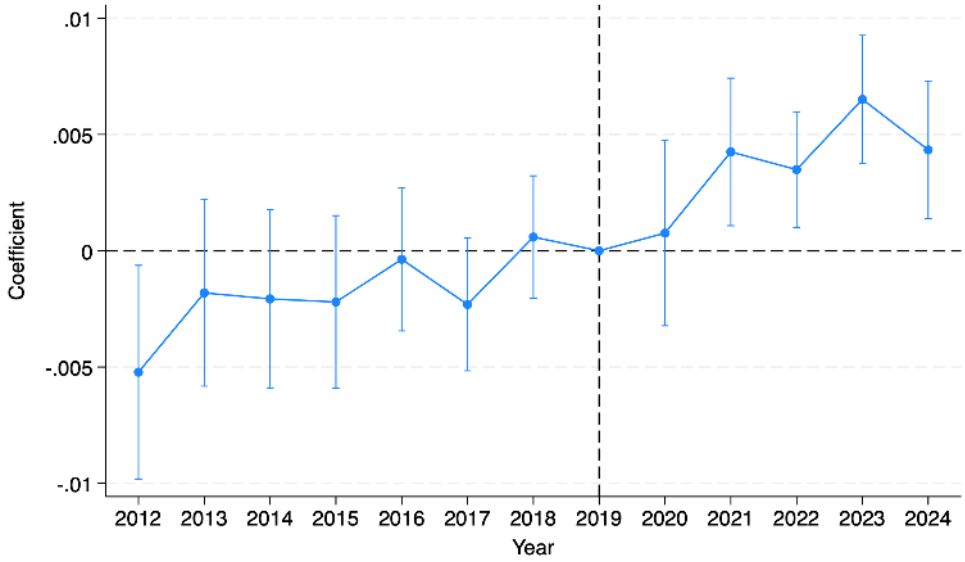
(b) Noncollege Mothers



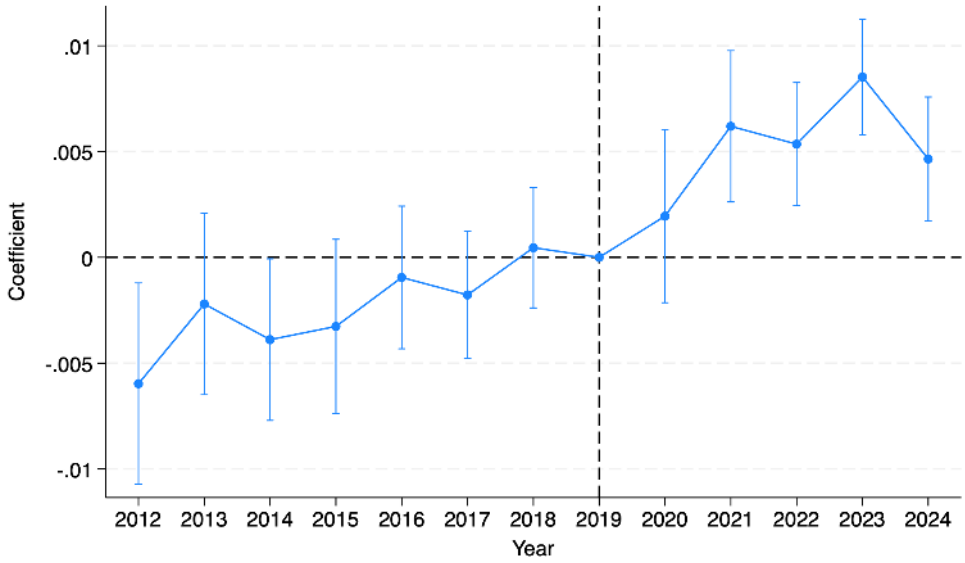
Notes: This figure displays the effects of WFH opportunities on employment for (a) college-educated mothers and (b) noncollege mothers. Mothers are defined as women with at least one own child under age 18 residing with the individual. Each panel plots the coefficients and corresponding 95% confidence intervals from Equation (3). The year 2019 is used as the reference point and normalized to zero. The WFH measure is standardized to have a mean of 0 and a standard deviation of 1. Standard errors are clustered at the MSA level.

Figure A4: Effects of WFH Opportunities on Full-Time Work

(a) Mothers



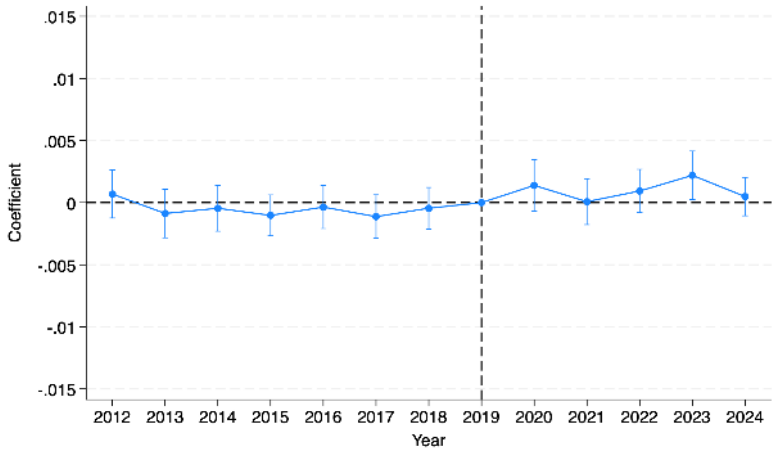
(b) College Mothers



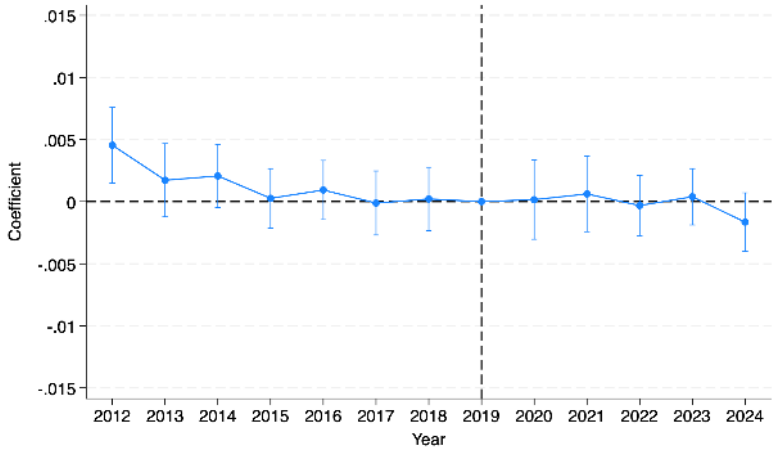
Notes: This figure displays the effects of WFH opportunities on full-time work for (a) mothers and (b) college-educated mothers. Mothers are defined as women with at least one own child under age 18 residing with the individual. Each panel plots the coefficients and corresponding 95% confidence intervals from Equation (3). The year 2019 is used as the reference point and normalized to zero. The WFH measure is standardized to have a mean of 0 and a standard deviation of 1. Standard errors are clustered at the MSA level.

Figure A5: Effects of WFH on Fathers with Young Children

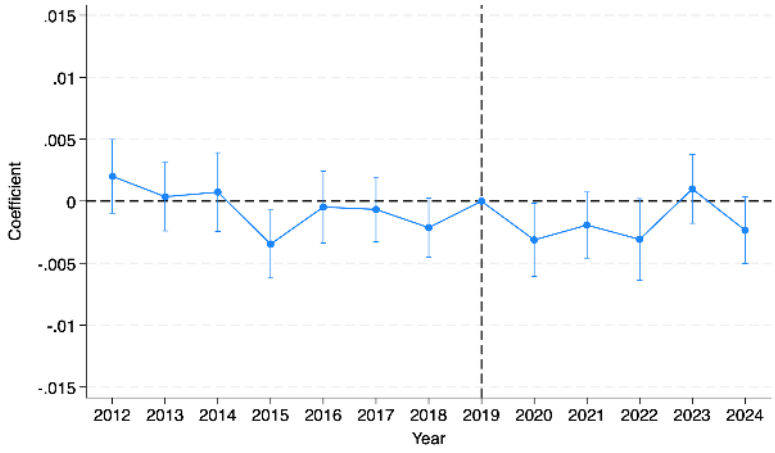
(a) Labor Force Participation



(b) Employment

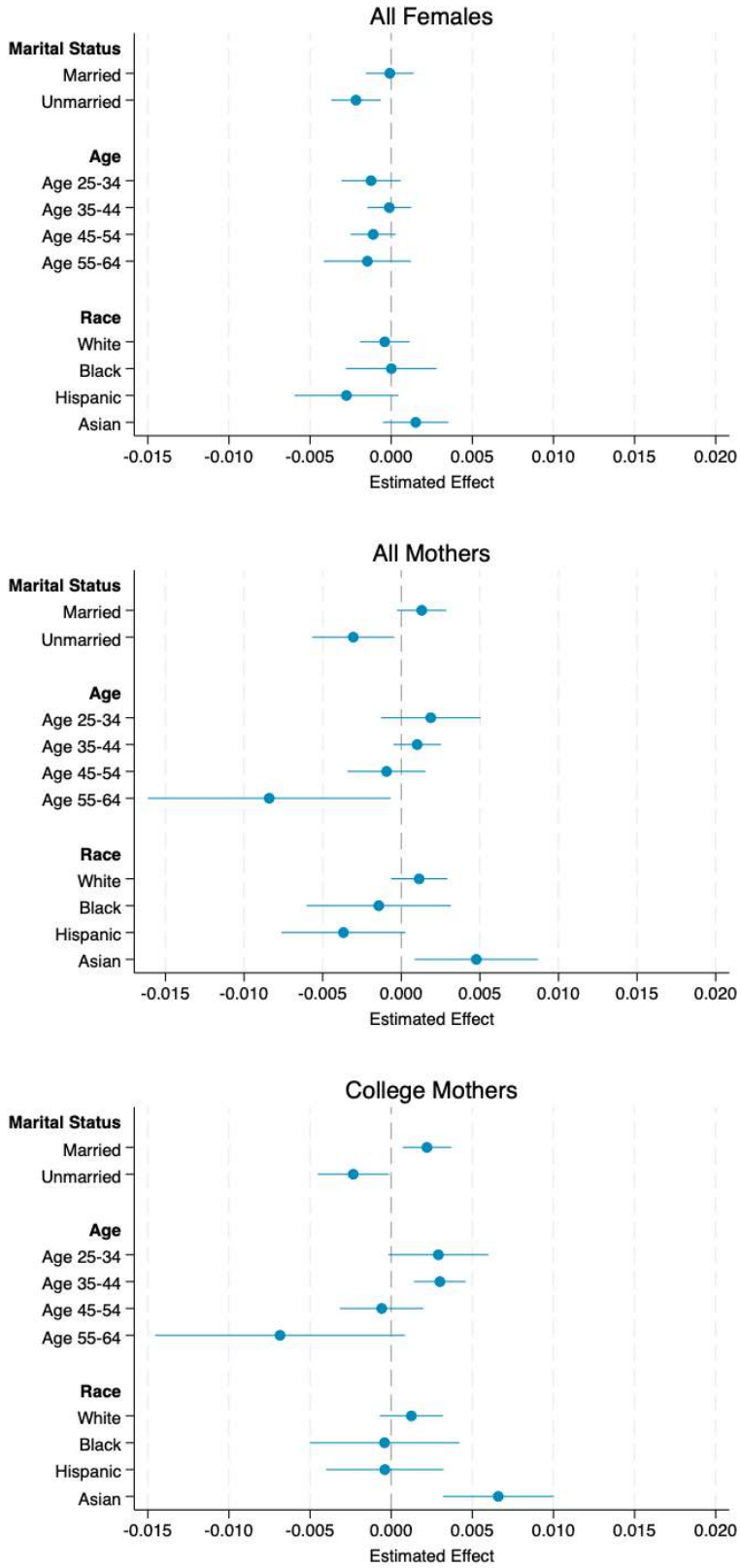


(c) Full-Time Work



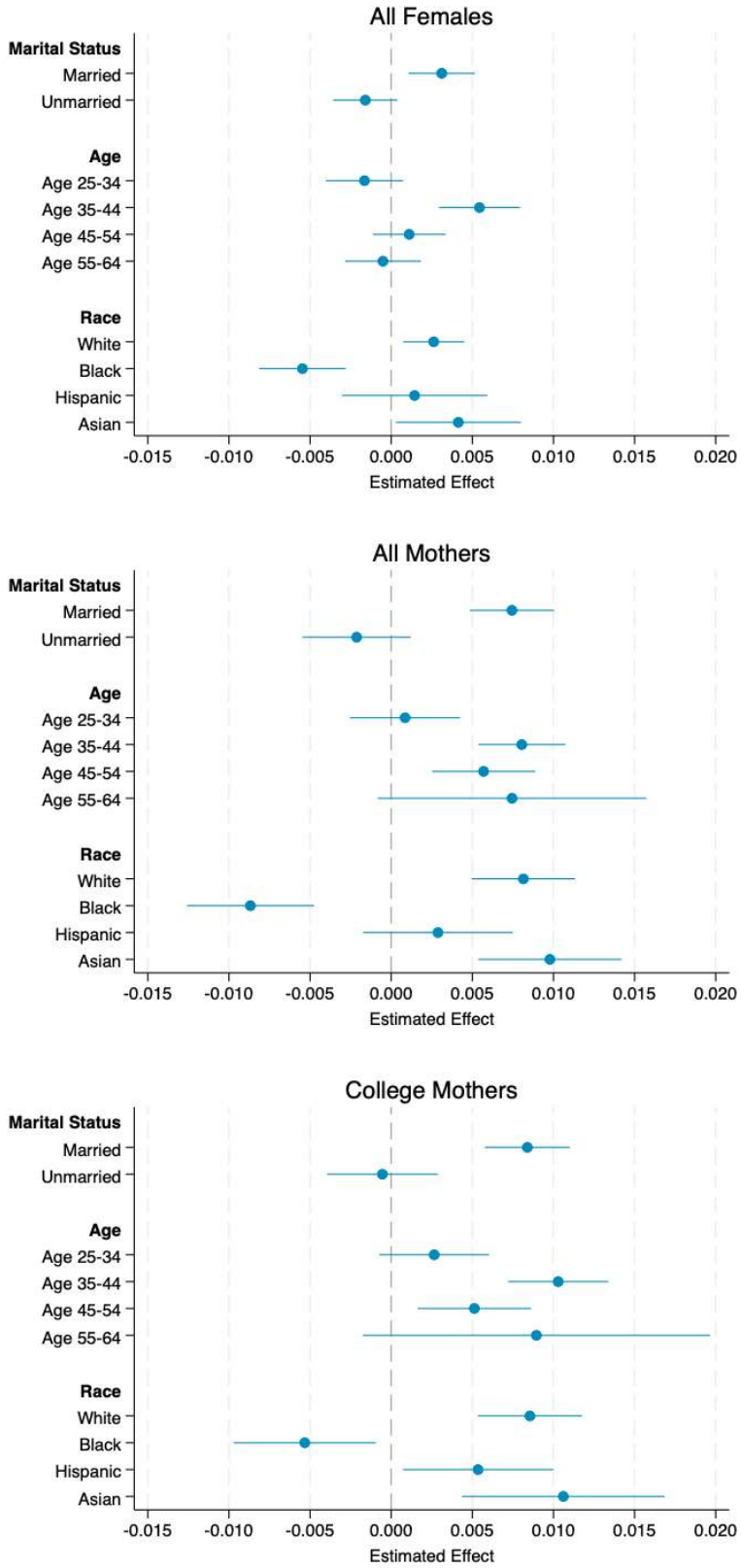
Notes: This figure displays the effects of WFH opportunities on (a) labor force participation, (b) employment, and (c) full-time work for fathers with young children. Fathers with young children are defined as men with at least one own child aged 4 and under residing with the individual. Each panel plots the coefficients and corresponding 95% confidence intervals from Equation (3). The year 2019 is used as the reference point and normalized to zero. The WFH measure is standardized to have a mean of 0 and a standard deviation of 1. Standard errors are clustered at the MSA level.

Figure A6: Heterogeneous Effects of WFH on Employment



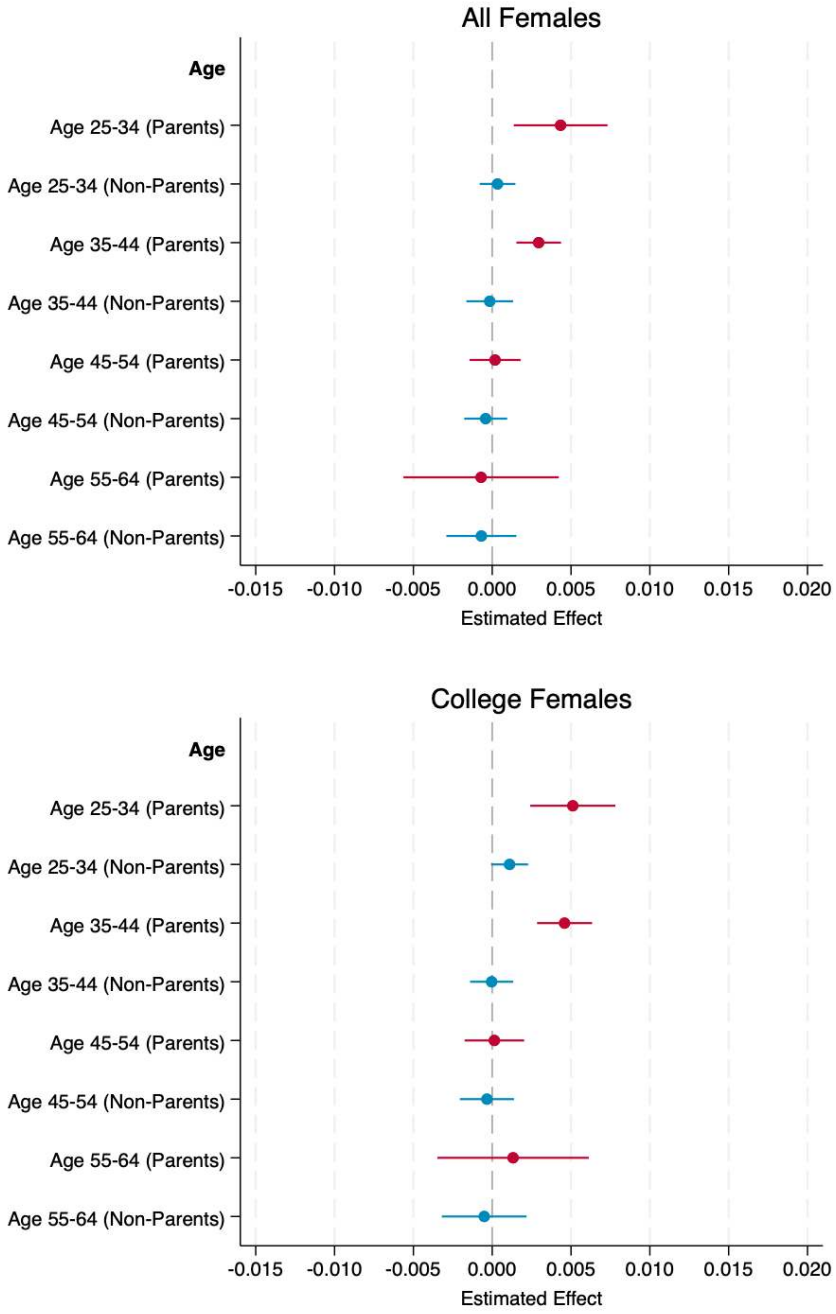
Notes: This figure plots the heterogeneous effects of WFH opportunities on employment by demographic groups. Each panel reports estimated coefficients and 95% confidence intervals from equation (2). Standard errors are clustered at the MSA level.

Figure A7: Heterogeneous Effects of WFH on Full-Time Work



Notes: This figure plots the heterogeneous effects of WFH opportunities on full-time work by demographic groups. Each panel reports estimated coefficients and 95% confidence intervals from equation (2). Standard errors are clustered at the MSA level.

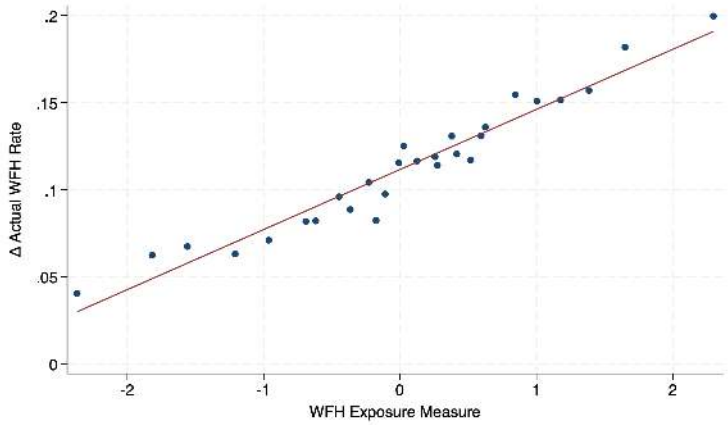
Figure A8: LFP Effects by Age and Parental Status



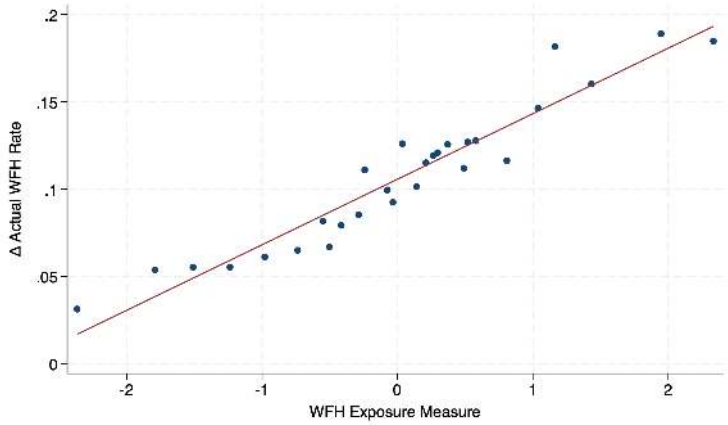
Notes: This figure examines the heterogeneous effects of WFH expansion on labor force participation by age group and parental status. Each panel plots the average post-pandemic effect of WFH opportunities on labor force participation for mothers (red) and non-mothers (blue) across four age groups: 25–34, 35–44, 45–54, and 55–64. An individual is classified as a parent if their youngest child is under age 18. The top panel reports estimates for all females and the bottom panel for college-educated females. Standard errors are clustered at the MSA level. All models include controls, year fixed effects, and MSA fixed effects.

Figure A9: Linearity of WFH Exposure Measure and Changes in Actual WFH Rates

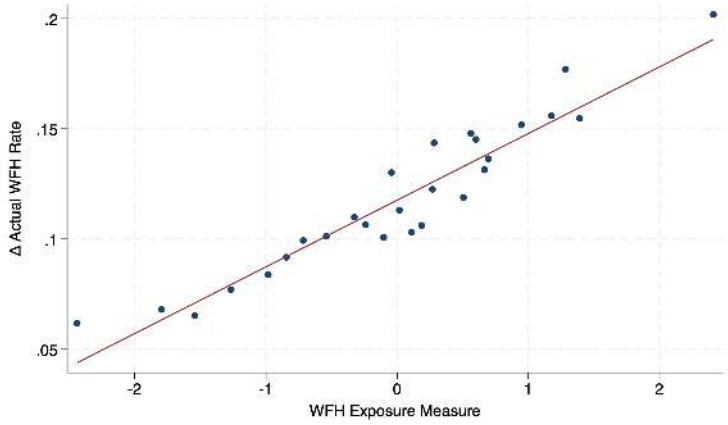
(a) Pooled Sample



(b) Male

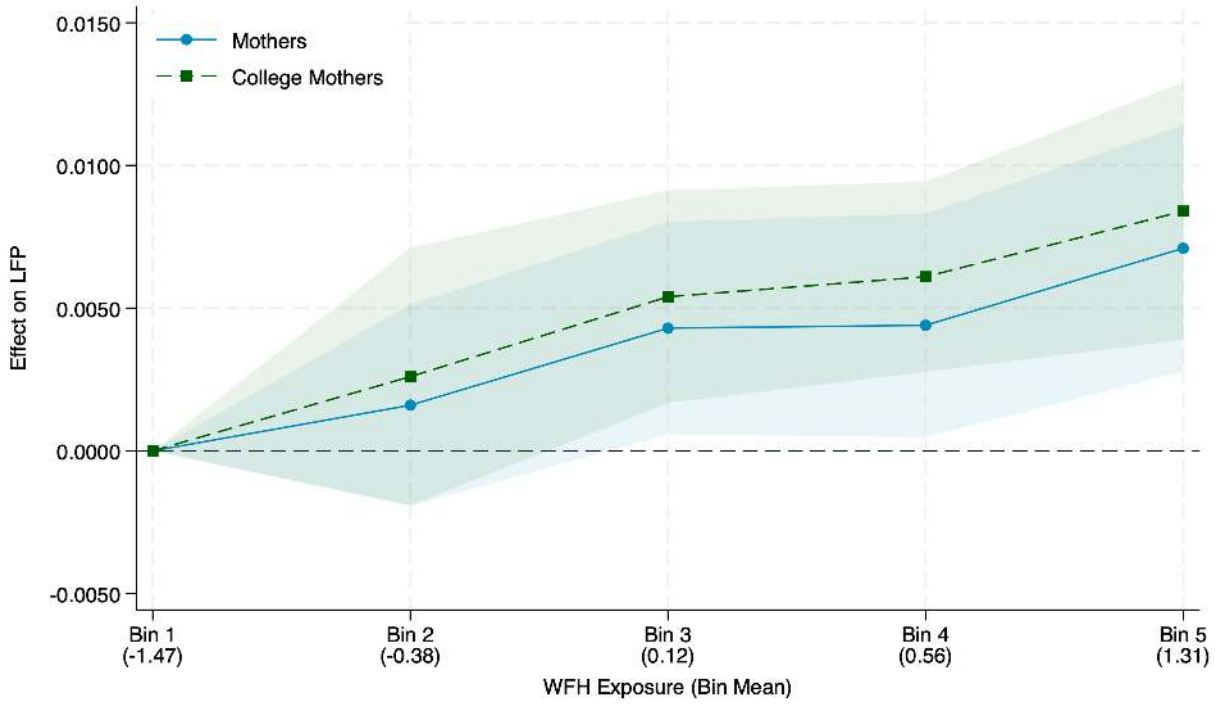


(c) Female



Notes: This figure plots the change in the actual WFH rate between the pre-pandemic (2012–2019) and post-pandemic (2020–2024) periods against the pre-pandemic WFH exposure measure. Panel (a) focuses on the pooled sample with the pooled WFH measure, panel (b) on the male sample with the male-specific WFH measure, and panel (c) on the female sample with the female-specific WFH measure. The WFH exposure measure is standardized to have a mean of zero and a standard deviation of one. The WFH measures are at the MSA level, and the actual WFH rates are therefore based on MSA-level averages and plotted using binned values.

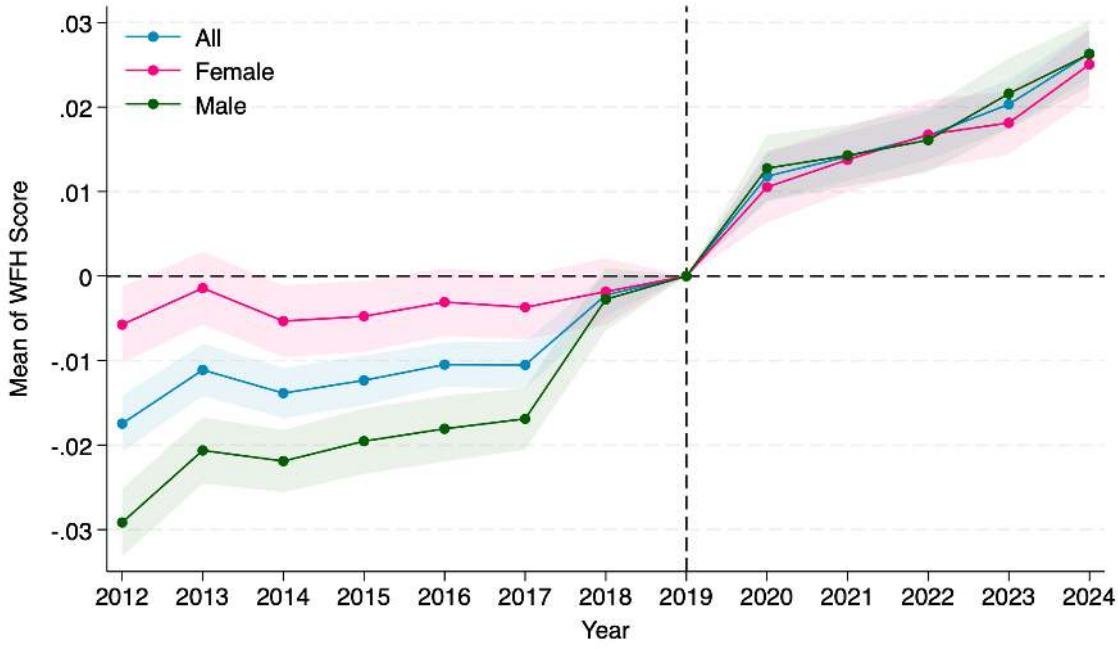
Figure A10: LFP Effects by Binned WFH Exposure



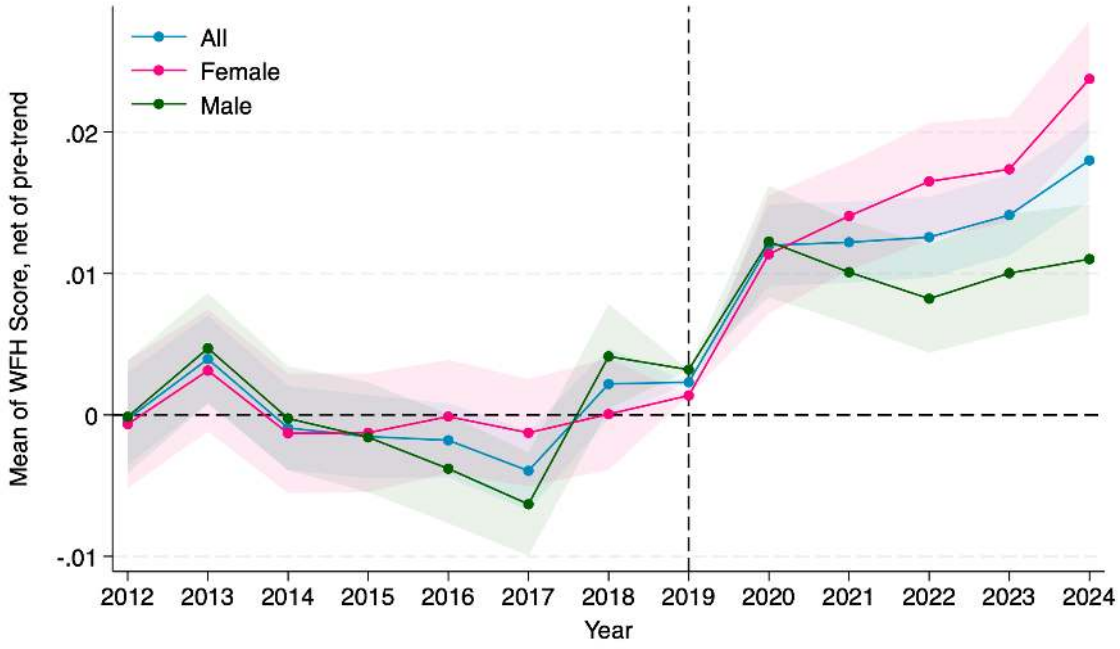
Notes: This figure examines the linearity assumption in the continuous difference-in-differences design. The WFH exposure index is divided into quintiles, with Bin 1 (lowest exposure) serving as the reference category. The plotted coefficients represent the average post-COVID labor force participation effect for each bin relative to Bin 1, estimated from a regression that replaces the continuous WFH score with bin indicators interacted with a post-2020 indicator. The x-axis labels report the mean WFH exposure score within each bin. Shaded areas represent 95% confidence intervals. Standard errors are clustered at the MSA level. All models include individual-level controls, year fixed effects, and MSA fixed effects.

Figure A11: Mean WFH Score of Employed Workers Over Time

(a) Raw Trajectory



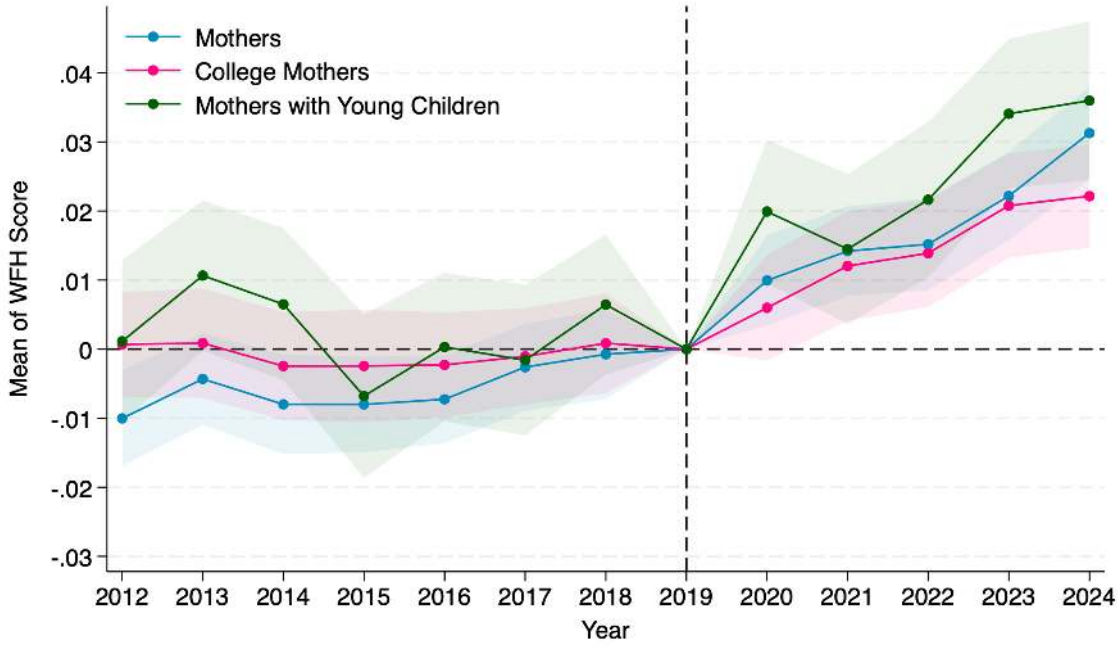
(b) Detrended Trajectory



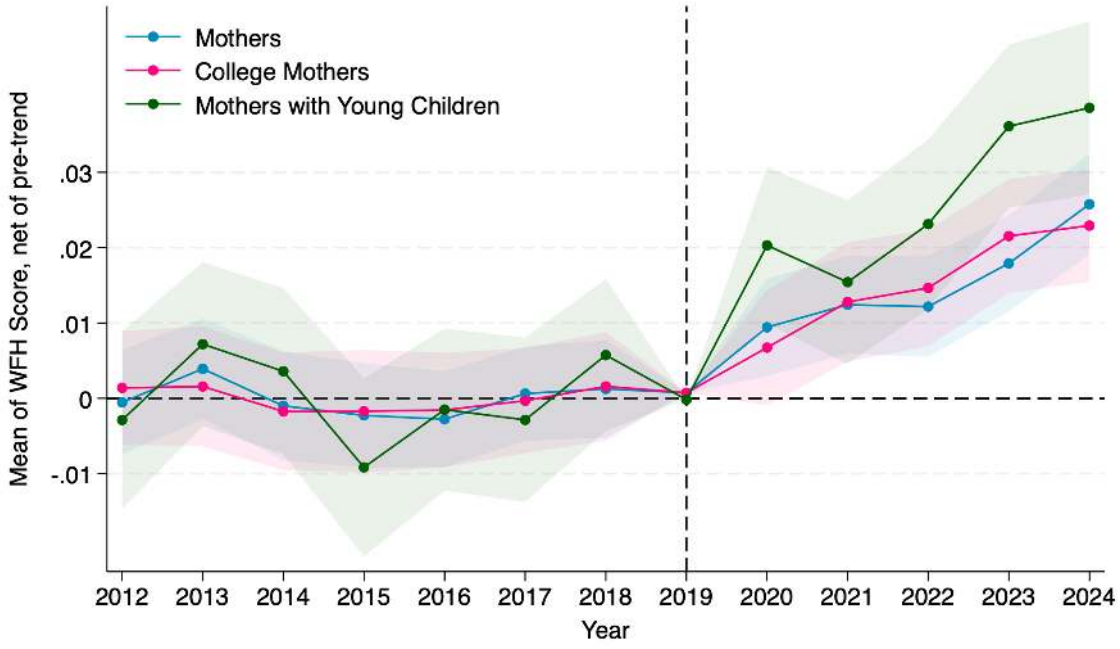
Notes: This figure shows how the mean WFH score among employed workers has evolved over time, separately for all employed workers, employed women, and employed men. Within each metropolitan area and year, the mean Dingel and Neiman (2020) teleworkability score is calculated across employed workers in the relevant group, then normalized within MSA by subtracting the 2019 value. Each panel plots the cross-MSA mean of this normalized score by year, with 95% confidence interval bands constructed from the cross-MSA standard error. Panel (a) reports the raw trajectory. Panel (b) subtracts the linear trend fitted on 2012–2019 from every year, so the pre-period is flat near zero by construction and post-2019 values represent the deviation from the extrapolated pre-pandemic trajectory.

Figure A12: Mean WFH Score of Employed Mothers Over Time

(a) Raw Trajectory

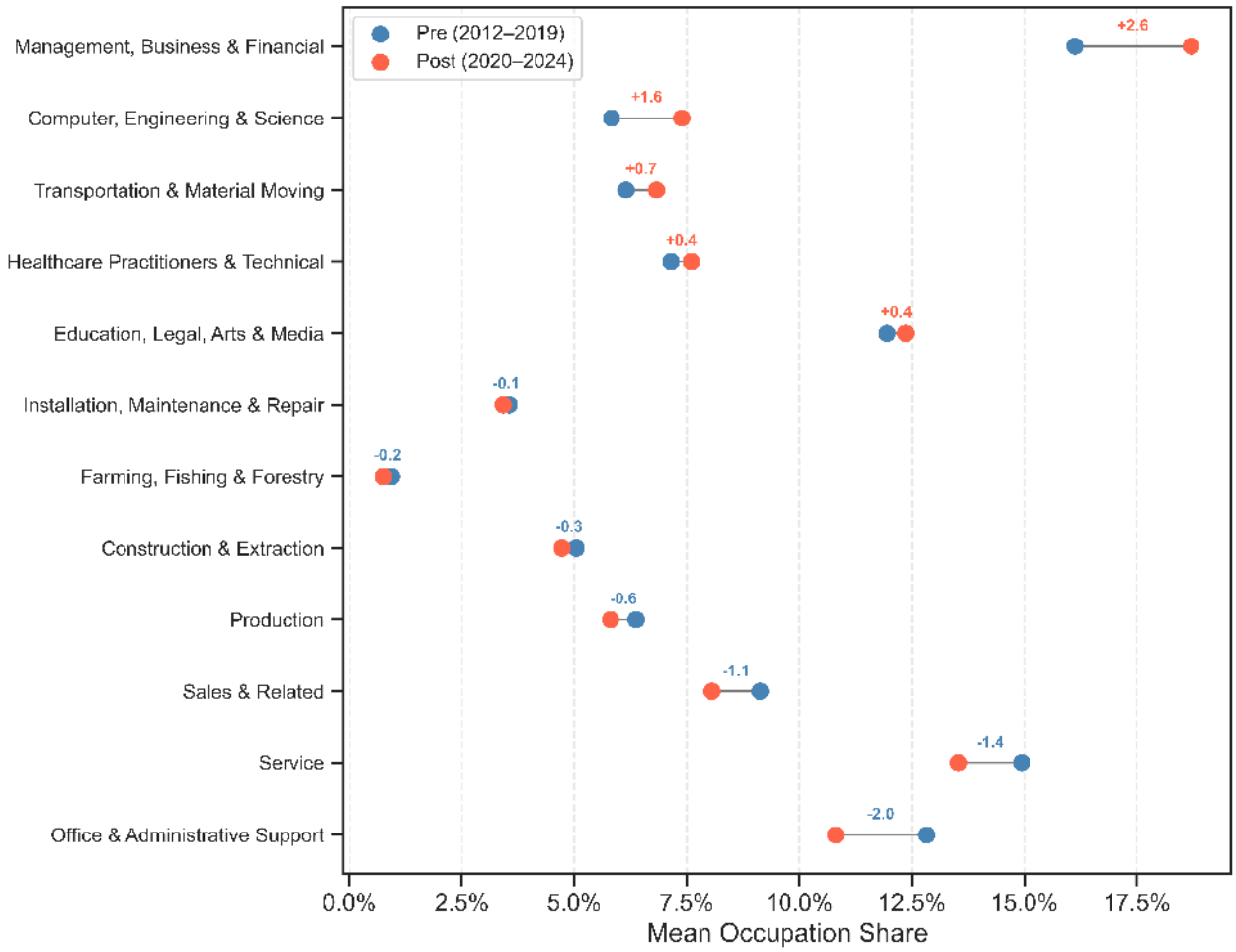


(b) Detrended Trajectory



Notes: This figure shows how the mean WFH score among employed mothers has evolved over time, separately for all employed mothers, college-educated mothers, and mothers with at least one child under age five. Within each metropolitan area and year, the mean Dingel and Neiman (2020) teleworkability score is calculated across employed workers in the relevant group, then normalized within MSA by subtracting the 2019 value. Each panel plots the cross-MSA mean of this normalized score by year, with 95% confidence interval bands constructed from the cross-MSA standard error. Panel (a) reports the raw trajectory. Panel (b) subtracts the linear trend fitted on 2012–2019 from every year, so the pre-period is flat near zero by construction and post-2019 values represent the deviation from the extrapolated pre-pandemic trajectory.

Figure A13: National Occupation Composition Changes



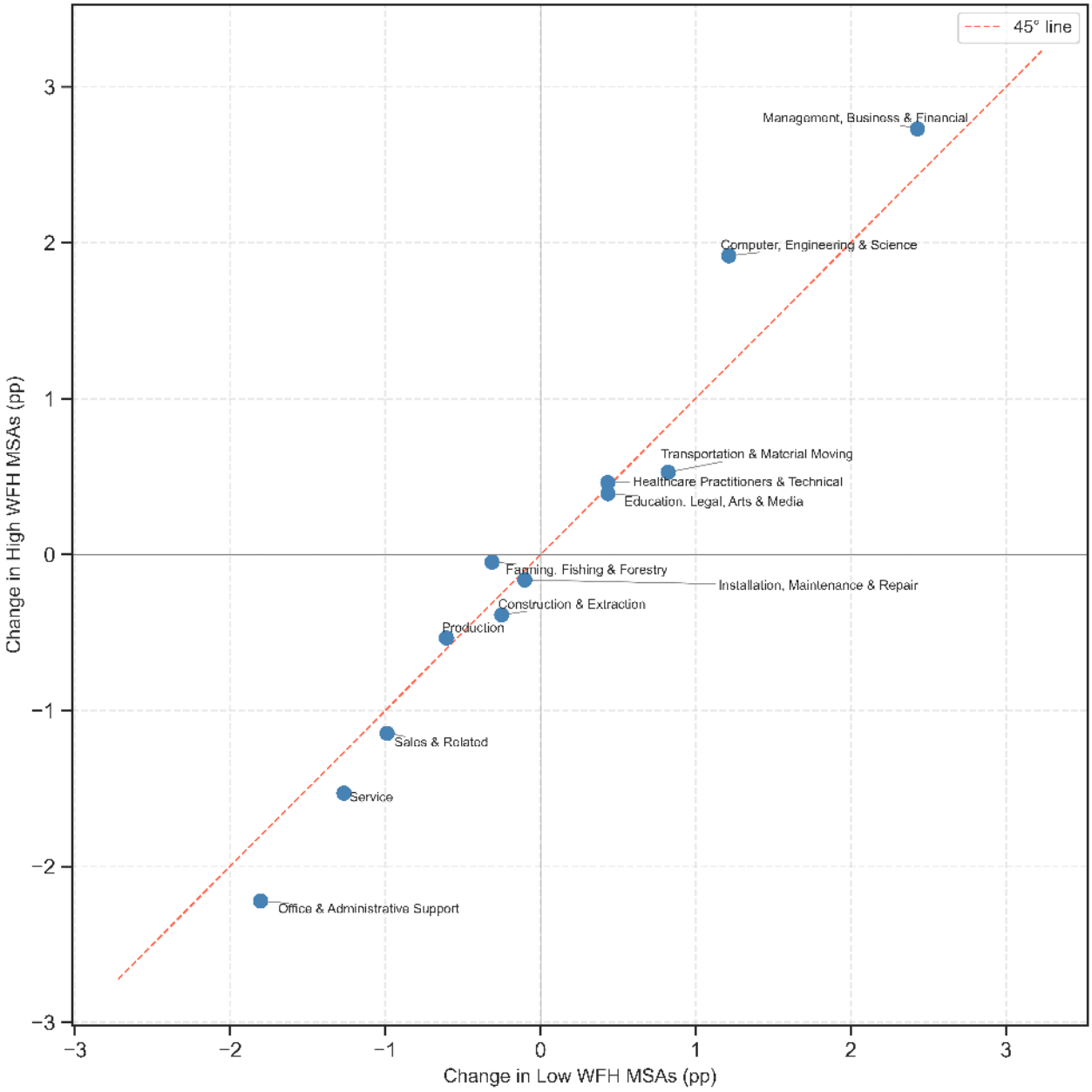
Notes: This figure plots the mean occupation share for each of twelve SOC intermediate groups across 242 MSAs, separately for the pre-pandemic (2012–2019) and post-pandemic (2020–2024) periods. Occupation shares are computed as the fraction of employed workers in each group within each MSA-year cell. Labels indicate the change in percentage points. The twelve groups follow the 2018 SOC intermediate aggregation (Table 5, BLS SOC 2018 User Guide); military occupations are excluded.

Figure A14: Occupation Composition by WFH Exposure Group



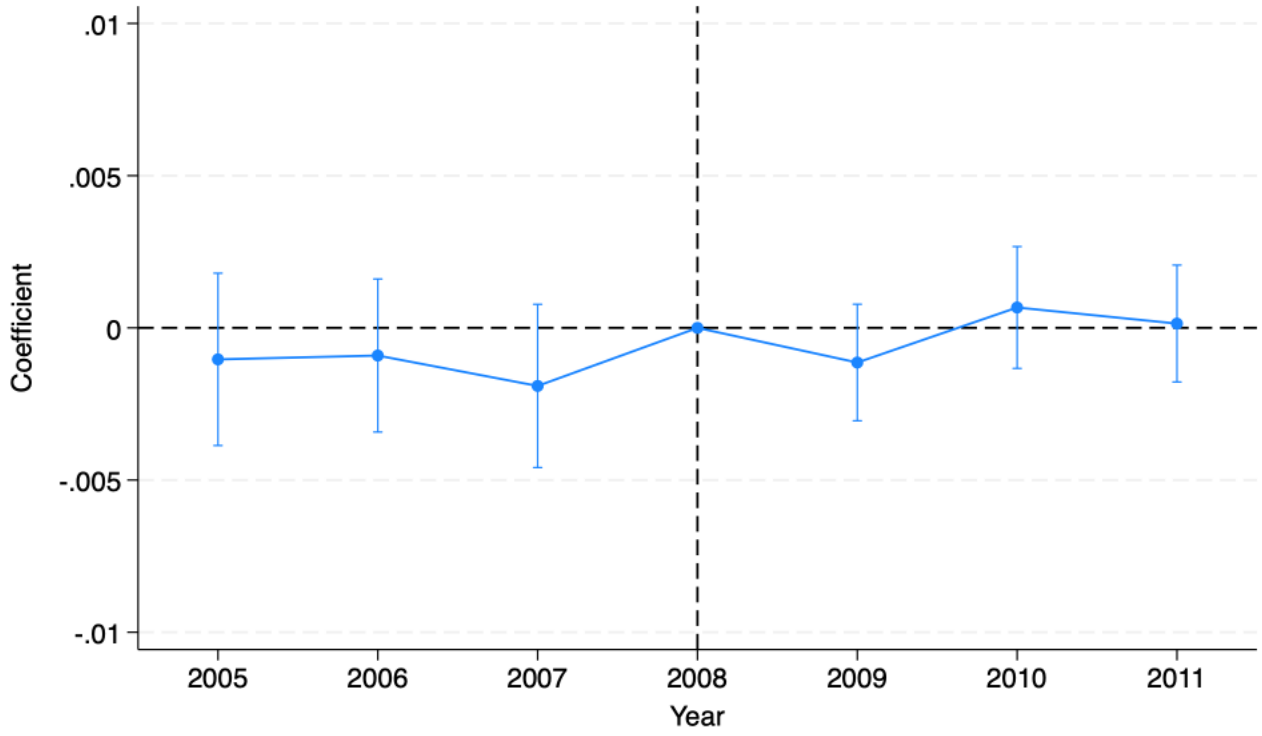
Notes: This figure replicates Figure A13 separately for low-WFH MSAs (below-median WFH exposure) and high-WFH MSAs (above-median WFH exposure). Occupation shares are computed as the fraction of employed workers in each group within each MSA-year cell, separately for the pre-pandemic (2012–2019) and post-pandemic (2020–2024) periods. Labels indicate the change in percentage points. The twelve groups follow the 2018 SOC intermediate aggregation (Table 5, BLS SOC 2018 User Guide); military occupations are excluded.

Figure A15: Occupation Share Changes: High vs. Low WFH MSAs



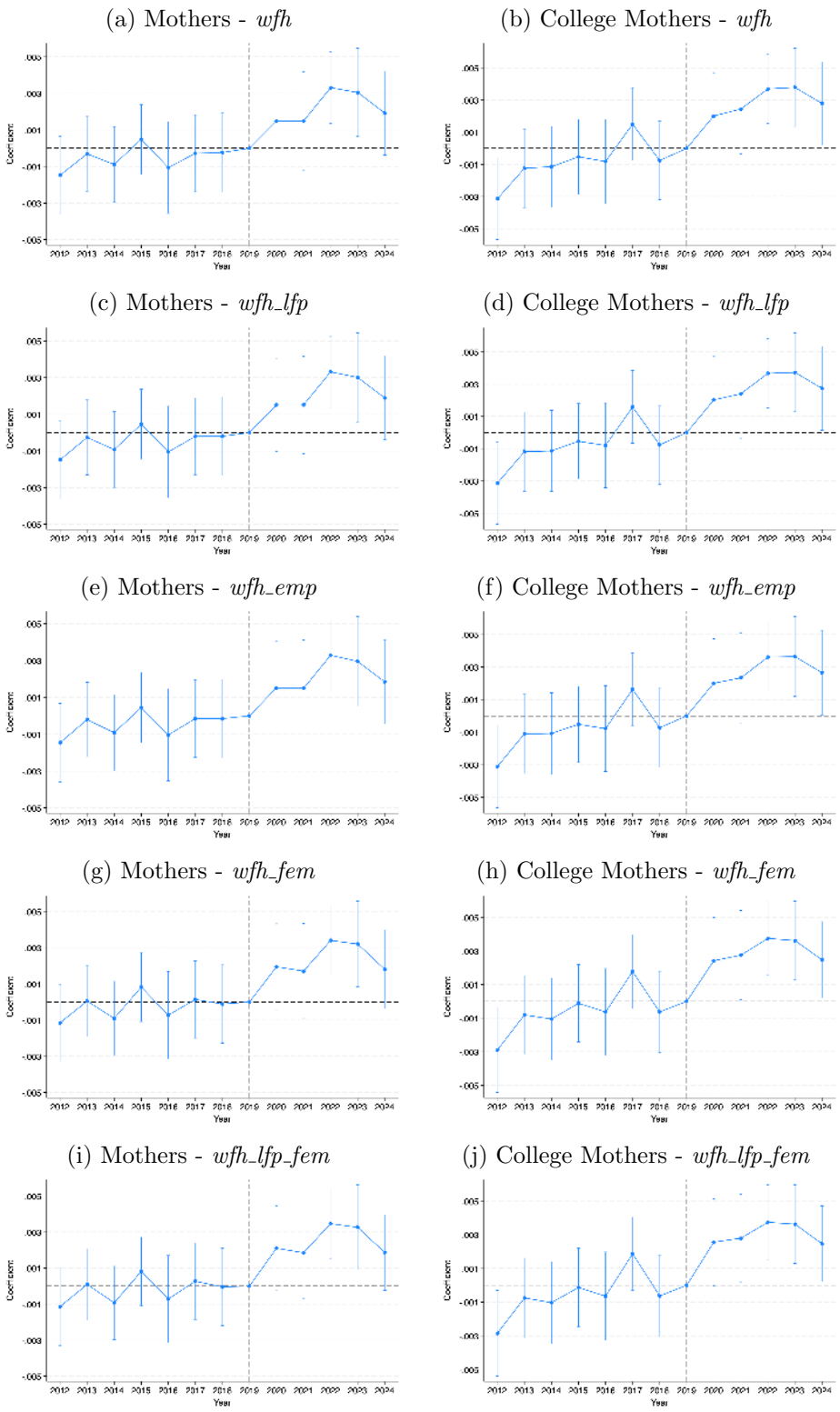
Notes: This figure plots the change in occupation share (pre-pandemic to post-pandemic) for each of twelve SOC intermediate groups, with the change in low-WFH MSAs (below-median WFH exposure) on the horizontal axis and the change in high-WFH MSAs (above-median WFH exposure) on the vertical axis. The 45-degree reference line indicates symmetric composition changes across the two groups. Points clustered along this line indicate that the pandemic affected occupation composition similarly regardless of WFH exposure.

Figure A16: Falsification Test of 2008 Financial Crisis



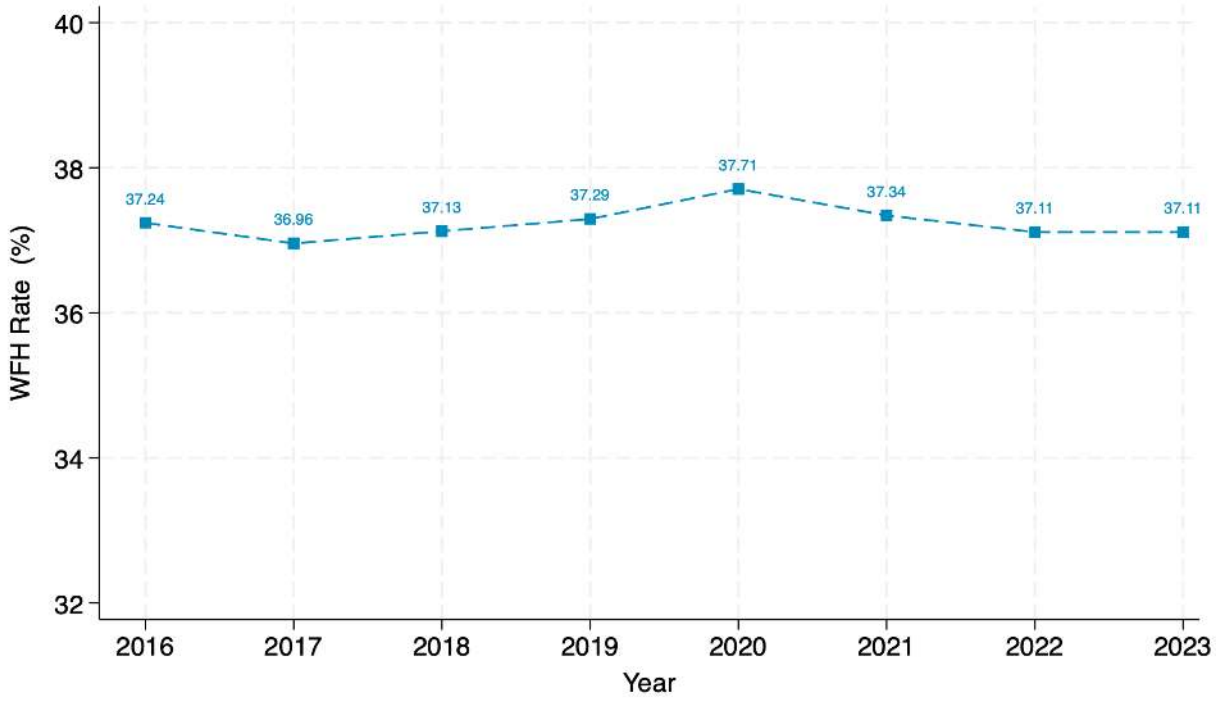
Notes: This figure replicates the event study design using the 2008 financial crisis as a placebo event to assess whether the estimated effects are specific to the COVID-19 recession. The sample uses ACS data from 2005–2011 for college-educated mothers. The outcome variable is labor force participation. The WFH exposure index is constructed following the same methodology as in the main analysis. All specifications include year fixed effects and MSA fixed effects. Standard errors are clustered at the MSA level.

Figure A17: Effects of WFH Opportunities on Female LFP: Robustness Using Alternative WFH Measures



Notes: This figure compares the effects of individual-level exposure to different WFH measures on female labor force participation for mothers (left column) and college mothers (right column). Each row corresponds to a different WFH measure. It displays the coefficients and corresponding 95% confidence intervals for the triple interaction terms from Equation (3). The year 2019, one year prior to the onset of the COVID-19 pandemic, is used as the reference point and normalized to zero. Standard errors are clustered at the MSA level, and each WFH measure is standardized to have a mean of 0 and a standard deviation of 1.

Figure A18: Stability of Dingel & Neiman WFH Measure



Notes: This figure replicates Dingel and Neiman's WFH measure using O*NET data from 2016 to 2023, showing the percentage of occupations classified as suitable for remote work across different years. The results indicate that the WFH rate remains consistently around 37%, aligning with the original analysis by Dingel and Neiman (2020). This stability underscores the robustness of the measure, ensuring that any observed effects are attributable to WFH potential rather than temporal changes in occupational classifications.

Appendix B Supplemental Tables

Table B1: MSA Characteristics Summary Statistics

	Above Median		Below Median	
Variable	Mean	SD	Mean	SD
Panel A: MSA Characteristics				
WFH Exposure Index	0.17	0.75	−1.65	0.57
Labor Force Participation	0.90	0.01	0.88	0.02
Employment	0.86	0.02	0.84	0.02
Full-Time Work	0.85	0.02	0.84	0.02
Female Share	0.50	0.01	0.50	0.01
White Share	0.72	0.15	0.70	0.21
College Share	0.66	0.05	0.55	0.06
Parents Share	0.37	0.03	0.37	0.04
Population (millions)	0.79	1.24	0.16	0.20
Median Household Income (thousands)	83.28	14.81	68.84	9.32
Panel B: Occupation Composition				
Computer, Engineering, and Science	0.08	0.03	0.05	0.01
Construction and Extraction	0.04	0.01	0.05	0.01
Education, Legal, Community Service, Arts, and Media	0.13	0.02	0.11	0.01
Farming, Fishing, and Forestry	0.00	0.00	0.01	0.03
Healthcare Practitioners and Technical	0.07	0.01	0.07	0.01
Installation, Maintenance, and Repair	0.03	0.01	0.04	0.01
Management, Business, and Financial	0.19	0.02	0.15	0.02
Office and Administrative Support	0.12	0.01	0.12	0.01
Production	0.05	0.02	0.07	0.03
Sales and Related	0.09	0.01	0.09	0.01
Service	0.13	0.02	0.16	0.03
Transportation and Material Moving	0.06	0.01	0.07	0.02
N	121		121	

Notes: This table presents MSA characteristics separately for above- and below-median WFH exposure groups. The WFH exposure index is standardized to have zero mean and unit variance. The median split is based on the pooled pre-period WFH exposure index rather than the gender-specific measures. The sample is restricted to working-age individuals aged 25–64, and individuals with military-specific occupations are excluded. Median household income is reported in thousands of 2019 US dollars, deflated using the CPI-U (Series CUUR0000SA0). Occupation shares are computed among employed workers and follow the 2018 SOC intermediate aggregation (Table 5, BLS SOC 2018 User Guide).

Table B2: Top Ten and Bottom Ten MSAs by WFH Exposure

MSA	WFH Index	Panel A: MSA Characteristics					Panel B: Occupation Composition				
		LFP	Female Share	College Share	Median Income	Pop	Mgmt, Bus, Sci, Arts	Nat Res, Constr, Maint	Prod, Trans, Moving	Sales Office	Service
Top Ten MSAs											
Bloomington, IL	2.36	0.91	0.49	0.69	82.57	0.07	0.49	0.07	0.10	0.22	0.12
Washington-Arlington-Alexandria, DC-VA-MD-WV	2.30	0.92	0.51	0.76	125.34	3.07	0.63	0.05	0.05	0.16	0.11
Ithaca, NY	2.12	0.88	0.51	0.79	75.12	0.04	0.61	0.05	0.06	0.14	0.13
Lawrence, KS	2.06	0.90	0.50	0.75	75.30	0.05	0.56	0.07	0.09	0.17	0.11
San Jose-Sunnyvale-Santa Clara, CA	1.93	0.91	0.47	0.77	149.38	0.94	0.61	0.05	0.07	0.15	0.11
San Francisco-Oakland-Hayward, CA	1.54	0.91	0.49	0.77	131.66	2.32	0.58	0.05	0.07	0.17	0.12
Ann Arbor, MI	1.49	0.90	0.50	0.82	92.15	0.16	0.62	0.05	0.07	0.16	0.10
State College, PA	1.48	0.90	0.48	0.62	79.74	0.06	0.50	0.09	0.11	0.17	0.13
Bridgeport-Stamford-Norwalk, CT	1.46	0.92	0.49	0.74	128.59	0.44	0.55	0.06	0.06	0.20	0.13
Trenton, NJ	1.44	0.91	0.50	0.70	109.80	0.17	0.55	0.05	0.09	0.19	0.12
Bottom Ten MSAs											
El Centro, CA	−2.58	0.88	0.49	0.50	57.87	0.06	0.28	0.14	0.13	0.20	0.25
Lake Havasu City-Kingman, AZ	−2.61	0.81	0.50	0.45	56.08	0.08	0.31	0.11	0.13	0.24	0.21
Yakima, WA	−2.61	0.87	0.49	0.41	63.36	0.10	0.31	0.20	0.16	0.17	0.15
Hickory-Lenoir-Morganton, NC	−2.62	0.88	0.49	0.53	61.09	0.18	0.36	0.09	0.23	0.19	0.13
Elkhart-Goshen, IN	−2.67	0.90	0.48	0.45	68.20	0.09	0.32	0.08	0.29	0.20	0.11
Bakersfield, CA	−2.69	0.88	0.48	0.46	65.89	0.34	0.33	0.19	0.14	0.18	0.16
Lima, OH	−2.79	0.88	0.50	0.51	65.95	0.04	0.34	0.09	0.22	0.19	0.15
Merced, CA	−2.87	0.88	0.48	0.43	62.34	0.11	0.30	0.19	0.17	0.18	0.16
Visalia-Porterville, CA	−2.89	0.88	0.47	0.43	60.20	0.18	0.30	0.21	0.14	0.18	0.17
Madera, CA	−2.98	0.87	0.47	0.44	69.01	0.06	0.32	0.20	0.14	0.18	0.16

Notes: This table reports the ten MSAs with the highest and lowest WFH exposure index values. The WFH exposure index is standardized to have zero mean and unit variance. The sample is restricted to working-age individuals aged 25–64, and individuals with military-specific occupations are excluded. Median household income is reported in thousands of 2019 US dollars, deflated using the CPI-U (Series CUUR0000SA0). Population is reported in millions. Occupation shares are computed among employed workers and follow the 2018 SOC high-level aggregation (Table 6, BLS SOC 2018 User Guide). The ranking is based on all 242 MSAs in the sample rather than restricting to the 100 largest by employment as in Dingel and Neiman (2020).

Table B3: WFH Measures Summary Statistics

Variable	Mean	SD	25th Percentile	75th Percentile	Unique N
wfh	0.4195	0.0514	0.3905	0.4488	242
wfh_emp	0.4282	0.0514	0.4041	0.4545	242
wfh_lfp	0.4226	0.0519	0.3967	0.4505	242
wfh_emp_female	0.4991	0.0406	0.4752	0.5228	242
wfh_lfp_female	0.4941	0.0415	0.4686	0.5186	242
wfh_emp_male	0.3648	0.0620	0.3311	0.3955	242

Notes: This table presents summary statistics for WFH measures constructed at the MSA level, averaged over the pre-period (2012–2019) using Equation (1). All measures apply ACS person weights and differ in their population focus: *wfh* includes all individuals; *wfh_emp* restricts to employed individuals; *wfh_lfp* restricts to individuals in the labor force; *wfh_emp_female* and *wfh_lfp_female* apply the same restrictions to females; and *wfh_emp_male* restricts to employed males. Each measure computes the weighted average teleworkability at the MSA level using occupation-level scores from Dingel and Neiman (2020). In the baseline analysis, *wfh_emp_female* is used for female-only regressions, *wfh_emp_male* for male-only regressions, and the pooled index *wfh_emp* for regressions using the full sample. All WFH measures are standardized to have a mean of zero and a standard deviation of one.

Table B4: Pre- and Post-Period WFH Exposure

	Pool	Male	Female
Stable (2022–2024)	0.928***	1.073***	0.928***
	(0.021)	(0.021)	(0.030)
R^2	0.915	0.917	0.799
F-statistic	2577.8	2648.7	951.6
Pooled (2020–2024)	1.041***	1.070***	0.946***
	(0.019)	(0.019)	(0.027)
R^2	0.926	0.932	0.840
F-statistic	3012.1	3266.2	1258.2
N	242	242	242

Notes: This table reports cross-sectional regressions of the post-period MSA WFH exposure index on the pre-period index, with one observation per MSA. The post-period index is constructed for two windows, 2022–2024 (stable) and 2020–2024 (pooled), following the same procedure used for the pre-period index. The Pool, Male, and Female columns use WFH indices constructed from the occupation distribution of all employed workers, employed men, and employed women, respectively. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B5: Placebo Test of WFH Exposure Indices

	(1)	(2)	(3)	(4)	(5)	(6)
	Males			Females		
	Placebo (Female Index)	Pooled	Baseline (Male Index)	Placebo (Male Index)	Pooled	Baseline (Female Index)
Coefficient	0.0320***	0.0347***	0.0356***	0.0315***	0.0310***	0.0289***
	(0.0022)	(0.0019)	(0.0018)	(0.0020)	(0.0019)	(0.0022)
F-statistic	233.76	237.04	235.47	175.80	177.03	181.79
Ctrl Mean	0.0475	0.0475	0.0475	0.0570	0.0570	0.0570
N	5934998	5934998	5934998	5522785	5522785	5522785

Notes: This table presents placebo first-stage results comparing the predictive power of gender-specific and pooled WFH exposure indices. Columns (1) and (4) are placebo checks that use the cross-gender index: the female index is applied to the male sample and the male index to the female sample. Columns (2) and (5) use the pooled index for both samples. Columns (3) and (6) are the baseline specifications using the matched gender-specific index. The similarity of estimates across columns reflects the high correlation among the three indices across MSAs (pairwise correlations of 0.984, 0.955, and 0.895, respectively), consistent with similar occupational structures faced by men and women within local labor markets. All specifications follow the same difference-in-differences design as Table 2, with MSA and year fixed effects and standard errors clustered at the MSA level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B6: First Stage by Age Group

	(1)	(2)	(3)	(4)
	Females			
	Age 25–34	Age 35–44	Age 45–54	Age 55–64
Coefficient	0.0333***	0.0306***	0.0275***	0.0228***
	(0.0025)	(0.0024)	(0.0021)	(0.0023)
F-statistic	194.21	203.31	171.14	196.78
Control Mean	0.0394	0.0612	0.0620	0.0651
N	1379081	1367091	1473609	1303004

Notes: This table presents pseudo first-stage estimates of the WFH exposure index on actual WFH take-up, estimated separately by age group for females. All specifications follow the same design as Table 2, with MSA and year fixed effects and standard errors clustered at the MSA level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B7: Effects of WFH Opportunities on Mothers: Robustness to the Choice of Control Variables

	Labor Force		Employment		Full-Time	
	(1)	(2)	(3)	(4)	(5)	(6)
	Mothers	College Mothers	Mothers	College Mothers	Mothers	College Mothers
Panel A: No Controls						
2022–2024	0.0031***	0.0037***	0.0008	0.0021***	0.0065***	0.0084***
	(0.0008)	(0.0007)	(0.0009)	(0.0007)	(0.0014)	(0.0015)
Panel B: Baseline Individual Controls						
2022–2024	0.0030***	0.0036***	0.0004	0.0019***	0.0064***	0.0083***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0014)
Panel C: Individual Controls × Year						
2022–2024	0.0016**	0.0029***	−0.0003	0.0016**	0.0050***	0.0076***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0013)
Panel D: MSA Controls						
2022–2024	0.0031***	0.0037***	0.0008	0.0021***	0.0065***	0.0084***
	(0.0008)	(0.0007)	(0.0009)	(0.0007)	(0.0014)	(0.0015)
Panel E: MSA Controls × Year						
2022–2024	0.0024**	0.0024**	0.0019	0.0017	−0.0023	−0.0030*
	(0.0011)	(0.0011)	(0.0013)	(0.0012)	(0.0015)	(0.0016)
Panel F: MSA Controls × Year (excl. income)						
2022–2024	0.0031***	0.0033***	0.0016*	0.0020**	0.0054***	0.0067***
	(0.0008)	(0.0008)	(0.0009)	(0.0008)	(0.0014)	(0.0015)
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	0.8690	0.8813	0.8295	0.8520	0.7184	0.7361
N	2,477,570	1,754,457	2,477,570	1,754,457	2,222,120	1,599,723

Notes: This table reports the average post-period effect (2022–2024) for mothers and college-educated mothers across six control specifications: no controls, baseline individual controls, individual controls interacted with year dummies, MSA-level controls, MSA-level controls interacted with year dummies, and MSA-level controls interacted with year dummies excluding median household income. All specifications include MSA and year fixed effects. Standard errors are clustered at the MSA level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B8: Effects of WFH Opportunities on Mothers: Additional Robustness Checks

	Labor Force		Employment		Full-Time	
	(1)	(2)	(3)	(4)	(5)	(6)
	Mothers	College Mothers	Mothers	College Mothers	Mothers	College Mothers
Panel A: Controlling for Pre-existing Linear Trends						
2022–2024	0.0026***	0.0025***	0.0023**	0.0029***	0.0044***	0.0057***
	(0.0008)	(0.0009)	(0.0009)	(0.0008)	(0.0010)	(0.0010)
Panel B: Controlling for COVID-19 Severity						
2022–2024	0.0030***	0.0036***	0.0004	0.0019***	0.0064***	0.0083***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0014)
Panel C: Controlling for School/Childcare Closure						
2022–2024	0.0030***	0.0036***	0.0004	0.0019***	0.0065***	0.0083***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0014)
Panel D: Controlling for COVID-19 Severity and School/Childcare Closure						
2022–2024	0.0030***	0.0036***	0.0004	0.0019***	0.0065***	0.0083***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0014)
Panel E: Industry-Based WFH Index (Dingel-Neiman NAICS Scores)						
2022–2024	0.0019***	0.0023***	-0.0003	0.0008	0.0055***	0.0070***
	(0.0006)	(0.0007)	(0.0008)	(0.0007)	(0.0015)	(0.0014)
Panel F: WFH Index Based on Prior-Year Residence						
2022–2024	0.0033***	0.0046***	0.0016	0.0033***	0.0036***	0.0053***
	(0.0009)	(0.0009)	(0.0010)	(0.0010)	(0.0011)	(0.0012)
Panel G: Excluding Older Workers (Ages 25–54 Only)						
2022–2024	0.0031***	0.0038***	0.0006	0.0021***	0.0064***	0.0082***
	(0.0007)	(0.0007)	(0.0008)	(0.0007)	(0.0014)	(0.0014)
Panel H: Shorter Pre-Period WFH Index (2017–2019)						
2022–2024	0.0030***	0.0037***	0.0004	0.0020***	0.0066***	0.0085***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0014)
Panel I: Triple-Difference						
2022–2024	0.0020***	0.0023***	0.0018**	0.0019**	0.0056***	0.0065***
	(0.0008)	(0.0008)	(0.0008)	(0.0008)	(0.0013)	(0.0015)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	0.8690	0.8813	0.8295	0.8520	0.7184	0.7361

Notes: This table reports the average post-period effect (2022–2024) for mothers and college-educated mothers across additional robustness checks. Each panel varies the baseline specification as described in the panel title. All specifications include MSA and year fixed effects. Standard errors are clustered at the MSA level and are shown in parentheses. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B9: Effects of WFH Opportunities on Mothers: Robustness Using Hansen et al. (2023) WFH Measure

	Labor Force		Employment		Full-Time	
	(1)	(2)	(3)	(4)	(5)	(6)
	Mothers	College Mothers	Mothers	College Mothers	Mothers	College Mothers
Panel A: Main Results						
Transition Effects	0.0012*	0.0023***	0.0003	0.0021**	0.0027***	0.0051***
(2020–2021)	(0.0006)	(0.0007)	(0.0009)	(0.0009)	(0.0010)	(0.0011)
Stable Effects	0.0023***	0.0030***	0.0004	0.0017**	0.0048***	0.0072***
(2022–2024)	(0.0006)	(0.0006)	(0.0008)	(0.0007)	(0.0011)	(0.0013)
Panel B: Pseudo First Stage						
2022–2024	0.0191***	0.0202***	0.0191***	0.0202***	0.0191***	0.0202***
	(0.0014)	(0.0014)	(0.0014)	(0.0014)	(0.0014)	(0.0014)
F-statistic	153.29	204.43	153.29	204.43	153.29	204.43
Panel C: Wald Estimator						
2022–2024	0.1204	0.1485	0.0209	0.0842	0.2513	0.3564
Relative effect (%)	13.83	16.81	2.51	9.86	34.39	47.70
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	0.8704	0.8833	0.8326	0.8543	0.7308	0.7472
N	3358185	2245432	3358185	2245432	3010150	2047497

Notes: This table reports results using a WFH measure constructed based on the occupational WFH scores from Hansen et al. (2023). This measure is based on job postings and captures the prevalence of both fully remote and hybrid work arrangements. Standard errors are clustered at the MSA level. All specifications include MSA and year fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B10: Effects of WFH Opportunities on LFP: Continuous vs. Binary Treatment

	Female		Mothers		College Mothers	
	Continuous	Binary	Continuous	Binary	Continuous	Binary
Transition (2020–2021)	0.0003	0.0001	0.0021***	0.0026*	0.0031***	0.0033***
	(0.0005)	(0.0009)	(0.0008)	(0.0013)	(0.0007)	(0.0011)
Stable (2022–2024)	0.0008*	0.0002	0.0030***	0.0031***	0.0036***	0.0033***
	(0.0005)	(0.0008)	(0.0007)	(0.0009)	(0.0007)	(0.0010)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes
Control Mean	0.8722	0.8722	0.8690	0.8690	0.8813	0.8813
N	6579056	6579056	2477570	2477570	1754457	1754457

Notes: This table compares estimates from the continuous treatment model against a binary treatment specification as a robustness check on the linearity assumption. The continuous model interacts the pre-determined MSA-level WFH exposure index with year indicators. The binary model replaces the continuous index with an indicator for whether the MSA's WFH exposure is above the sample median. Binary model coefficients are scaled by the difference in mean WFH exposure between above- and below-median MSAs (1.526) to make them comparable in magnitude to the continuous estimates. Standard errors for the binary model are scaled accordingly. Standard errors clustered at the MSA level are reported in parentheses. All models include individual-level controls, year fixed effects, and MSA fixed effects. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B11: WFH on Labor Market Outcomes: Comparison Across Geographic Units

	Labor Force		Employment		Full-Time	
	(1)	(2)	(3)	(4)	(5)	(6)
	Mothers	College Mothers	Mothers	College Mothers	Mothers	College Mothers
Panel A: Metropolitan Statistical Areas						
2020–2021	0.0021***	0.0031***	0.0007	0.0026**	0.0044***	0.0065***
	(0.0008)	(0.0007)	(0.0013)	(0.0011)	(0.0011)	(0.0011)
2022–2024	0.0030***	0.0036***	0.0004	0.0019***	0.0064***	0.0083***
	(0.0007)	(0.0007)	(0.0009)	(0.0007)	(0.0013)	(0.0014)
Control Mean	0.8690	0.8813	0.8295	0.8520	0.7184	0.7361
N	2477570	1754457	2477570	1754457	2222120	1599723
Panel B: Commuting Zones 1990						
2020–2021	0.0001	0.0012**	-0.0028***	-0.0011	0.0007	0.0033***
	(0.0005)	(0.0005)	(0.0009)	(0.0008)	(0.0009)	(0.0010)
2022–2024	0.0022***	0.0031***	-0.0003	0.0015***	0.0048***	0.0073***
	(0.0004)	(0.0004)	(0.0005)	(0.0005)	(0.0010)	(0.0011)
Control Mean	0.8634	0.8827	0.8260	0.8562	0.7230	0.7443
N	6200811	4188522	6200811	4188522	5561545	3839649
Panel C: Commuting Zones 2020						
2020–2021	0.0001	0.0013**	-0.0025***	-0.0006	0.0008	0.0036***
	(0.0005)	(0.0006)	(0.0009)	(0.0008)	(0.0010)	(0.0010)
2022–2024	0.0026***	0.0031***	0.0006	0.0018***	0.0049***	0.0074***
	(0.0005)	(0.0005)	(0.0005)	(0.0006)	(0.0012)	(0.0013)
Control Mean	0.8641	0.8827	0.8260	0.8558	0.7207	0.7424
N	4931424	3339758	4931424	3339758	4420858	3058300
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table compares estimates of the effect of WFH opportunities on labor force participation across three geographic unit definitions: MSAs (Panel A), commuting zones based on the 1990 delineation following Dorn (2009) (Panel B), and commuting zones based on the 2020 delineation from the Economic Research Service (Panel C). All specifications include year fixed effects, and geographic unit fixed effects. Standard errors are clustered at the geographic unit level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table B12: WFH on Labor Market Outcomes: Comparison Across Geographic Units (Weighted)

	Labor Force		Employment		Full-Time	
	(1)	(2)	(3)	(4)	(5)	(6)
	Mothers	College Mothers	Mothers	College Mothers	Mothers	College Mothers
Panel A: Metropolitan Statistical Areas						
2020–2021	0.0016	0.0035***	0.0004	0.0030**	0.0016	0.0037***
	(0.0010)	(0.0010)	(0.0017)	(0.0013)	(0.0014)	(0.0014)
2022–2024	0.0026***	0.0033***	-0.0000	0.0013	0.0047***	0.0073***
	(0.0009)	(0.0009)	(0.0012)	(0.0008)	(0.0015)	(0.0015)
Control Mean	0.8762	0.8878	0.8337	0.8565	0.7277	0.7472
N	2477570	1754457	2477570	1754457	2222120	1599723
Panel B: Commuting Zones 1990						
2020–2021	-0.0010	0.0009	-0.0046***	-0.0023**	-0.0003	0.0024**
	(0.0007)	(0.0007)	(0.0012)	(0.0010)	(0.0009)	(0.0010)
2022–2024	0.0014***	0.0025***	-0.0016**	0.0005	0.0032***	0.0059***
	(0.0005)	(0.0006)	(0.0007)	(0.0006)	(0.0011)	(0.0012)
Control Mean	0.8702	0.8874	0.8300	0.8586	0.7315	0.7525
N	6200811	4188522	6200811	4188522	5561545	3839649
Panel C: Commuting Zones 2020						
2020–2021	-0.0008	0.0011	-0.0041***	-0.0015	-0.0007	0.0023**
	(0.0008)	(0.0007)	(0.0012)	(0.0011)	(0.0010)	(0.0011)
2022–2024	0.0015**	0.0022***	-0.0011	0.0005	0.0029**	0.0057***
	(0.0006)	(0.0007)	(0.0008)	(0.0007)	(0.0013)	(0.0014)
Control Mean	0.8708	0.8874	0.8299	0.8581	0.7297	0.7510
N	4931424	3339758	4931424	3339758	4420858	3058300
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fixed Effects	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table compares estimates of the effect of WFH opportunities on labor force participation across three geographic unit definitions: MSAs (Panel A), commuting zones based on the 1990 delineation following Dorn (2009) (Panel B), and commuting zones based on the 2020 delineation from the Economic Research Service (Panel C). All estimates are weighted by ACS person weights. All specifications include individual-level controls, year fixed effects, and geographic unit fixed effects. Standard errors are clustered at the geographic unit level. * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Appendix C Analysis Using DDD Specification

While the standard two-way fixed effects difference-in-differences model provides a transparent framework for the estimation, incorporating a Difference-in-Differences-in-Differences (DDD) specification offers additional analytical advantages and addresses potential confounding factors. The key insight is that if males do not exhibit corresponding effects from WFH opportunities, they serve as an excellent control group for identifying female-specific responses. The lack of significant effects among males would constitute an important placebo test, alleviating concerns that the observed effects are driven by broader economic trends, regional shocks, or methodological artifacts rather than representing genuine increases in female labor force participation. However, if males also experience labor market effects from WFH opportunities, perhaps through different mechanisms or with different magnitudes, the triple-difference approach provides crucial advantages by allowing us to isolate the differential impact on women relative to men. In addition, this specification directly addresses concerns raised in the recent continuous-treatment DiD literature (Callaway et al., 2024): by treating men as effectively never treated under the assumption that men's labor supply is not affected by the WFH opportunity measure, the triple-difference approach provides a valid within-MSA control group that does not rely on lesser-treated units as counterfactuals.

Specifically, I estimate the following triple-difference specification:

$$\begin{aligned} \mathrm{Y}_{imt} = & \alpha_1 \cdot \mathit{Female}_i + \alpha_2 \cdot (\mathit{Female}_i \times \mathrm{WFH}_{m,\mathrm{pre}}) \\ & + \kappa \cdot (\mathit{Female}_i \times \mathit{Post}_t) \\ & + \lambda \cdot (\mathit{Female}_i \times \mathrm{WFH}_{m,\mathrm{pre}} \times \mathit{Post}_t) \\ & + \mathbf{X}'_{it}\beta + \tau_{mt} + \mu_{imt} \end{aligned} \tag{4}$$

Here, *Female* is a dummy variable indicating whether the individual i is female. *Post* is an indicator denoting the post-pandemic period covering the years of 2020 to 2024. The MSA-by-year fixed effects τ_{mt} account for baseline differences across MSA and any time-varying factors common to all individuals within a given MSA. These fixed effects absorb the direct effect of WFH potential, direct post-pandemic effects, and their two-way interaction $WFH_{m,pre} \times Post_t$. The remaining terms in the specification capture additional variation as follows. First, the $Female_i$ term reflects the baseline difference in labor force participation between women and men. Second, the interaction term $Female_i \times WFH_{m,pre}$ isolates how pre-pandemic WFH potential influences female labor force participation, independent of time. Third, $Female_i \times Year_t$ captures time-varying differences in labor force participation trends

for women relative to men.

Lastly, the triple interaction term $Female_i \times WFH_{m,pre} \times Year_t$ estimates the differential impact of WFH potential on female labor force participation over time relative to men, which is the primary coefficient of interest in this analysis. Additionally, the vector of individual-level controls X_{it} includes variables such as age, race, marital status, educational attainment, and the presence of children under age 5 to adjust for other potential confounding factors. Standard errors are clustered at the MSA-by-year level.

Table B8 Panel I presents the triple difference estimates comparing the effects of WFH opportunities on mothers relative to fathers. The results confirm significantly stronger positive effects on mother labor market outcomes, with college mothers showing particularly robust responses across all outcomes. Importantly, these triple difference findings closely mirror the main two-way DID results, confirming that the observed female labor force participation gains represent gender-specific benefits for women rather than general labor market improvements affecting both men and women.

Appendix D Dingel and Neiman WFH Measure

Dingel and Neiman (2020) developed a measure to classify occupations based on their feasibility to be performed entirely from home. The foundation of their classification system relies on two comprehensive surveys from O*NET, a database sponsored by the US Department of Labor that provides detailed information about nearly 1,000 occupations. The first survey, called the Work Context Questionnaire, examines physical and social factors that shape the nature of work, such as interpersonal relationships, physical conditions, and structural job characteristics. The second survey, the Generalized Work Activities Questionnaire, focuses on general job behaviors that occur across multiple jobs, including how workers process information, interact with others, engage in mental tasks, and produce work outputs. These surveys are quite robust, with the median occupation receiving responses from 26 and 25 workers respectively for the Work Context and Generalized Work Activities questionnaires.

The authors developed a systematic approach to classify jobs as unable to be performed at home based on specific criteria from these surveys. From the Work Context survey, they identified seven key conditions that would make remote work impossible. These include using email less than monthly, dealing with violent people weekly, working outdoors daily, having weekly exposure to diseases or infections, experiencing minor injuries weekly, spending the majority of time walking or running, or requiring protective equipment for most of the workday. Similarly, from the Generalized Work Activities survey, they identified eight critical activities that, if rated as "very important," would preclude working from home. These activities include performing physical tasks, handling objects, controlling machines, operating vehicles, working directly with the public, and various types of equipment maintenance and inspection.

To validate their classification, Dingel and Neiman manually reviewed and scored occupations, achieving strong alignment between their algorithmic and manual assessments. They further integrated these classifications with employment data from the Bureau of Labor Statistics (BLS), enabling them to calculate weighted measures of work-from-home feasibility. These weights reflected both the prevalence and wages of each occupation, ensuring the measure accurately captured the economic significance of remote work potential.

The final WFH measure assigned a binary classification to each occupation, indicating whether it could be performed entirely from home. These binary scores were then aggregated across industries, metropolitan areas, and national employment data to estimate the share of jobs suitable for remote work. Their analysis revealed that approximately 37% of U.S. jobs could be done entirely from home, with substantial variation across cities, industries, and income levels. By mapping U.S. occupational codes to international standards, they extended

their methodology to other countries, highlighting disparities in remote work feasibility between high- and low-income economies.

Stability of the Measure

I employ Dingel and Neiman's WFH measure at the occupational level by linking the scores to individuals in the dataset based on their reported occupation codes. However, the O*NET database, which forms the basis of Dingel and Neiman's measure, undergoes regular updates to reflect the evolving nature of work and maintain its relevance in a dynamic labor market. These updates, conducted multiple times annually, incorporate new data on job requirements, worker attributes, and workplace conditions derived from comprehensive surveys such as the Work Context Questionnaire and the Generalized Work Activities Questionnaire. Additionally, frequent revisions ensure alignment with updated national occupational standards, such as the Standard Occupational Classification (SOC) system, enhancing consistency and accuracy in occupational data.

It is crucial to assess whether the WFH measure remains stable over time, as this ensures that any observed effects arise from the potential for remote work rather than changes in the underlying occupational classifications. While the O*NET database is updated multiple times each year, I focus on the major updates typically released each August to evaluate the temporal stability of the WFH measure.

Figure A18 presents a replication of Dingel and Neiman's WFH construction using O*NET data from 2016 to 2023. The results indicate that the share of occupations classified as suitable for remote work consistently remains around 37% across all years, with negligible fluctuations, which aligns with the results reported in Dingel and Neiman's original analysis. This consistency demonstrates the robustness of the measure and ensures that the analysis effectively isolates the effects of work-from-home potential without being confounded by temporal variations in occupational classifications. Such stability is critical for evaluating the long-term implications of remote work opportunities on labor market outcomes.

Appendix E Constructing PUMA-to-CZ Crosswalks

Appendix E.1 Overview

Commuting zones (CZs) are clusters of counties linked by commuting flows, originally developed by Tolbert and Sizer (1996) to delineate local labor markets that cover the entire United States. Two CZ vintages are currently available: CZ 1990, the long-standing definition built from 1990 commuting flows and popularized by Autor et al. (2019); Autor and Dorn (2013), and CZ 2020, a new definition recently released by the USDA Economic Research Service based on the methodology of Fowler (2024).

Public survey data such as the ACS usually do not directly identify commuting zones. PUMAs are the smallest geographic units identified in the ACS, and researchers must map PUMAs to CZs using crosswalks. The most widely used approach in the economics literature follows Autor and Dorn (2013) and Autor et al. (2019), which provides crosswalks from PUMAs to CZ 1990 with population-share allocation factors for PUMA vintages up to 2010.

A complication is that the Census Bureau redraws PUMA boundaries every ten years based on population information from the most recent decennial census. Under this redrawing, PUMAs may split, merge, disappear, or newly emerge. This poses a challenge for maintaining a consistent geographic unit when a sample spans two or more PUMA vintages. For example, my sample straddles two PUMA vintages: the 2012–2021 ACS uses 2010 PUMA definitions and the 2022–2024 ACS uses 2020 PUMA definitions. No widely used crosswalk currently exists for mapping PUMA 2020 to either CZ vintage.

Motivated by the absence of a widely used PUMA 2020 crosswalk, I construct mappings for both CZ vintages covering PUMA vintages 1990 through 2020. For CZ 1990, I first replicate the Autor et al. (2019); Autor and Dorn (2013) crosswalks for PUMAs 1990, 2000, and 2010 to verify that my pipeline reproduces their published files, then extend to PUMA 2020 by chaining through PUMA 2010 using the IPUMS PUMA 2010-to-PUMA 2020 equivalency file. For CZ 2020, which is built on 2020-era county geographies, I apply the same county-as-bridge logic: I first construct a direct PUMA 2020-to-CZ 2020 crosswalk through the 2020 county, then extend to PUMA 2010 by chaining through PUMA 2020.

Two ideas unify the construction of the crosswalks. First, the allocation factor in each crosswalk is a population share: for a (PUMA, CZ) pair, it equals the share of the PUMA's population residing in that CZ, measured from the decennial census that defined the PUMA vintage. For example, a PUMA 2020 that splits 70/30 across two CZ 2020s generates two rows in the crosswalk with factors 0.70 and 0.30. The allocation factor therefore has a clean probabilistic interpretation: it is the probability that a person drawn at random from the PUMA lives in that CZ. Second, when the PUMA vintage and the CZ vintage share

a common county base, I merge directly through the county; when they do not, I chain through an intermediate PUMA vintage using the IPUMS PUMA equivalency files. In the latter case, the allocation factor for a (source PUMA, CZ) pair is the product of the source-PUMA-to-intermediate-PUMA population share and the intermediate-PUMA-to-CZ population share, summed over all intermediate PUMAs that the source PUMA overlaps. This sum is the share of the source PUMA's population residing in that CZ.

The subsections document the construction and usage of each crosswalk.

Appendix E.2 PUMA 1990 → CZ 1990

Approach. I construct a crosswalk from PUMA 1990 to CZ 1990 using the 1990 county as the bridge and 1990 census population shares as allocation factors.

Inputs.

- `geocorr1990.csv`: Geocorr 1990 PUMA 1990-to-county 1990 allocation file, weighted by 1990 census population; source: Missouri Census Data Center, Geocorr 1990 (`https://mcdc.missouri.edu/applications/geocorr1990.html`).
- `counties90_cz90.csv`: county 1990-to-CZ 1990 mapping using the ERS90 classification (Tolbert and Sizer, 1996); obtained from the Penn State Commuting Zones repository (Fowler and Jensen, 2020), `https://sites.psu.edu/psucz/data/`.

Steps.

1. Load the Geocorr 1990 file. Each row is a (PUMA 1990, county 1990) intersection with an allocation factor equal to the share of the PUMA 1990's population residing in that county 1990.
2. Merge the county 1990-to-CZ 1990 mapping on the five-digit county FIPS code.
3. Collapse to the (PUMA 1990, CZ 1990) level by summing the PUMA 1990-to-county 1990 allocation factors over all counties 1990 belonging to the same CZ 1990. The resulting sum is the PUMA 1990-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 1990's population residing in that CZ 1990.
4. Verify that the allocation factors sum to one within each PUMA 1990.

Output and Usage. The crosswalk is saved as `puma1990_cz1990_crosswalk.csv` and `puma1990_cz1990_crosswalk.dta`, with columns `puma1990`, `cz1990`, and `afactor`. The `puma1990` column is a six-character string formed by concatenating the two-digit state FIPS code with the four-digit 1990 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The crosswalk matches Autor and Dorn (2013)'s published `cw_puma1990_czone.dta` exactly: 100 percent of PUMA–CZ pairs appear in both files, with a mean absolute difference of 0.00009 in allocation factors and a maximum of 0.002 driven by rounding. This replication confirms that the ERS90 codes used here are identical to Autor and Dorn's `czone` codes.

Description. I obtain PUMA 1990-to-county 1990 population shares from the Missouri Census Data Center's Geocorr tool, which queries the MABLE (Master Area Block Level Equivalency) database. I select PUMA A (5% Sample) as the source geography, county as the target geography, and 1990 census population as the weighting variable. For every (PUMA 1990, county 1990) intersection, the extract reports an allocation factor equal to the share of the PUMA 1990's population residing in that county 1990, computed directly from block-level populations within MABLE. By construction, these shares sum to one within each PUMA 1990.

The county-to-CZ mapping uses the ERS90 classification, which is the 1990 commuting zone delineation originally constructed by Tolbert and Sizer (1996) at the USDA Economic Research Service through hierarchical cluster analysis of county-to-county commuter flows from the 1990 Census journey-to-work data. I obtain the file from the Penn State Commuting Zones repository assembled by Fowler and Jensen (2020), which distributes the ERS90 delineation together with delineations from other agencies and with fit statistics for each commuting zone. Because counties 1990 nest exactly within CZs 1990, assigning each (PUMA 1990, county 1990) row to a CZ 1990 is a simple merge on the five-digit county FIPS code. I then collapse to the (PUMA 1990, CZ 1990) level by summing the PUMA 1990-to-county 1990 allocation factors over all counties 1990 sharing the same CZ 1990. The resulting `afactor` is the share of the PUMA 1990's population residing in that CZ 1990, interpretable as the probability that a person drawn at random from the PUMA 1990 lives in that CZ 1990. This construction involves no further approximation beyond the block-level weighting already contained in the Geocorr input. PUMAs 1990 do not nest within counties 1990, which is why the Geocorr step is needed in the first place: a single PUMA 1990 can span multiple counties 1990, and those counties 1990 may in turn belong to different CZs 1990.

Appendix E.3 PUMA 2000 → CZ 1990

Approach. I construct a crosswalk from PUMA 2000 to CZ 1990 using the 2000 county as the bridge and 2000 census population shares as allocation factors. Because the CZ 1990 file uses 1990-vintage county FIPS codes, I first reconcile the small number of county code changes that occurred between 1990 and 2000 before merging.

Inputs.

- `geocorr2000.csv`: Geocorr 2000 PUMA 2000-to-county 2000 allocation file, weighted by 2000 census population; source: Missouri Census Data Center, Geocorr 2000 (https://mcdc.missouri.edu/applications/geocorr2000.html).
- `counties90_cz90.csv`: county 1990-to-CZ 1990 mapping using the ERS90 classification (Tolbert and Sizer, 1996); obtained from the Penn State Commuting Zones repository (Fowler and Jensen, 2020), https://sites.psu.edu/psucz/data/.

Steps.

1. Load the Geocorr 2000 file. Each row is a (PUMA 2000, county 2000) intersection with an allocation factor equal to the share of the PUMA 2000's population residing in that county 2000.
2. Identify county 2000 FIPS codes that do not appear in the county 1990-to-CZ 1990 mapping and remap them to their 1990 equivalents: Denali Borough, AK (02068); Skagway-Hoonah-Angoon, AK (02232); Yakutat City and Borough, AK (02282); and Miami-Dade County, FL (12086). These four remappings are documented in the Census Bureau's county code change files (https://www.census.gov/programs-surveys/geography/technical-documentation/county-changes.January_2020.html).
3. Merge the county 1990-to-CZ 1990 mapping on the reconciled five-digit county FIPS code.
4. Collapse to the (PUMA 2000, CZ 1990) level by summing the PUMA 2000-to-county 2000 allocation factors over all counties 2000 belonging to the same CZ 1990. The resulting sum is the PUMA 2000-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 2000's population residing in that CZ 1990.
5. Verify that the allocation factors sum to one within each PUMA 2000.

Output and Usage. The crosswalk is saved as `puma2000_cz1990_crosswalk.csv` and `puma2000_cz1990_crosswalk.dta`, with columns `puma2000`, `cz1990`, and `afactor`. The `puma2000` column is a six-character string formed by concatenating the two-digit state FIPS code with the four-digit 2000 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The crosswalk matches Autor et al. (2019)'s published `cw_puma2000_czone.dta` exactly: 2,880 PUMA–CZ pairs appear in both files with no pairs unique to either, and the mean absolute difference in allocation factors is 0.000087 (maximum 0.002, driven by rounding).

Description. I obtain PUMA 2000-to-county 2000 population shares from the Missouri Census Data Center's Geocorr 2000 tool, which queries the MABLE (Master Area Block Level Equivalency) database. I select PUMA (5% Sample) as the source geography, county as the target geography, and 2000 census population as the weighting variable. For every (PUMA 2000, county 2000) intersection, the extract reports an allocation factor equal to the share of the PUMA 2000's population residing in that county 2000, computed directly from block-level populations within MABLE. By construction, these shares sum to one within each PUMA 2000.

Merging the Geocorr 2000 file with the county 1990-to-CZ 1990 mapping is not a direct join on county FIPS because county boundaries and FIPS codes changed slightly between 1990 and 2000. Four counties appear in the Geocorr 2000 file under codes that are not present in the 1990-vintage CZ file. Denali Borough, AK (02068) was carved out of Yukon-Koyukuk (02290) in 1990; I remap 02068 to 02290 so that Denali's 2000 population is assigned to the CZ 1990 containing its pre-1990 parent. Skagway-Hoonah-Angoon, AK (02232) and Yakutat City and Borough, AK (02282) were both created in 1992 when the former Skagway-Yakutat-Angoon borough (02231) was partitioned; I remap both 02232 and 02282 to 02231. Miami-Dade County, FL (12086) was renamed from Dade County (12025) in 1997 with no change in territory; I remap 12086 to 12025. Two additional 1990-vintage codes appear in the CZ file but not in the Geocorr 2000 file: Yellowstone National Park, MT (30113), which was dissolved in 1997 and absorbed into Gallatin and Park counties, and South Boston City, VA (51780), which reverted to Halifax County in 1995. These simply drop out of the merge because no PUMA 2000 row references them. The remapping is documented in the Census Bureau's county code change files.

After the merge, I collapse to the (PUMA 2000, CZ 1990) level by summing the PUMA 2000-to-county 2000 allocation factors over all counties 2000 sharing the same CZ 1990. The

resulting `afactor` is the share of the PUMA 2000's population residing in that CZ 1990, interpretable as the probability that a person drawn at random from the PUMA 2000 lives in that CZ 1990. This construction involves no further approximation beyond the block-level weighting already contained in the Geocorr input. The final `afactor` is rounded to three decimal places.

Appendix E.4 PUMA 2010 → CZ 1990

Approach. I construct a crosswalk from PUMA 2010 to CZ 1990 by chaining through PUMA 2000, using the PUMA 2010-to-PUMA 2000 equivalency and the PUMA 2000-to-CZ 1990 crosswalk built in Appendix Appendix E.3.

Inputs.

- `puma2k_puma2010.csv`: PUMA 2010-to-PUMA 2000 equivalency file, weighted by 2010 census population; source: Missouri Census Data Center (https://mcdc.missouri.edu/geography/PUMAs.html).
- `puma2000_cz1990_crosswalk.csv`: population-weighted PUMA 2000-to-CZ 1990 crosswalk constructed in Appendix Appendix E.3.

Steps.

1. Load the PUMA 2010-to-PUMA 2000 equivalency file. Each row is a (PUMA 2010, PUMA 2000) pair with an allocation factor (`AFACT2`) equal to the share of the PUMA 2010's 2010 census population residing in that PUMA 2000.
2. Load the PUMA 2000-to-CZ 1990 crosswalk, which reports for each (PUMA 2000, CZ 1990) pair the share of the PUMA 2000's population residing in that CZ 1990.
3. Merge the two crosswalks on the six-character `puma2000` code.
4. Compute the combined allocation factor as the product of the PUMA 2010-to-PUMA 2000 factor and the PUMA 2000-to-CZ 1990 factor.
5. Collapse to the (PUMA 2010, CZ 1990) level by summing the combined allocation factors over all PUMA 2000 intermediaries sharing the same CZ 1990. The resulting sum is the PUMA 2010-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 2010's population residing in that CZ 1990.
6. Verify that the allocation factors sum to one within each PUMA 2010.

Output and Usage. The crosswalk is saved as `puma2010_cz1990_crosswalk.csv` and `puma2010_cz1990_crosswalk.dta`, with columns `puma2010`, `cz1990`, and `afactor`. The `puma2010` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2010 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The crosswalk matches Autor et al. (2019)'s published `cw_puma2010_czone.dta` closely: 4,077 PUMA–CZ pairs appear in both files with no pairs unique to either, and the mean absolute difference in allocation factors is 0.000178 (maximum 0.001564). The small residual reflects compounded rounding across the two-step allocation and is consistent with the three-decimal precision applied at each step.

Description. PUMA 2010 boundaries are aligned with the 2010 decennial census geography, while CZ 1990 boundaries are aligned with the 1990 decennial census geography. Rather than map PUMA 2010 directly to 1990-vintage counties, I chain through PUMA 2000, for which I have already built a crosswalk to CZ 1990 in Appendix Appendix E.3, and I rely on the Missouri Census Data Center's PUMA 2010-to-PUMA 2000 equivalency file to bridge the first link.

The PUMA 2010-to-PUMA 2000 equivalency file is published by the Missouri Census Data Center. For each (PUMA 2010, PUMA 2000) pair, it reports the share of the PUMA 2010's 2010 census population residing in that PUMA 2000, computed directly from census block-level populations within the MABLE (Master Area Block Level Equivalency) database. By construction, these shares sum to one within each PUMA 2010. I then merge this file with the PUMA 2000-to-CZ 1990 crosswalk from Appendix Appendix E.3, which is itself built from block-level populations.

The combined allocation factor for a (PUMA 2010, PUMA 2000, CZ 1990) triple is the product of the PUMA 2010-to-PUMA 2000 share and the PUMA 2000-to-CZ 1990 share. Summing over PUMA 2000 intermediaries within a common CZ 1990 yields the share of the PUMA 2010's population residing in that CZ 1990, which is the `afactor`. The final `afactor` is rounded to three decimal places.

Appendix E.5 PUMA 2020 → CZ 1990

Approach. I construct a crosswalk from PUMA 2020 to CZ 1990 by chaining through PUMA 2010, using the PUMA 2010-to-PUMA 2020 equivalency from IPUMS and the PUMA 2010-to-CZ 1990 crosswalk built in Appendix Appendix E.4.

Inputs.

- `PUMA2010_PUMA2020_crosswalk.xls`: IPUMS PUMA 2010-to-PUMA 2020 equivalency file; source: IPUMS USA, PUMA composition files (https://usa.ipums.org/usa/volii/pumas20.shtml).
- `puma2010_cz1990_crosswalk.csv`: population-weighted PUMA 2010-to-CZ 1990 crosswalk constructed in Appendix Appendix E.4.

Steps.

1. Load the IPUMS PUMA 2010-to-PUMA 2020 equivalency file. Each row is a (PUMA 2010, PUMA 2020) pair with a variable `pPUMA20_Pop20` equal to the percentage of the PUMA 2020's 2020 population residing in that PUMA 2010. Divide by 100 to obtain the allocation factor.
2. Exclude Puerto Rico PUMAs, which are not covered by CZ 1990.
3. Load the PUMA 2010-to-CZ 1990 crosswalk, which reports for each (PUMA 2010, CZ 1990) pair the share of the PUMA 2010's population residing in that CZ 1990.
4. Merge the two crosswalks on the seven-character `puma2010` code.
5. Compute the combined allocation factor as the product of the PUMA 2020-to-PUMA 2010 factor and the PUMA 2010-to-CZ 1990 factor.
6. Collapse to the (PUMA 2020, CZ 1990) level by summing the combined allocation factors over all PUMA 2010 intermediaries sharing the same CZ 1990. The resulting sum is the PUMA 2020-to-CZ 1990 allocation factor (`afactor`), which represents the share of the PUMA 2020's population residing in that CZ 1990.
7. Verify that the allocation factors sum to one within each PUMA 2020.

Output and Usage. The crosswalk is saved as `puma2020_cz1990_crosswalk.csv` and `puma2020_cz1990_crosswalk.dta`, with columns `puma2020`, `cz1990`, and `afactor`. The `puma2020` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2020 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. I report the internal consistency check: the allocation factors sum to one within each of the 2,462 mainland PUMAs 2020 (mean 0.999998, minimum 0.998, maximum 1.004). The residual spread reflects compounded rounding across the two chained allocation steps.

Description. PUMA 2020 boundaries are aligned with the 2020 decennial census geography, while CZ 1990 boundaries are aligned with the 1990 decennial census geography. Rather than map PUMA 2020 directly to 1990-vintage counties, I chain through PUMA 2010, for which I have already built a crosswalk to CZ 1990 in Appendix Appendix E.4. The IPUMS equivalency file bridges the first link. The effective full chain is PUMA 2020 → PUMA 2010 → PUMA 2000 → CZ 1990, with no cross-vintage county remapping at any step.

The IPUMS equivalency file reports, for each (PUMA 2010, PUMA 2020) pair, the share of the PUMA 2020's 2020 population residing in that PUMA 2010, which sums to one within each PUMA 2020 by construction. I then merge this file with the PUMA 2010-to-CZ 1990 crosswalk from Appendix Appendix E.4. The combined allocation factor for a (PUMA 2020, PUMA 2010, CZ 1990) triple is the product of the PUMA 2020-to-PUMA 2010 share and the PUMA 2010-to-CZ 1990 share. Summing over PUMA 2010 intermediaries within a common CZ 1990 yields the share of the PUMA 2020's population residing in that CZ 1990, which is the `afactor`. The final `afactor` is rounded to three decimal places.

Appendix E.6 PUMA 2020 → CZ 2020

Approach. I construct a crosswalk from PUMA 2020 to CZ 2020, using the 2020 county as the bridge.

Inputs.

- `geocorr2022_2601502573.csv`: Geocorr 2022 PUMA 2020-to-county 2020 allocation file, weighted by 2020 census population; source: Missouri Census Data Center, Geocorr 2022 (https://mcdc.missouri.edu/applications/geocorr2022.html).
- `commuting-zones-2020.csv`: county 2020-to-CZ 2020 mapping. The file places the 3,222 counties and county-equivalents in the United States and Puerto Rico into 598 distinct labor markets, following the Fowler (2024) methodology with an added contiguity constraint; source: USDA Economic Research Service (https://www.ers.usda.gov/data-products/commuting-zones-and-labor-market-areas).

Steps.

1. Load the Geocorr 2022 file and drop rows with missing PUMA codes. Each remaining row is a (PUMA 2020, county 2020) intersection with an allocation factor equal to the share of the PUMA 2020's population residing in that county 2020.

2. Standardize county FIPS codes to five-digit strings in both files.

3. Merge the county 2020-to-CZ 2020 mapping onto the Geocorr file on the five-digit county FIPS code.

4. Collapse to the (PUMA 2020, CZ 2020) level by summing the PUMA 2020-to-county 2020 allocation factors over all counties 2020 belonging to the same CZ 2020. The resulting sum is the PUMA 2020-to-CZ 2020 allocation factor (`afactor`), which represents the share of the PUMA 2020's population residing in that CZ 2020.

5. Verify that the allocation factors sum to one within each PUMA 2020.

Output and Usage. The crosswalk is saved as `puma2020_cz2020_crosswalk.csv` and `puma2020_cz2020_crosswalk.dta`, with columns `puma2020`, `cz2020`, and `afactor`. The `puma2020` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2020 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The allocation factors sum to one within each of the 2,486 PUMAs 2020 (mean 0.999996, minimum 0.998, maximum 1.002). The residual spread reflects rounding in the collapse to three decimal places.

Description. PUMA 2020 boundaries are aligned with 2020 decennial census geography, and CZ 2020 is defined over 2020 counties, which include Connecticut planning regions and Puerto Rico municipios as county-equivalents. The 2020 county therefore serves as a natural bridge, and no county-code harmonization is required.

The Geocorr 2022 file reports, for each (PUMA 2020, county 2020) pair, the share of the PUMA's 2020 population residing in that county, computed from 2020 census block populations. These shares sum to one within each PUMA by construction. Because every county nests entirely within a single CZ under the ERS delineation, summing the PUMA-to-county allocation factor over all counties within a CZ yields the share of the PUMA 2020's population residing in that CZ 2020, which is the `afactor`.

Appendix E.7 PUMA 2010 → CZ 2020

Approach. I construct a crosswalk from PUMA 2010 to CZ 2020 by chaining through PUMA 2020, using the IPUMS PUMA 2010-to-PUMA 2020 equivalency and the PUMA 2020-to-CZ 2020 crosswalk built in Appendix Appendix E.6.

Inputs.

- `PUMA2010_PUMA2020_crosswalk.xls`: IPUMS PUMA 2010-to-PUMA 2020 equivalency file; source: IPUMS USA, PUMA composition files (https://usa.ipums.org/usa/volii/pumas20.shtml).
- `puma2020_cz2020_crosswalk.csv`: PUMA 2020-to-CZ 2020 crosswalk constructed in Appendix Appendix E.6.

Steps.

1. Load the IPUMS PUMA 2010-to-PUMA 2020 equivalency file. Each row is a (PUMA 2010, PUMA 2020) pair with a variable `pPUMA10_Pop20` equal to the percentage of the PUMA 2010's 2020 population residing in that PUMA 2020. Divide by 100 to obtain the allocation factor.
2. Load the PUMA 2020-to-CZ 2020 crosswalk, which reports for each (PUMA 2020, CZ 2020) pair the share of the PUMA 2020's population residing in that CZ 2020.
3. Merge the two crosswalks on the seven-character `puma2020` code.
4. Compute the combined allocation factor as the product of the PUMA 2010-to-PUMA 2020 factor and the PUMA 2020-to-CZ 2020 factor.
5. Collapse to the (PUMA 2010, CZ 2020) level by summing the combined allocation factors over all PUMA 2020 intermediaries sharing the same CZ 2020. The resulting sum is the PUMA 2010-to-CZ 2020 allocation factor (`afactor`), which represents the share of the PUMA 2010's population residing in that CZ 2020.
6. Verify that the allocation factors sum to one within each PUMA 2010.

Output and Usage. The crosswalk is saved as `puma2010_cz2020_crosswalk.csv` and `puma2010_cz2020_crosswalk.dta`, with columns `puma2010`, `cz2020`, and `afactor`. The `puma2010` column is a seven-character string formed by concatenating the two-digit state FIPS code with the five-digit 2010 PUMA code, both zero-padded. The `afactor` column is rounded to three decimal places.

Validation. The allocation factors sum to one within each of the 2,378 PUMAs 2010 (mean 0.999993, minimum 0.998, maximum 1.002). The residual spread reflects compounded rounding across the two chained allocation steps.

Description. PUMA 2010 boundaries are aligned with the 2010 decennial census geography, while CZ 2020 is defined over 2020 counties. Rather than map PUMA 2010 directly to 2020-vintage counties, I chain through PUMA 2020, for which I have already built a crosswalk to CZ 2020 in Appendix Appendix E.6. The IPUMS equivalency file bridges the first link.

The IPUMS equivalency file reports, for each (PUMA 2010, PUMA 2020) pair, the share of the PUMA 2010's 2020 population residing in that PUMA 2020, which sums to one within each PUMA 2010 by construction. I then merge this file with the PUMA 2020-to-CZ 2020 crosswalk from Appendix Appendix E.6. The combined allocation factor for a (PUMA 2010, PUMA 2020, CZ 2020) triple is the product of the PUMA 2010-to-PUMA 2020 share and the PUMA 2020-to-CZ 2020 share. Summing over PUMA 2020 intermediaries within a common CZ 2020 yields the share of the PUMA 2010's population residing in that CZ 2020, which is the `afactor`. The final `afactor` is rounded to three decimal places.